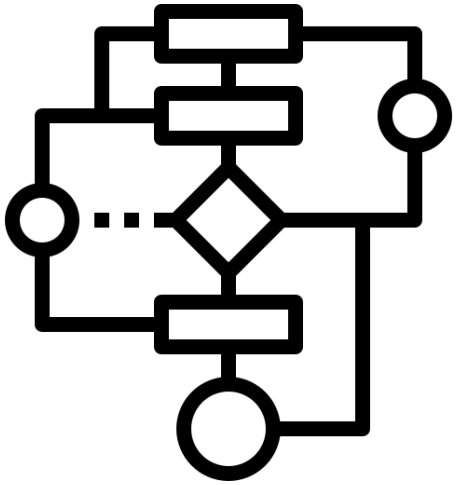

CS 310: Algorithms

Lecture 9

Instructor: Naveed Anwar Bhatti

Slides by Kevin Wayne (**heavily modified by Naveed Bhatti**).
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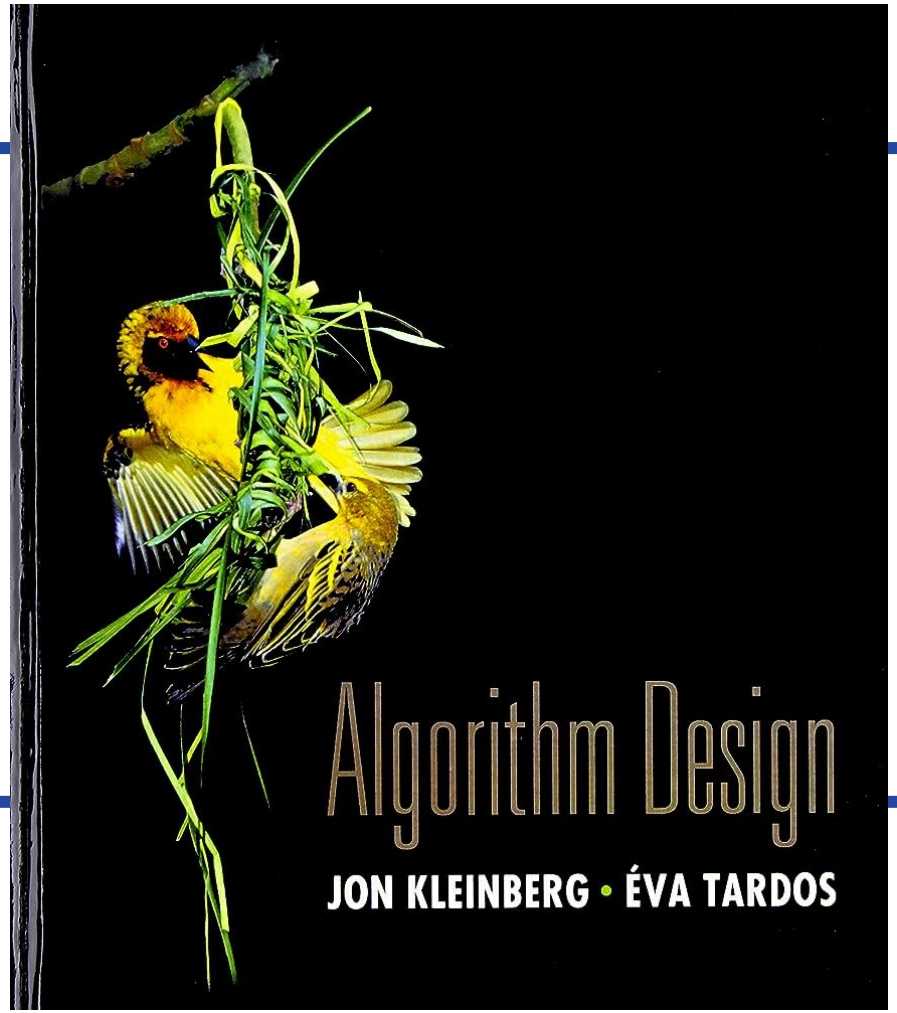

Chapter 3:
Graphs

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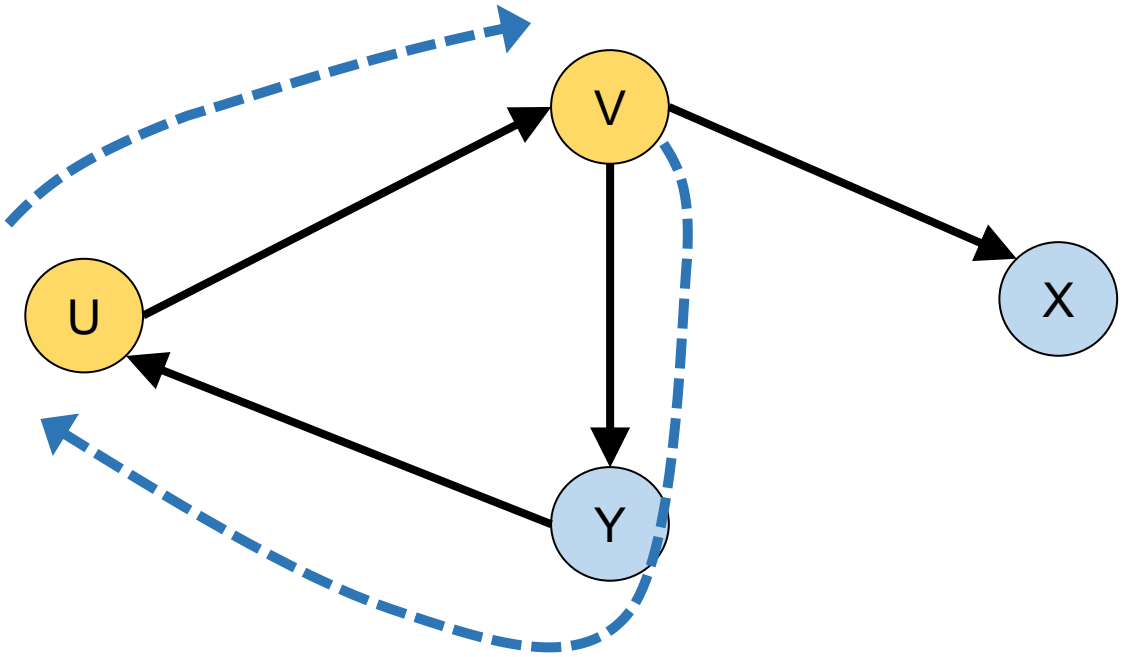
Section 3.5:
Connectivity in Directed Graphs

Strong Connectivity

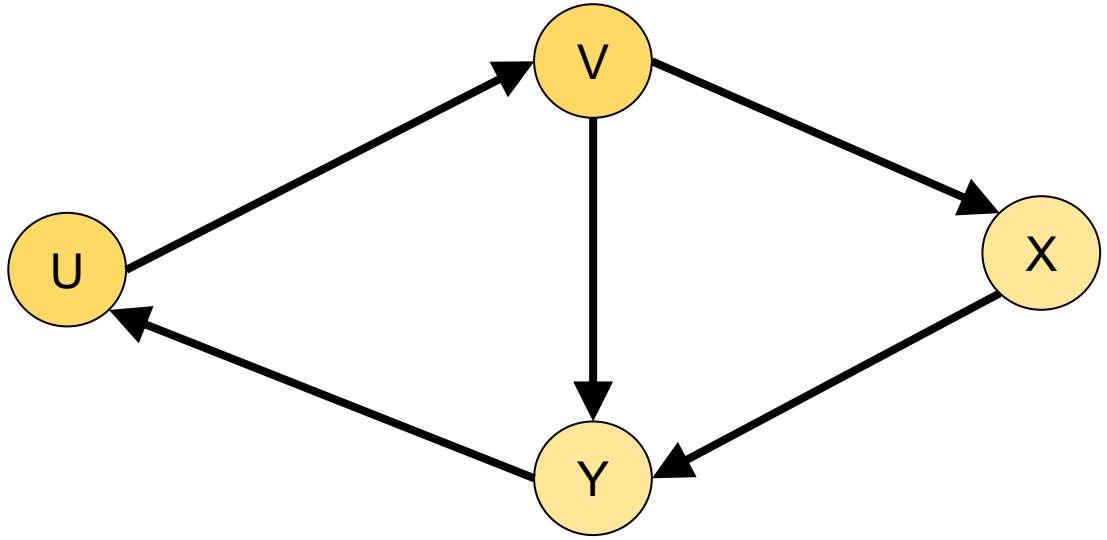
- **Def:** Node ***u*** and ***v*** are mutually reachable if there is a path from ***u*** to ***v*** and also a path from ***v*** to ***u***.



Strong Connectivity

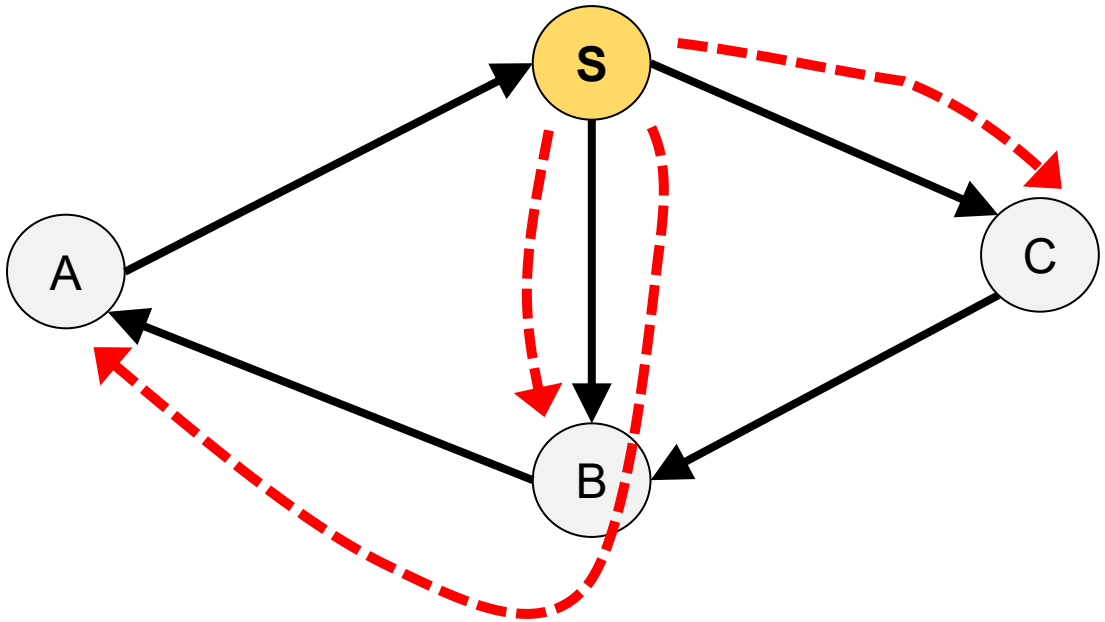
- **Def:** Node ***u*** and ***v*** are mutually reachable if there is a path from ***u*** to ***v*** and also a path from ***v*** to ***u***.

- **Def:** A graph is strongly connected if every pair of nodes is mutually reachable.



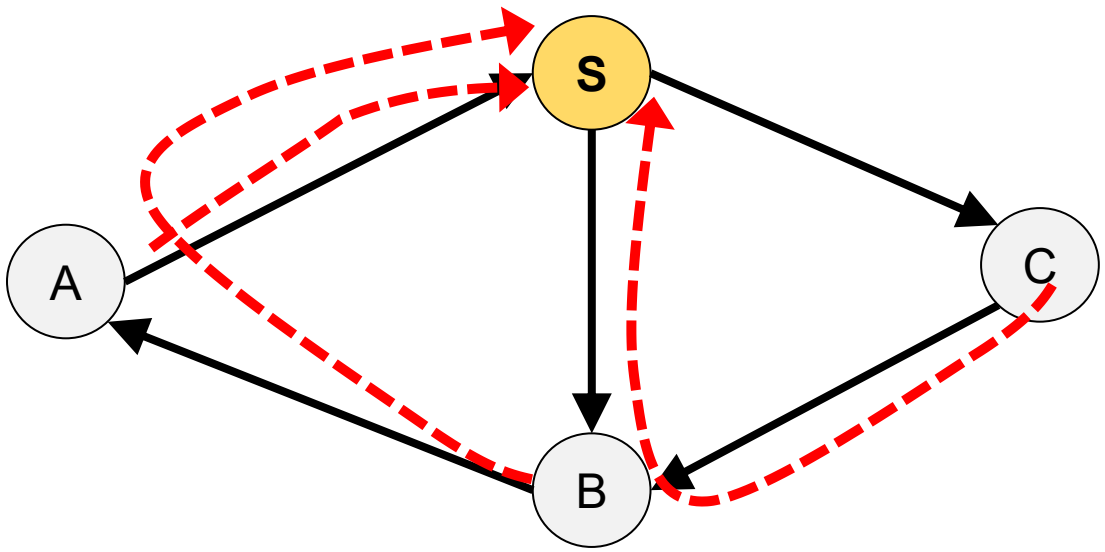
Strong Connectivity

- **Lemma.** Let ***s*** be any node in graph ***G***. ***G*** is ***strongly connected*** *iff* every node is reachable from ***s***, and ***s*** is reachable from every node.



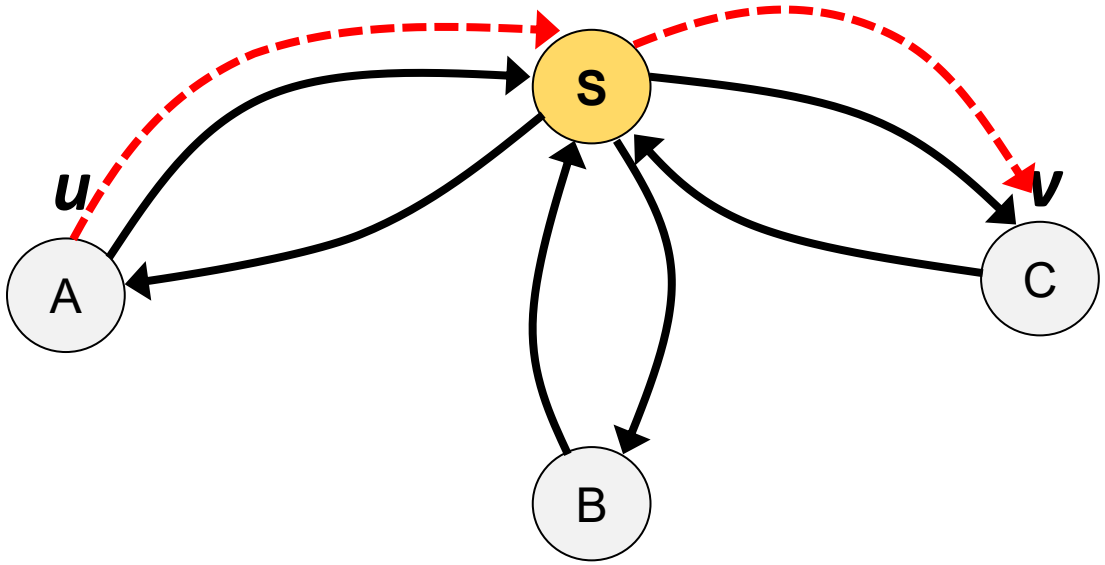
Strong Connectivity

- **Lemma.** Let ***s*** be any node in graph ***G***. ***G*** is ***strongly connected*** *iff* every node is reachable from ***s***, and ***s*** is reachable from every node**.**



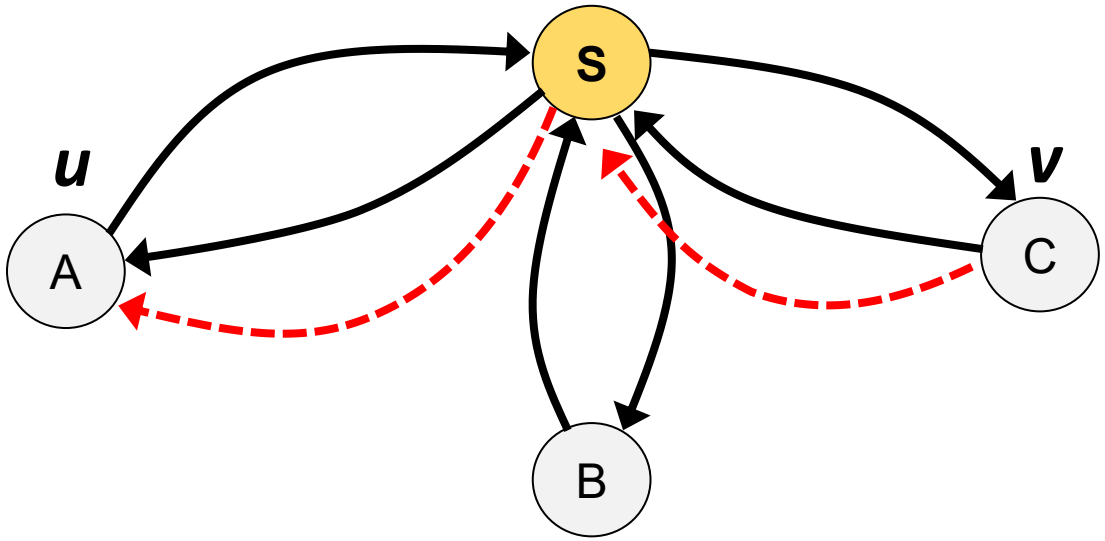
Strong Connectivity

- **Proof.** If every node is reachable from ***s*** and ***s*** is reachable from every node, then for any two nodes ***u*** and ***v*** in ***G***:
  - There is a path from **u** to **s**
  - And another from **s** to **v**
  - Combining these, ***u*** can reach ***v*** through **s**



Strong Connectivity

- **Proof.** If every node is reachable from ***s*** and ***s*** is reachable from every node, then for any two nodes ***u*** and ***v*** in ***G***:
  - There is a path from **u** to **s**
  - And another from **s** to **v**
  - Combining these, ***u*** can reach ***v*** through **s**
  - Similarly, ***v*** can reach ***u*** through ***s***

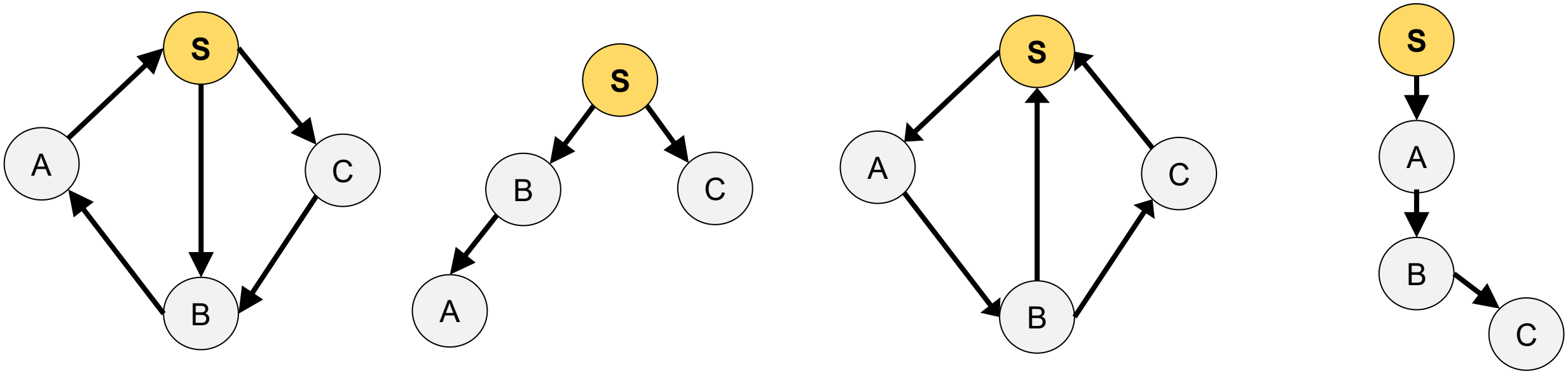


Strong Connectivity: Algorithm

- Pick any node s.
- Run BFS from s in G.
- Run BFS from s in G^{rev}

reverse orientation of every edge in **G**

- Return true ***iff*** all nodes reached in both BFS executions.
- Correctness follows immediately from previous lemma.



Strong Connectivity: Live Poll 1

What is the complexity of this algo?

A. $O(n)$
B. $O(n + m)$
C. $O(n^2)$
D. $O(m^2)$
E. None of above

Scan the QR code to vote or go to https://forms.office.com/r/Ne2tEuqWXa

Strong Connectivity: Live Poll 1

Only people in my organization can respond, Record name

1. What is the complexity of this algo?

O(n)	4%
O($n + m$)	64%
O(n^2)	16%
O(m^2)	4%
None of above	12%

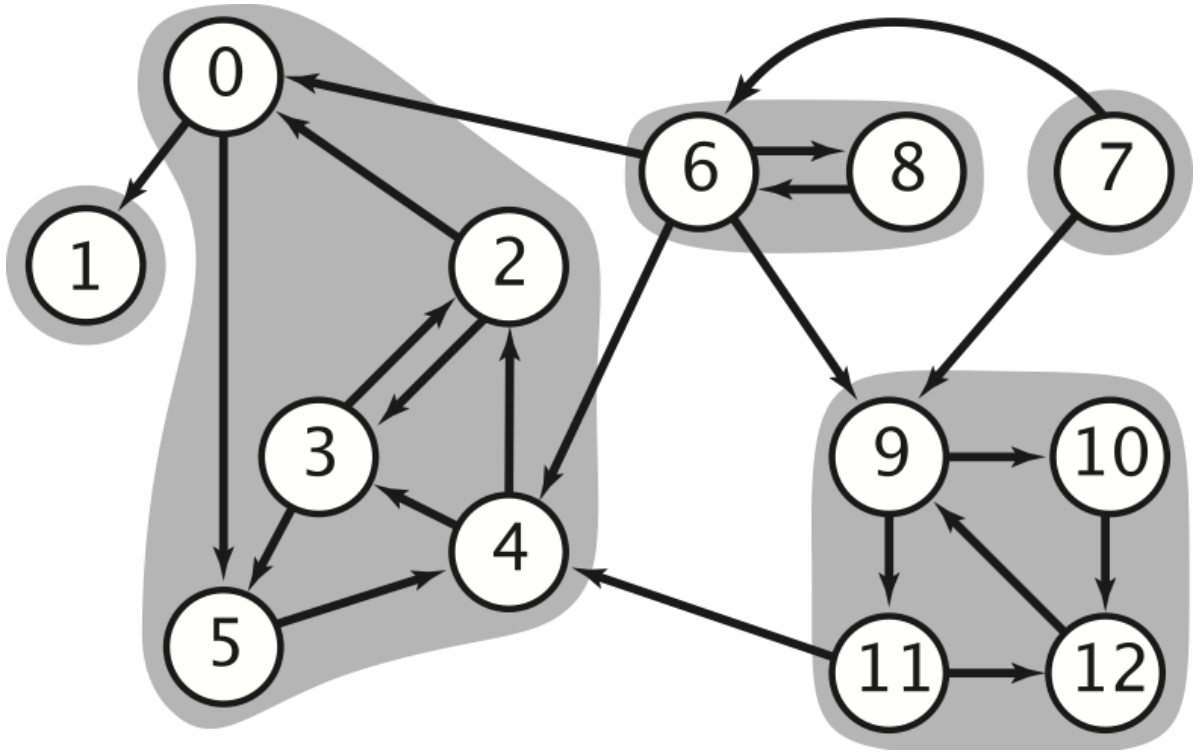
Scan the QR code to vote or go to https://forms.office.com/r/Ne2tEuqWXa

25 responses

1/1

Strong components

Def: A strong component is a maximal subset of mutually reachable nodes.



Algorithm for finding strong components in a directed graph

STRONG-COMPONENTS(G)

1 Call DFS(**G**) to compute finishing times $\boldsymbol{u.f}$ for each vertex $\boldsymbol{u}$

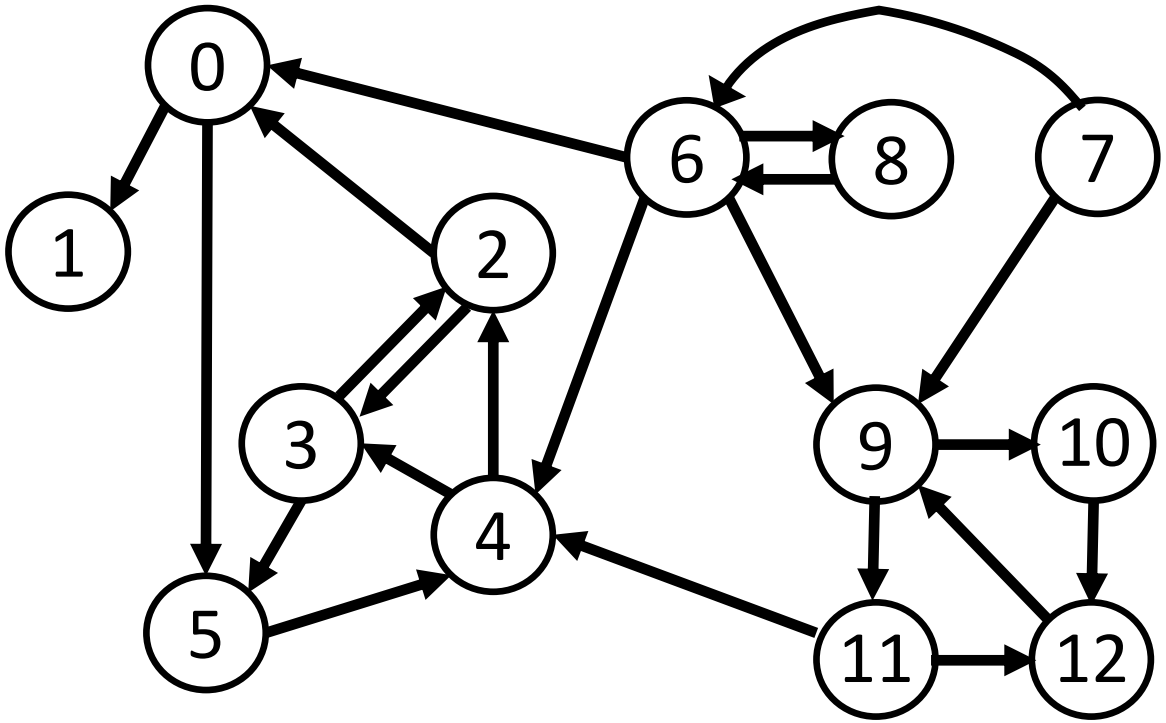
2 Compute $\boldsymbol{G^{reverse}}$

3 Call DFS($\boldsymbol{G^{reverse}}$), but in the main loop of DFS, consider the vertices in order of decreasing $\boldsymbol{u.f}$

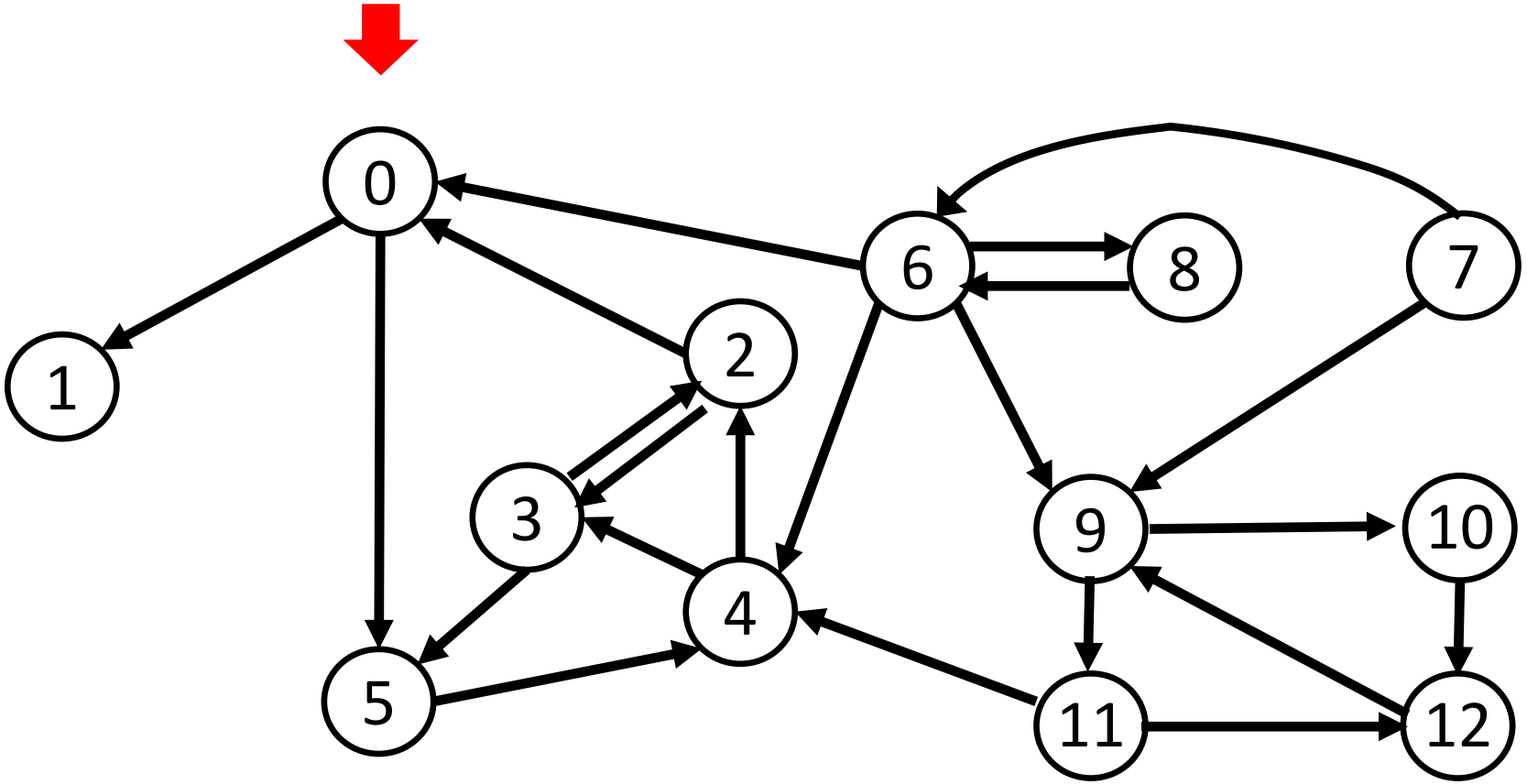
4 Output the vertices of each tree in the depth-first forest formed in line 3 as a separate strong component

Strong components

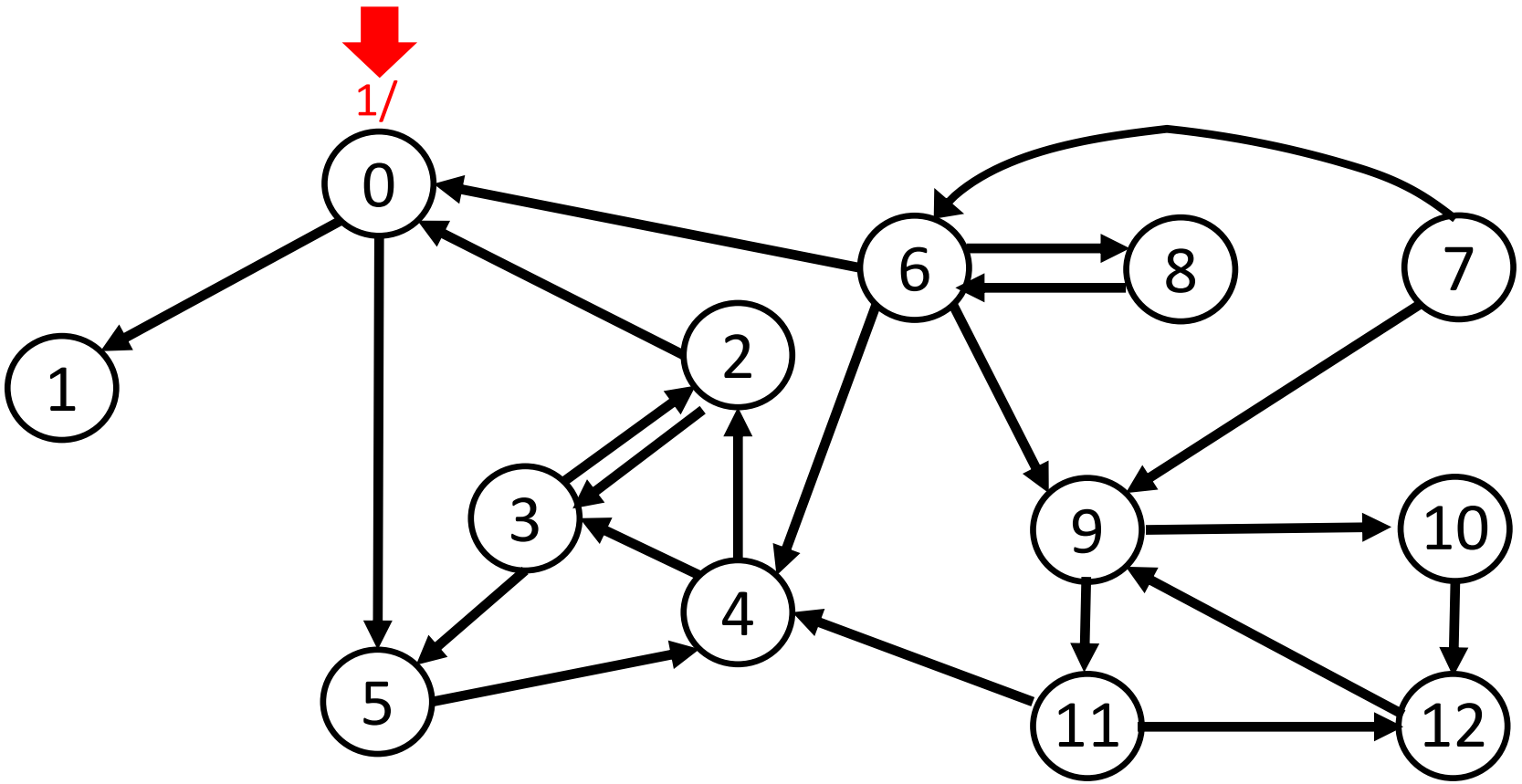
Def: A strong component is a maximal subset of mutually reachable nodes.



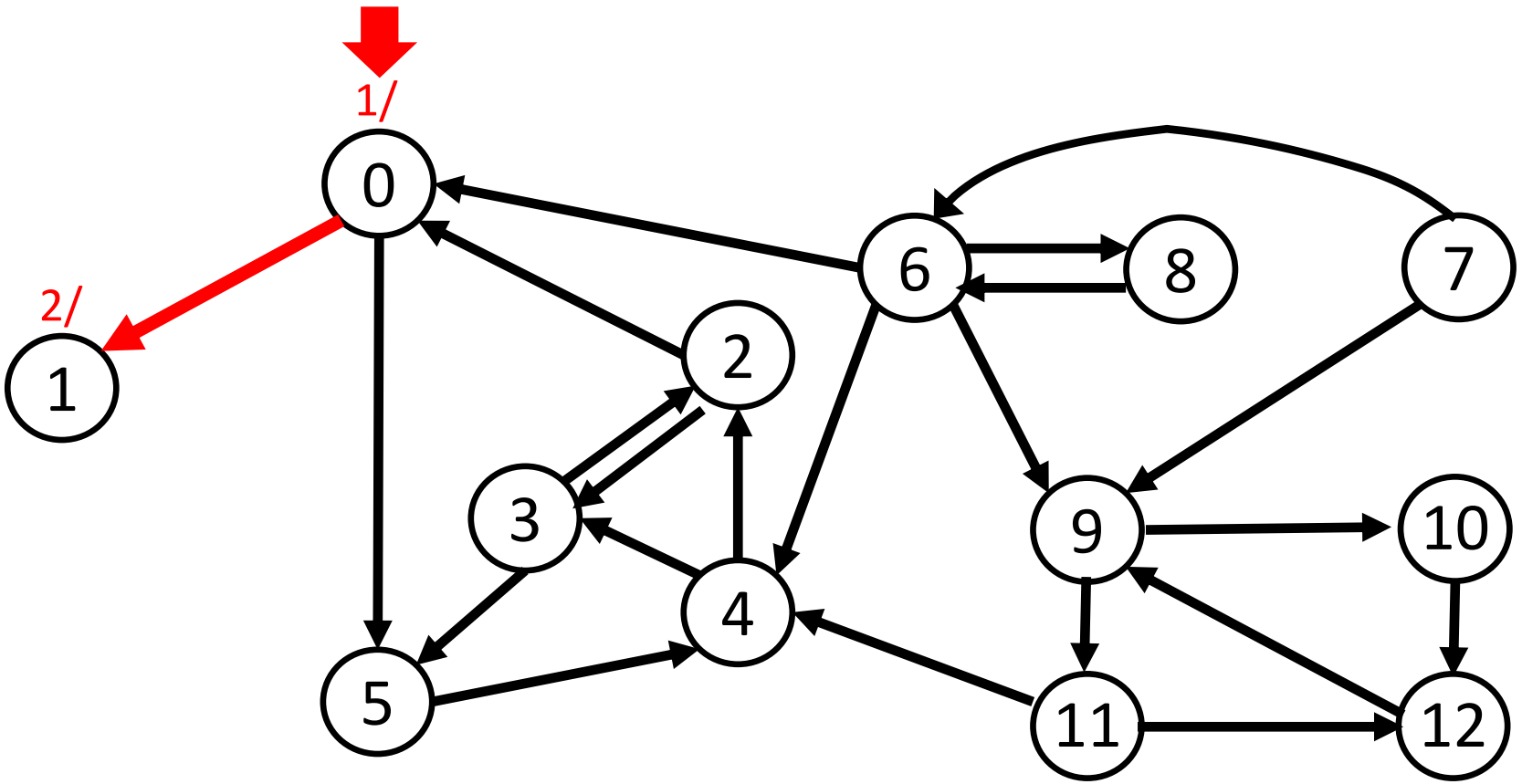
Strong components



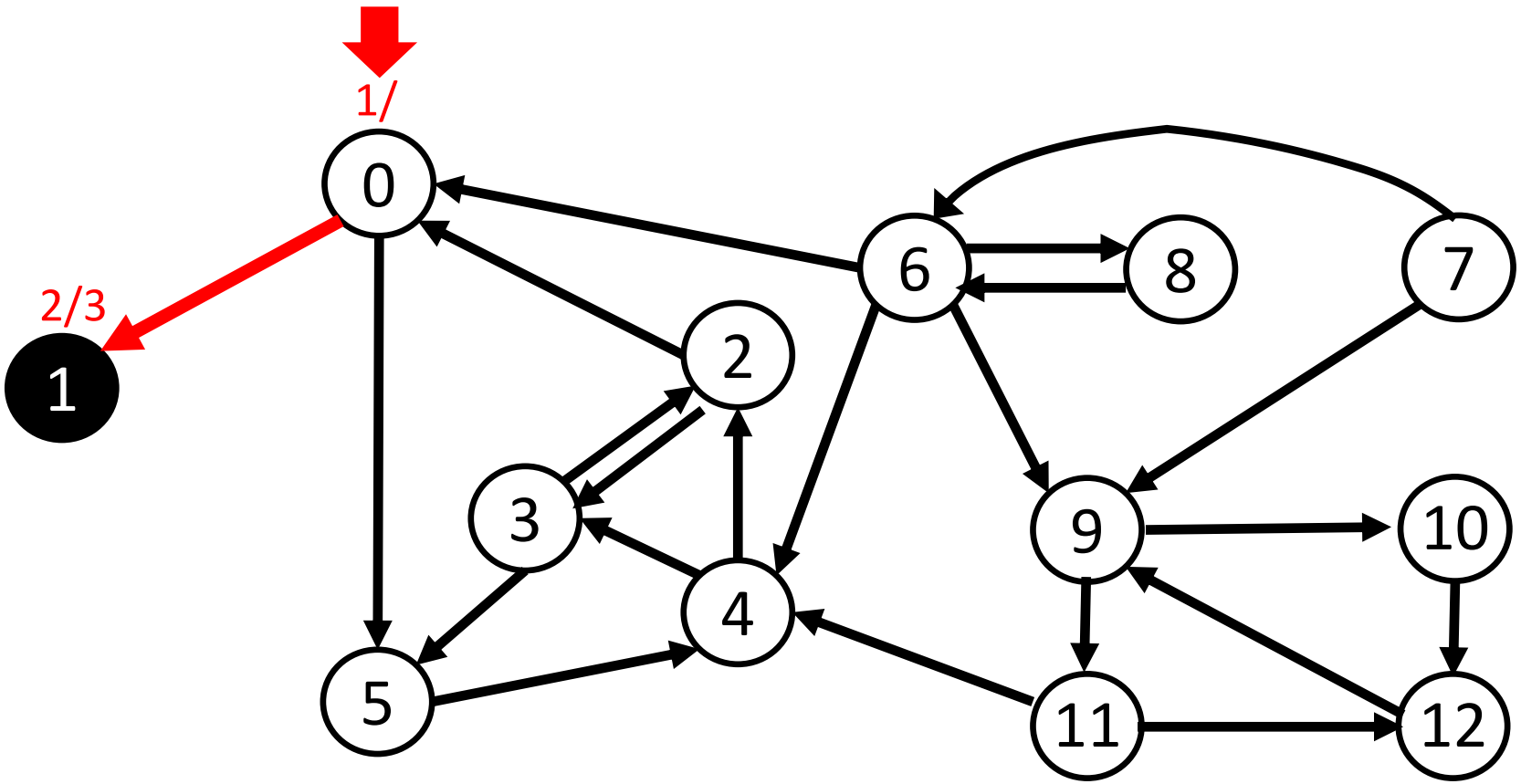
Strong components



Strong components



Strong components



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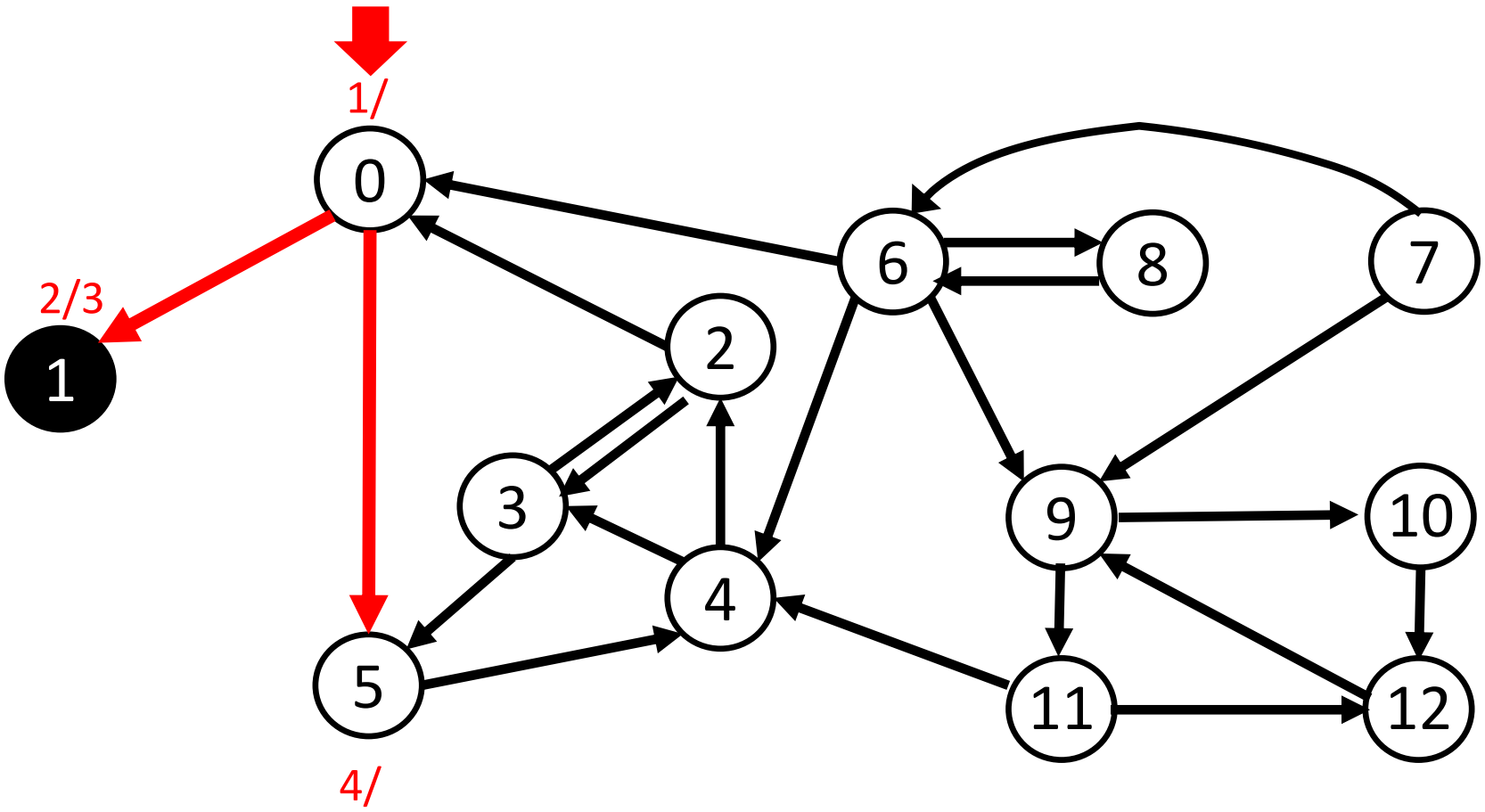

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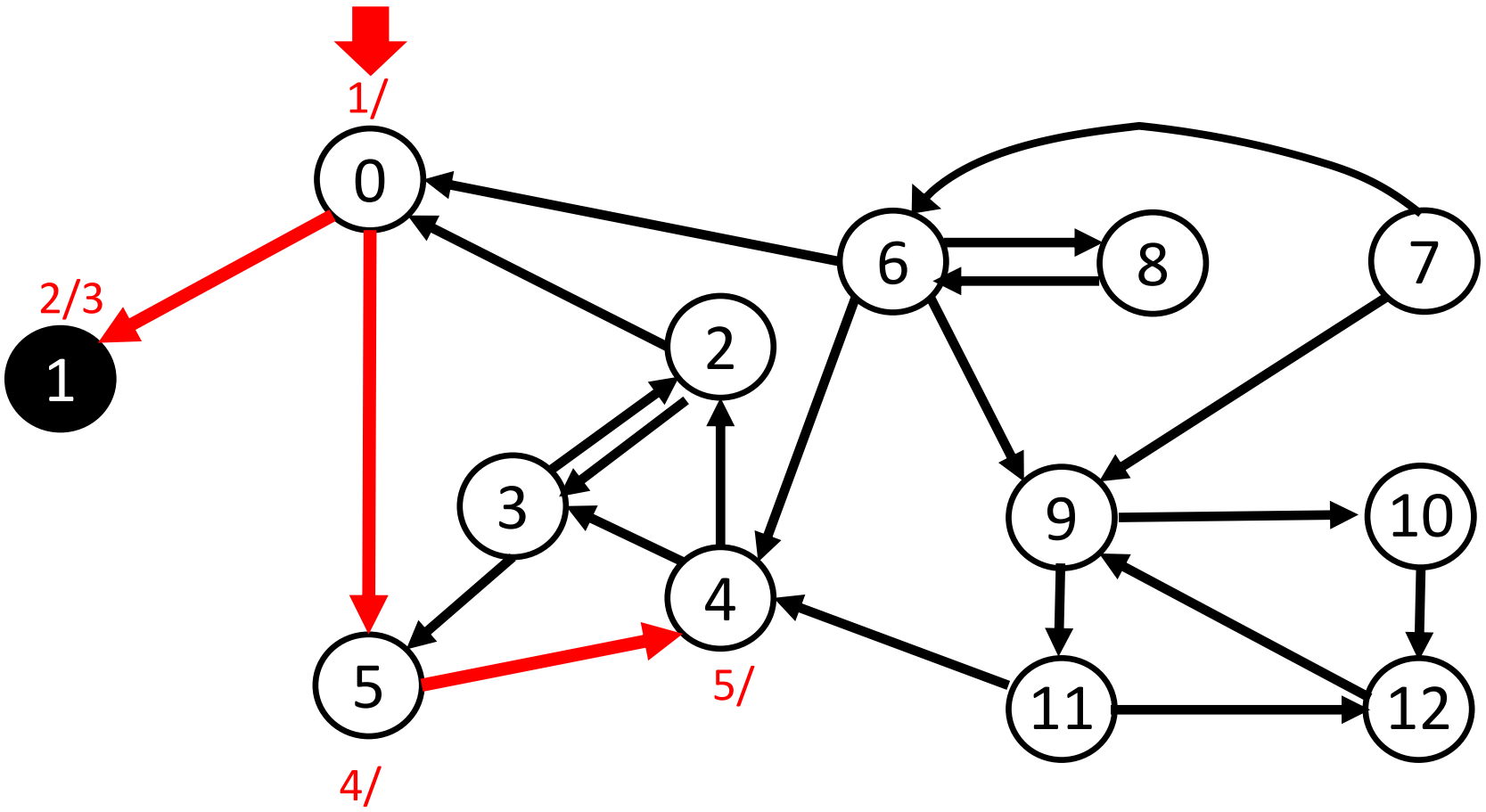
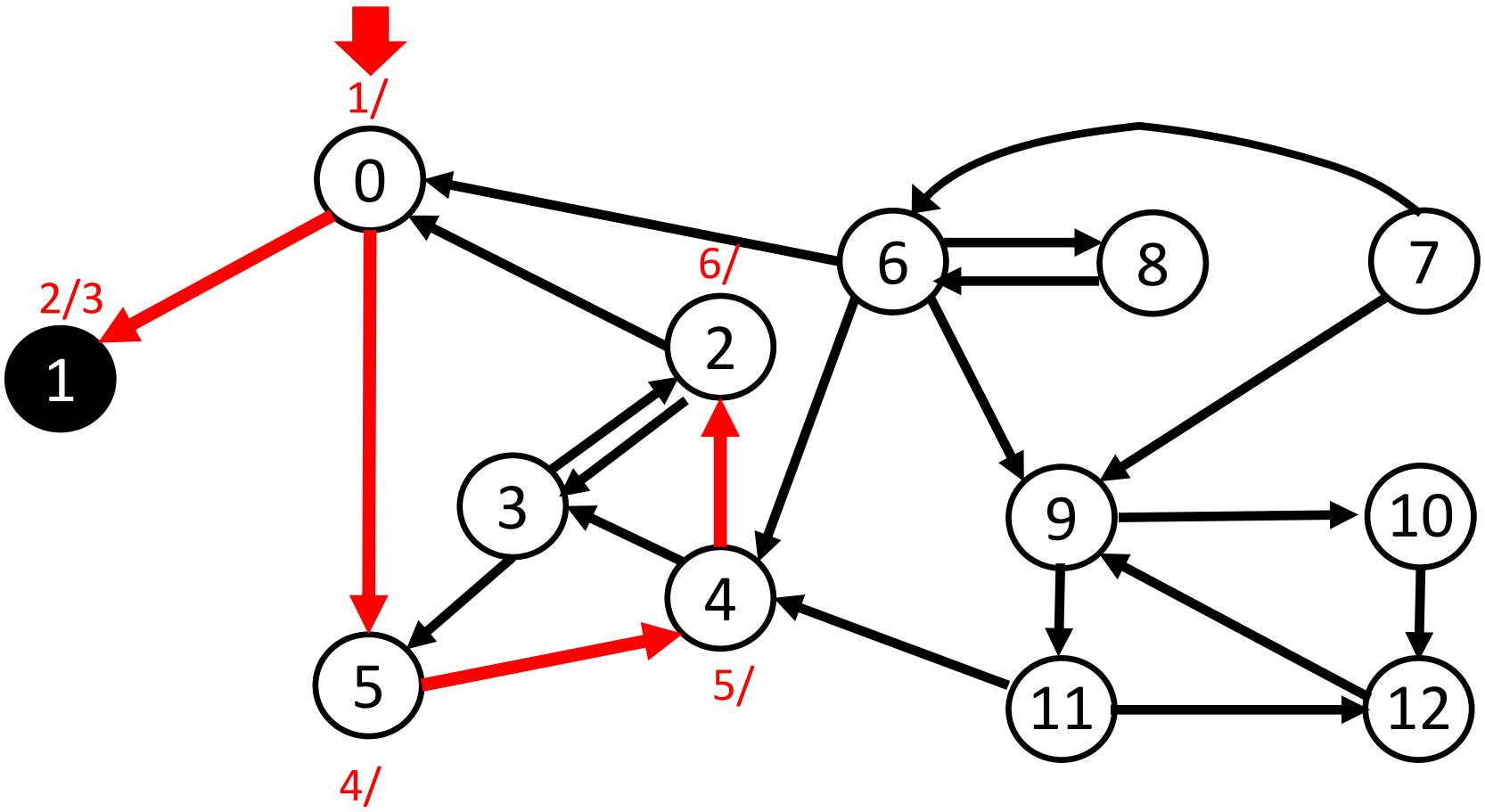


Strong components



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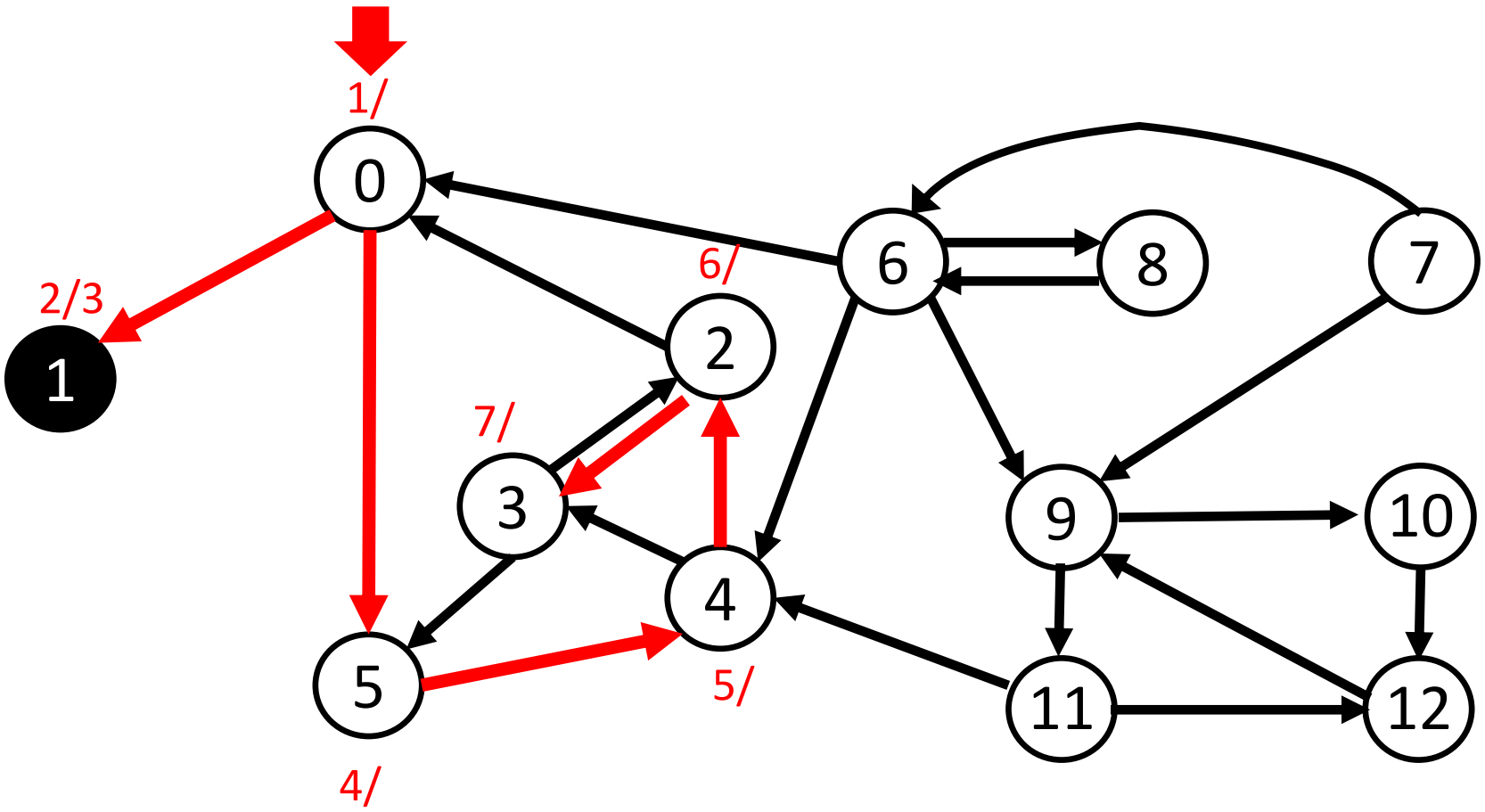

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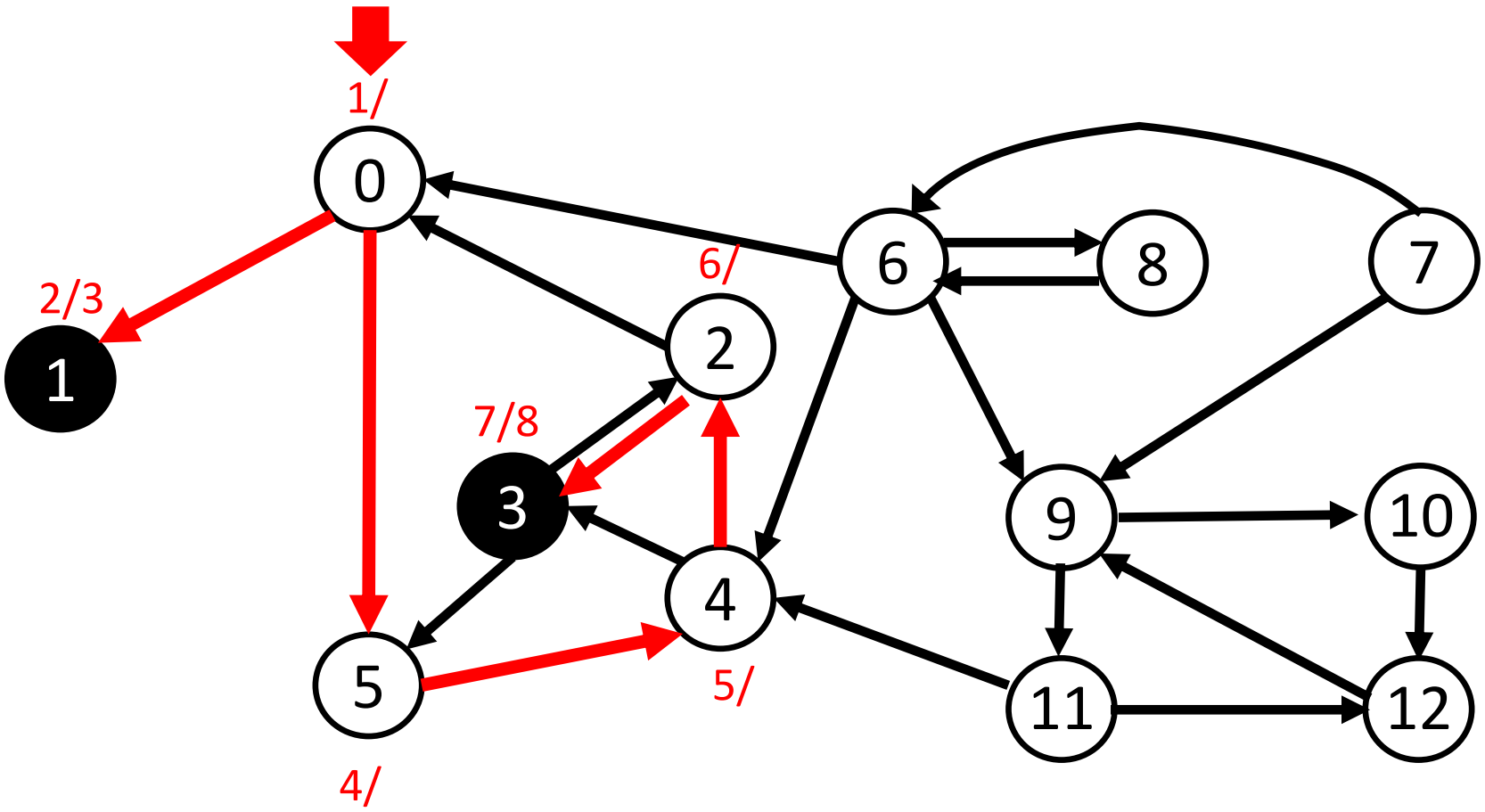

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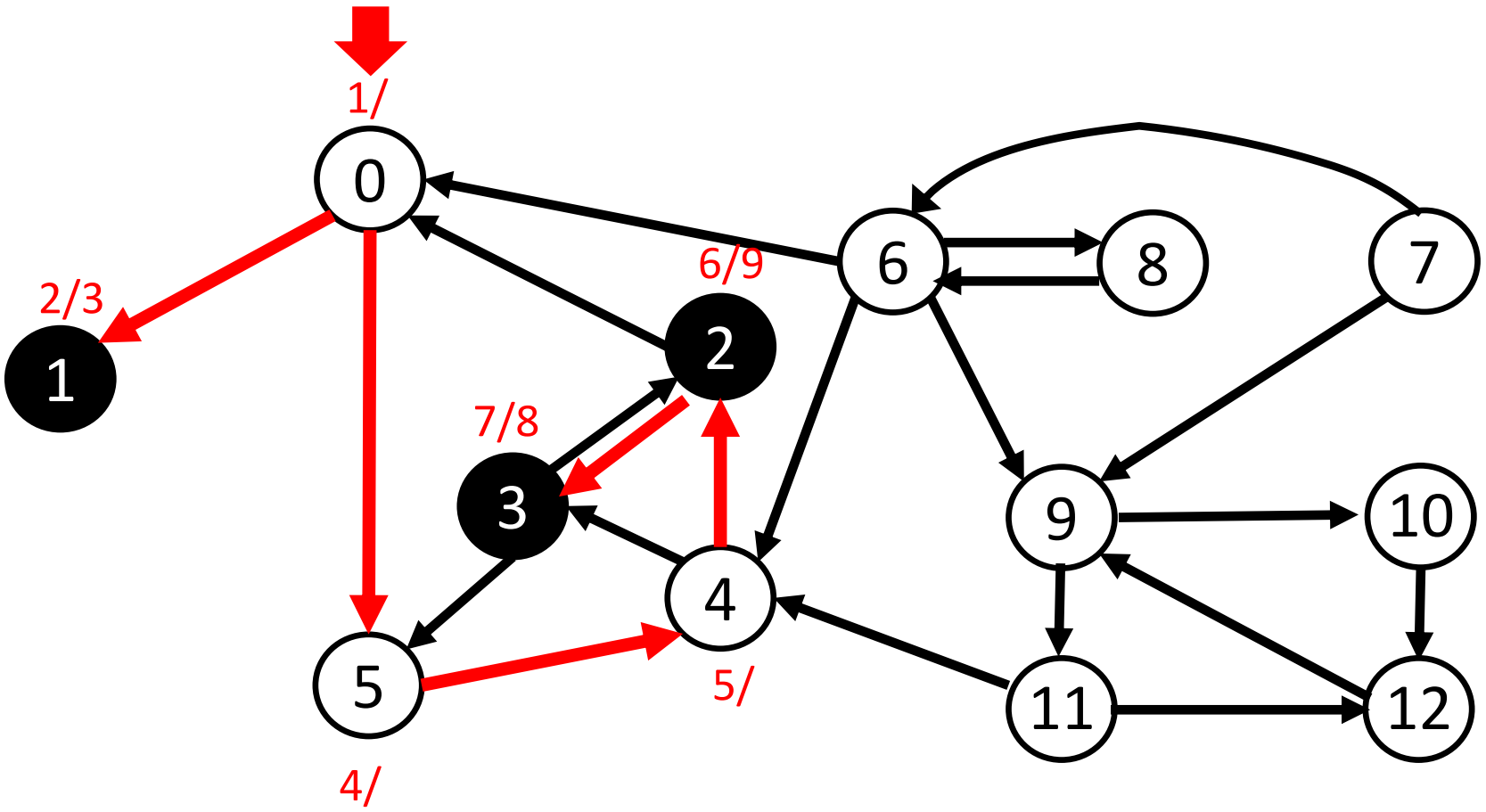

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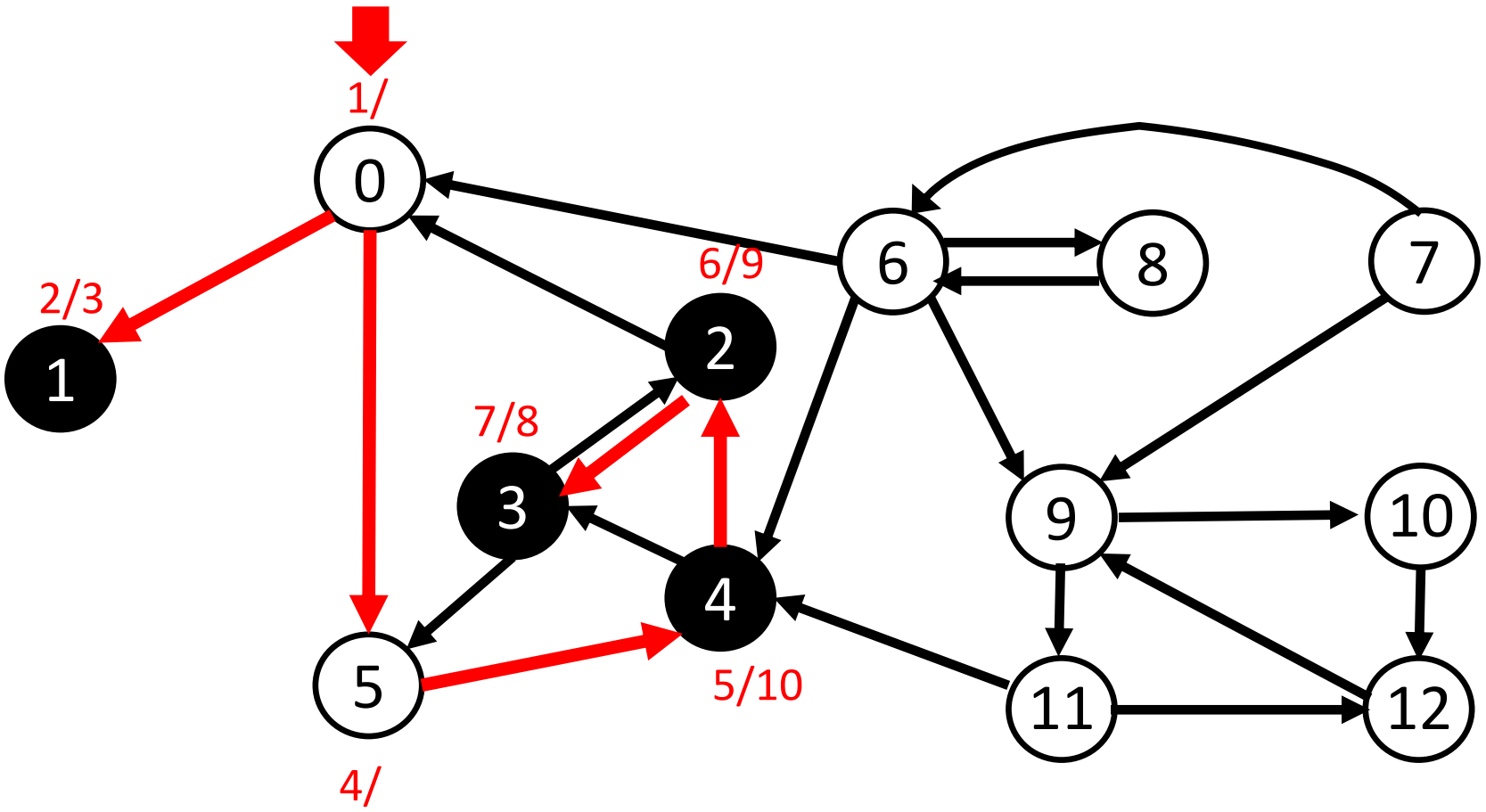

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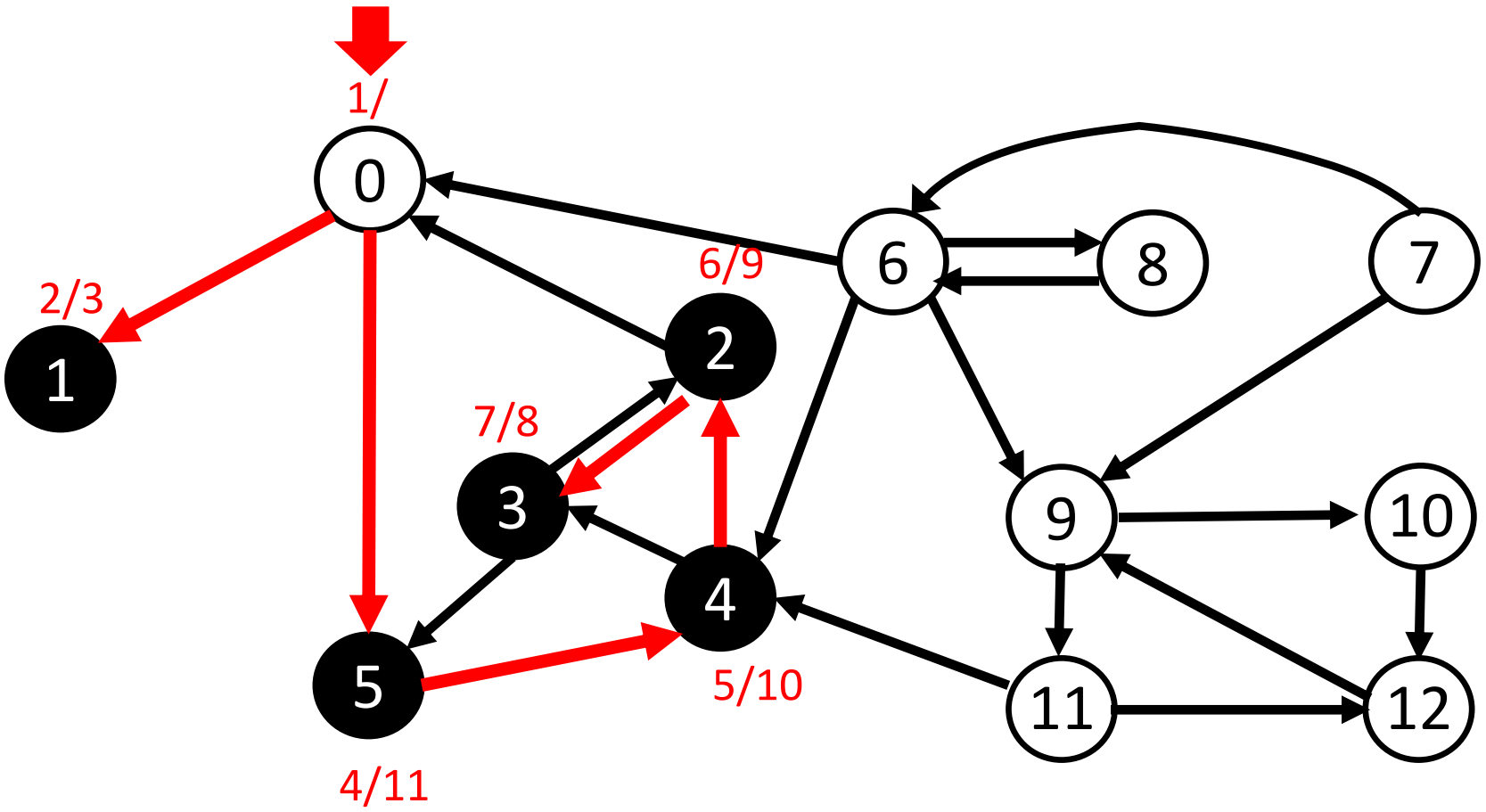

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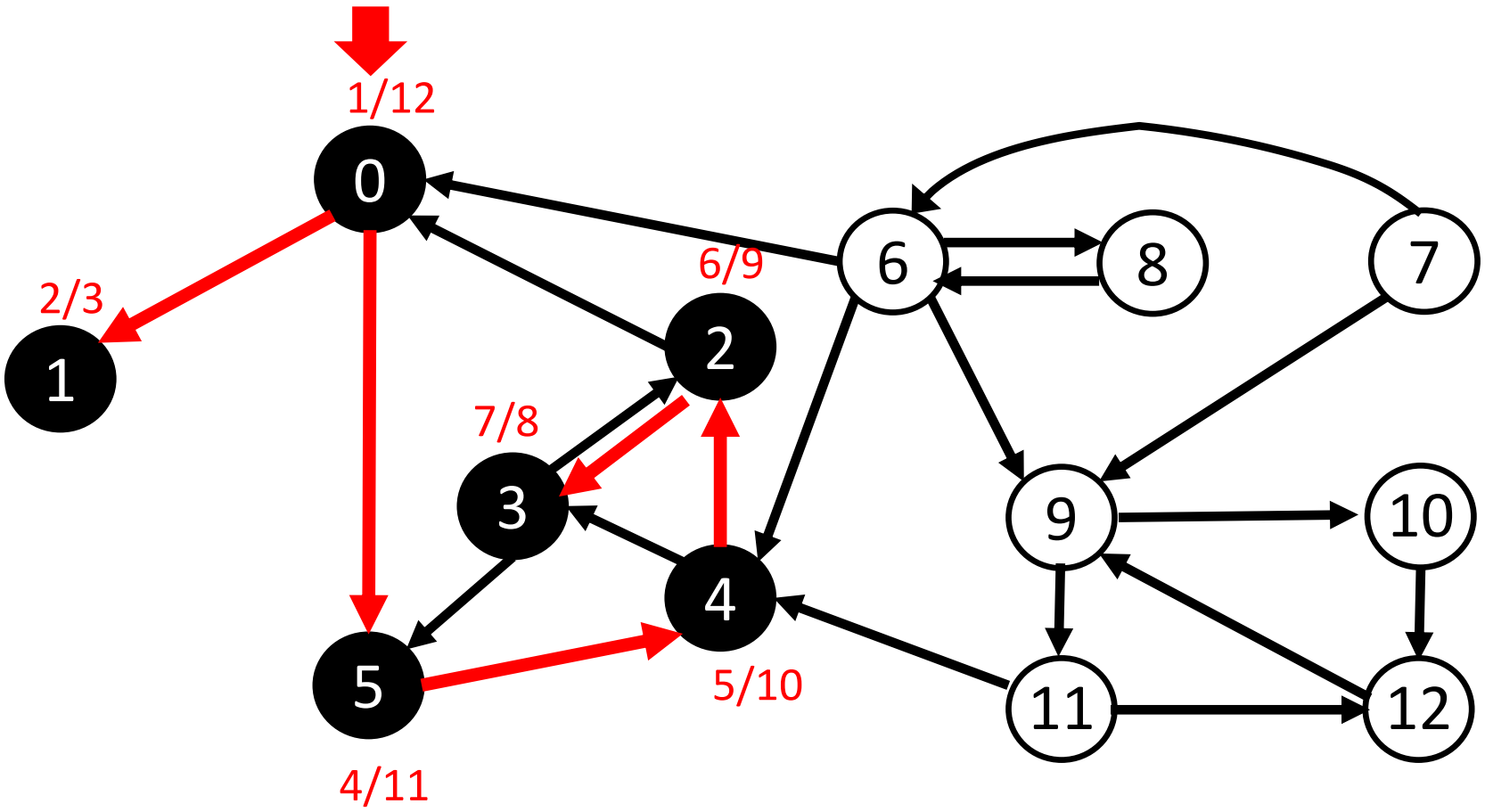

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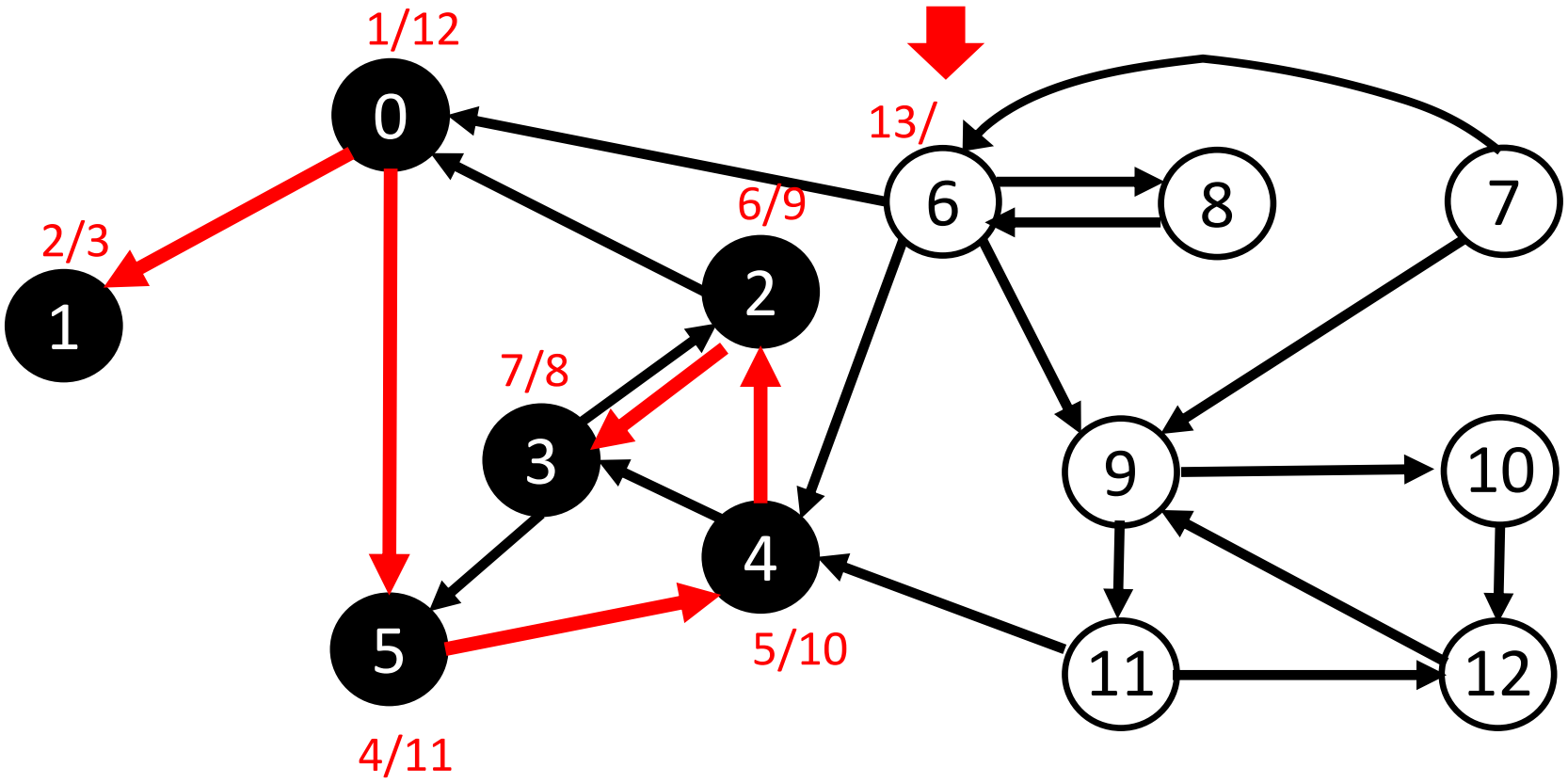

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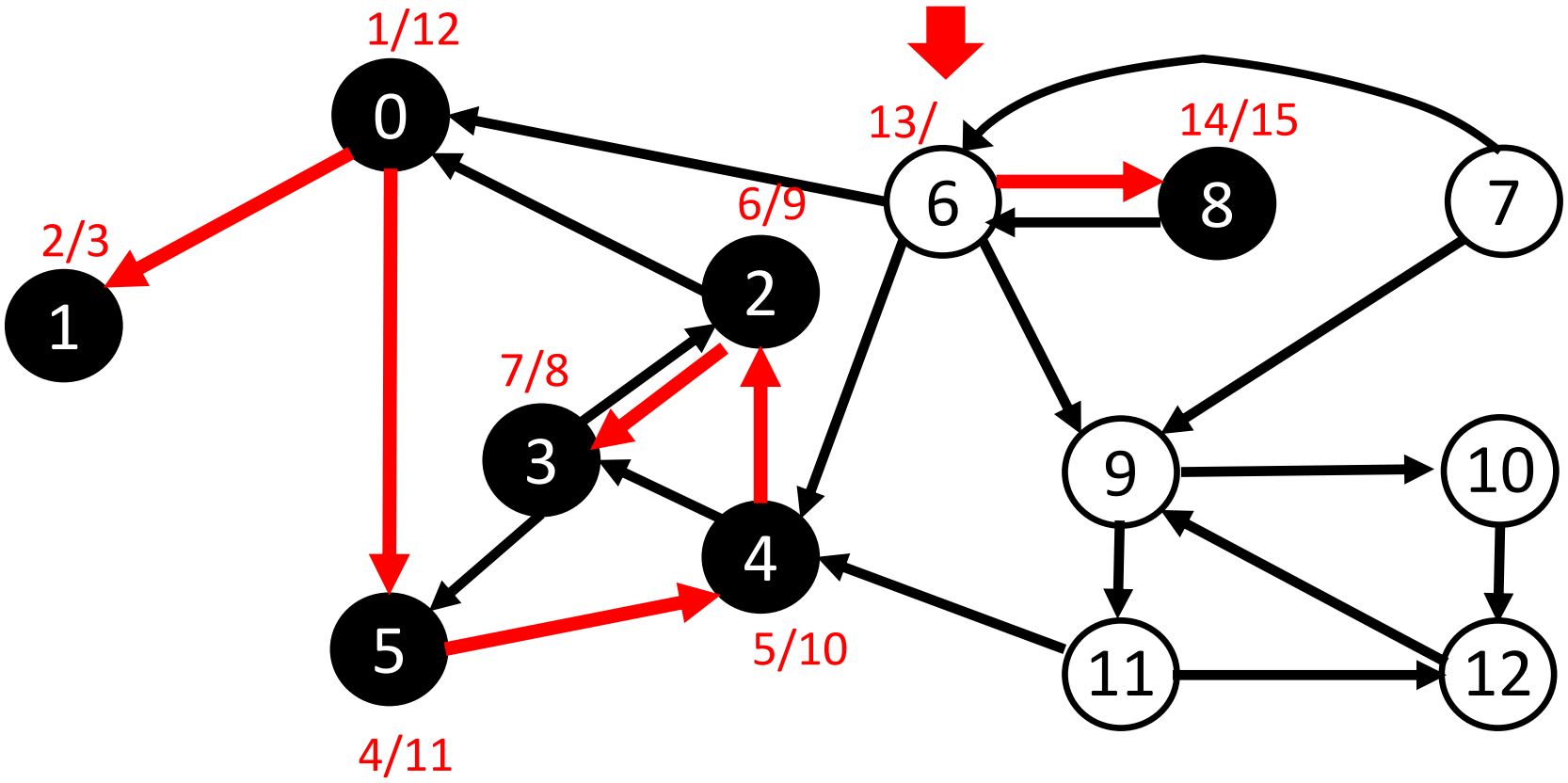

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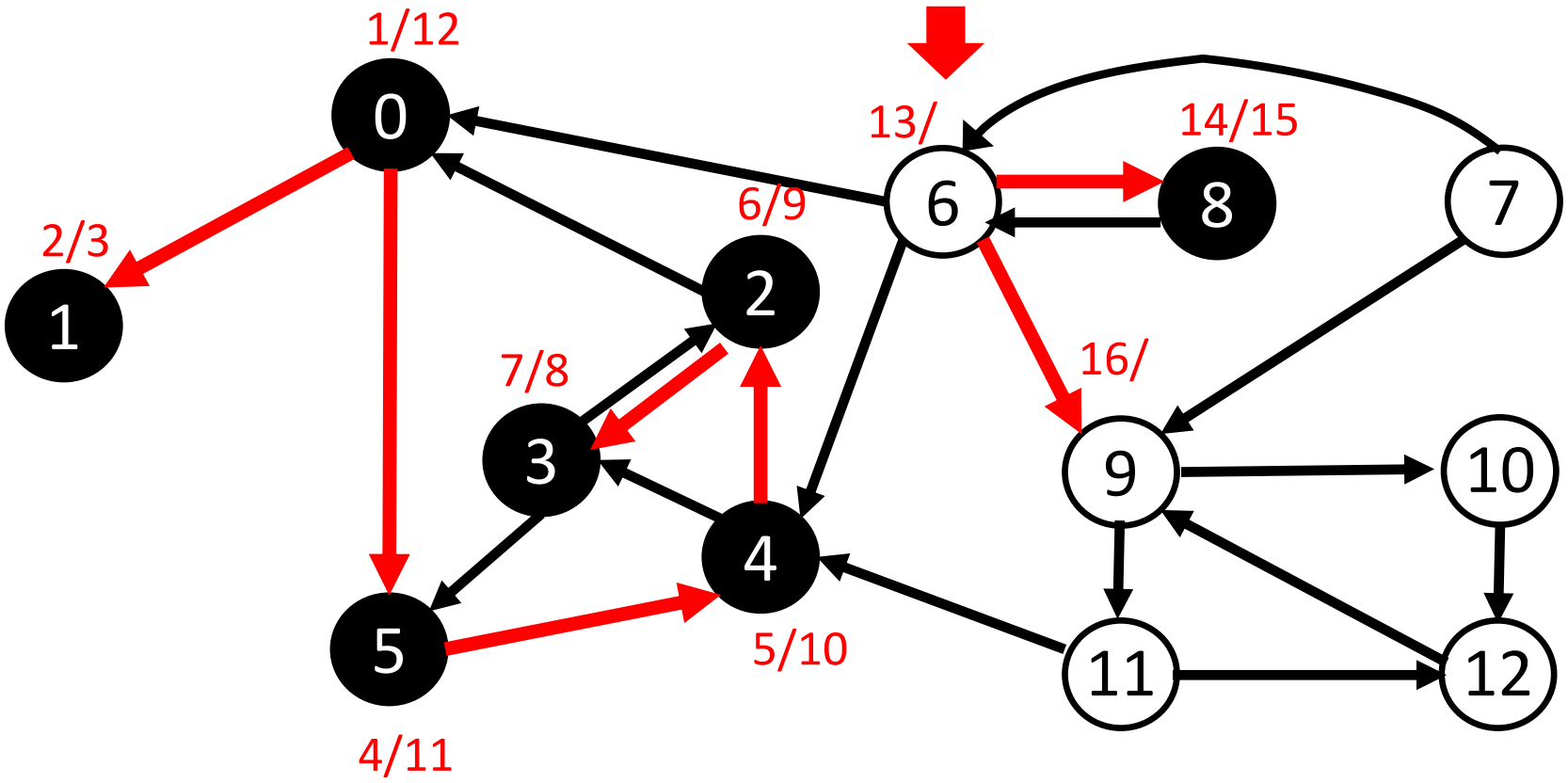

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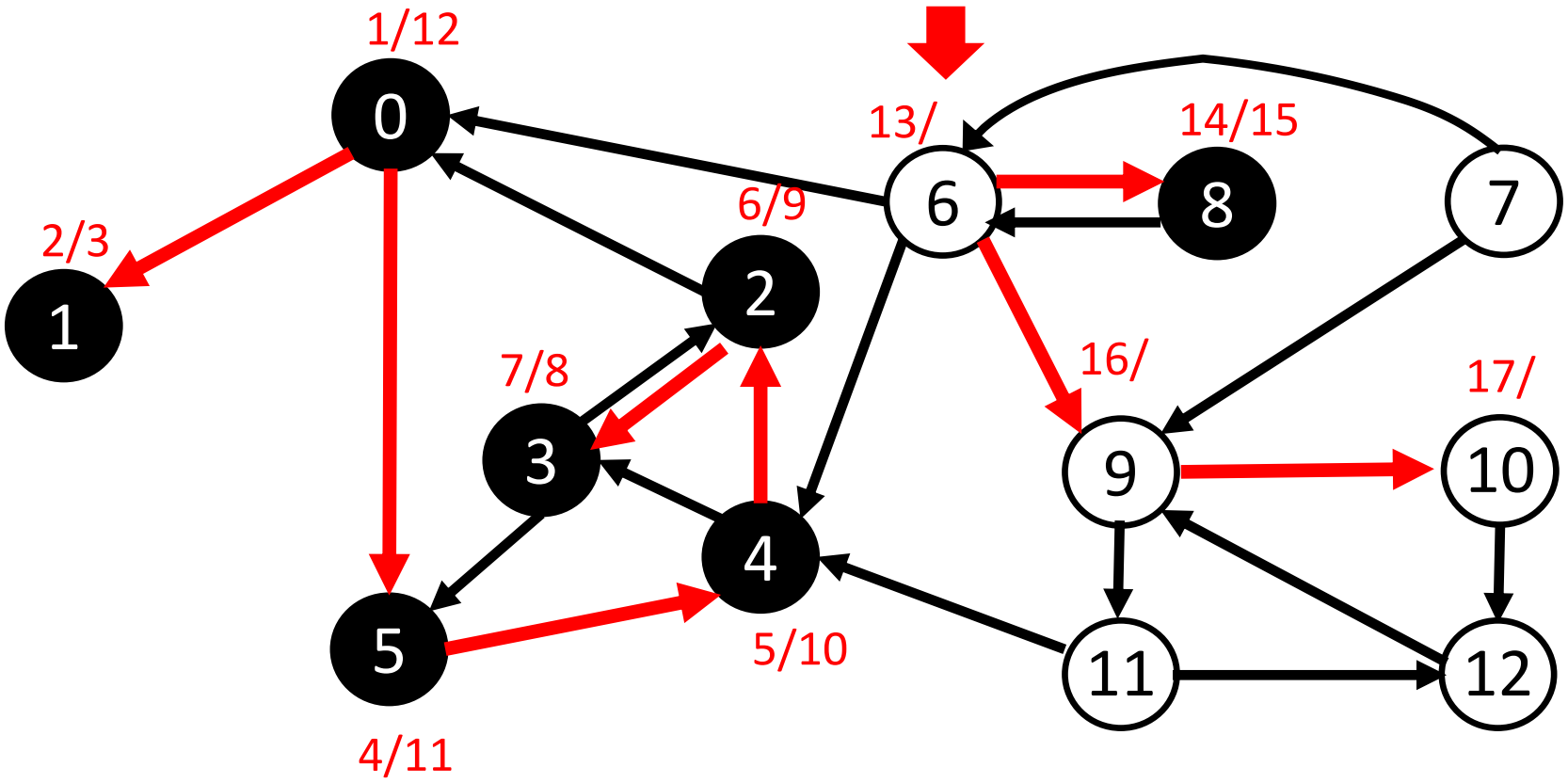

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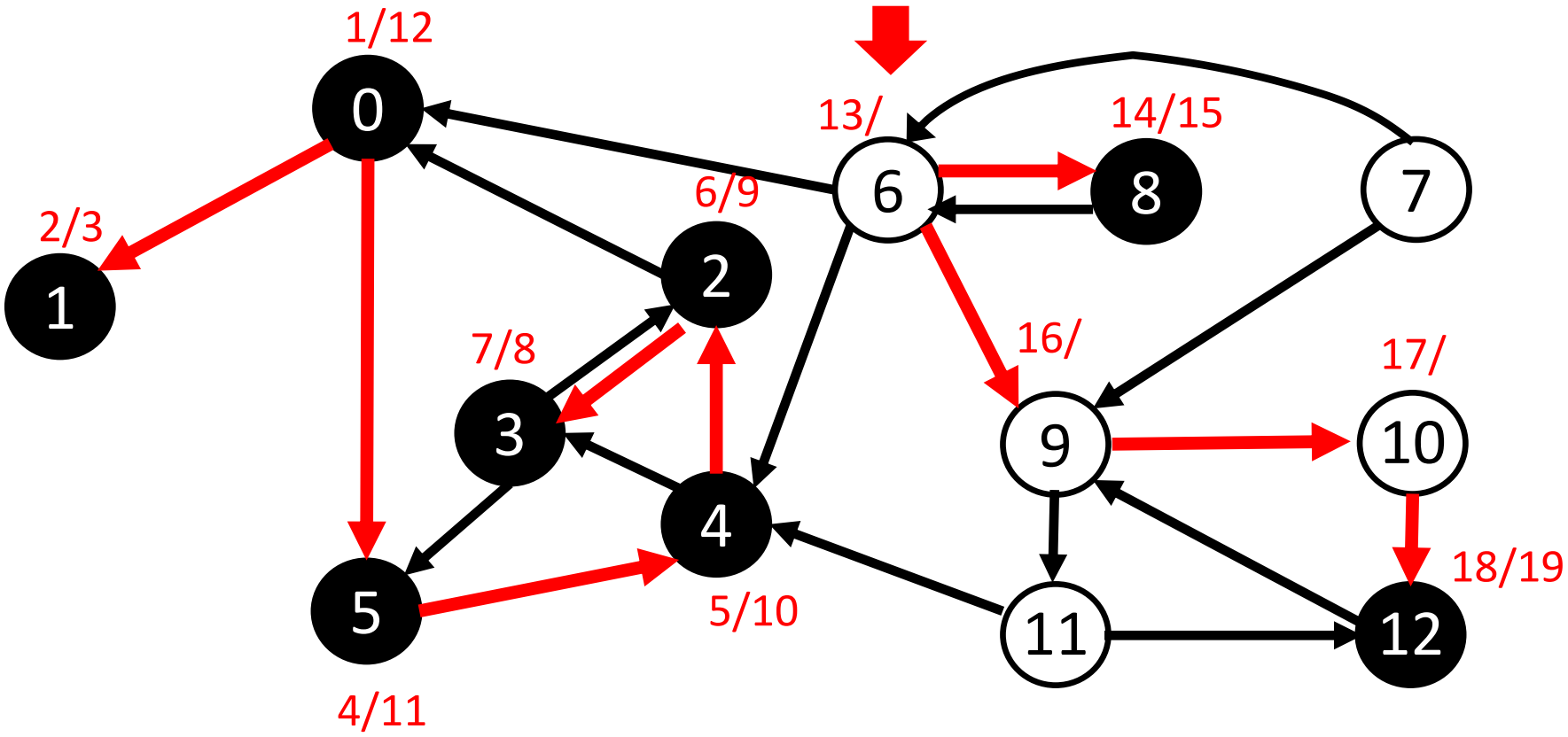

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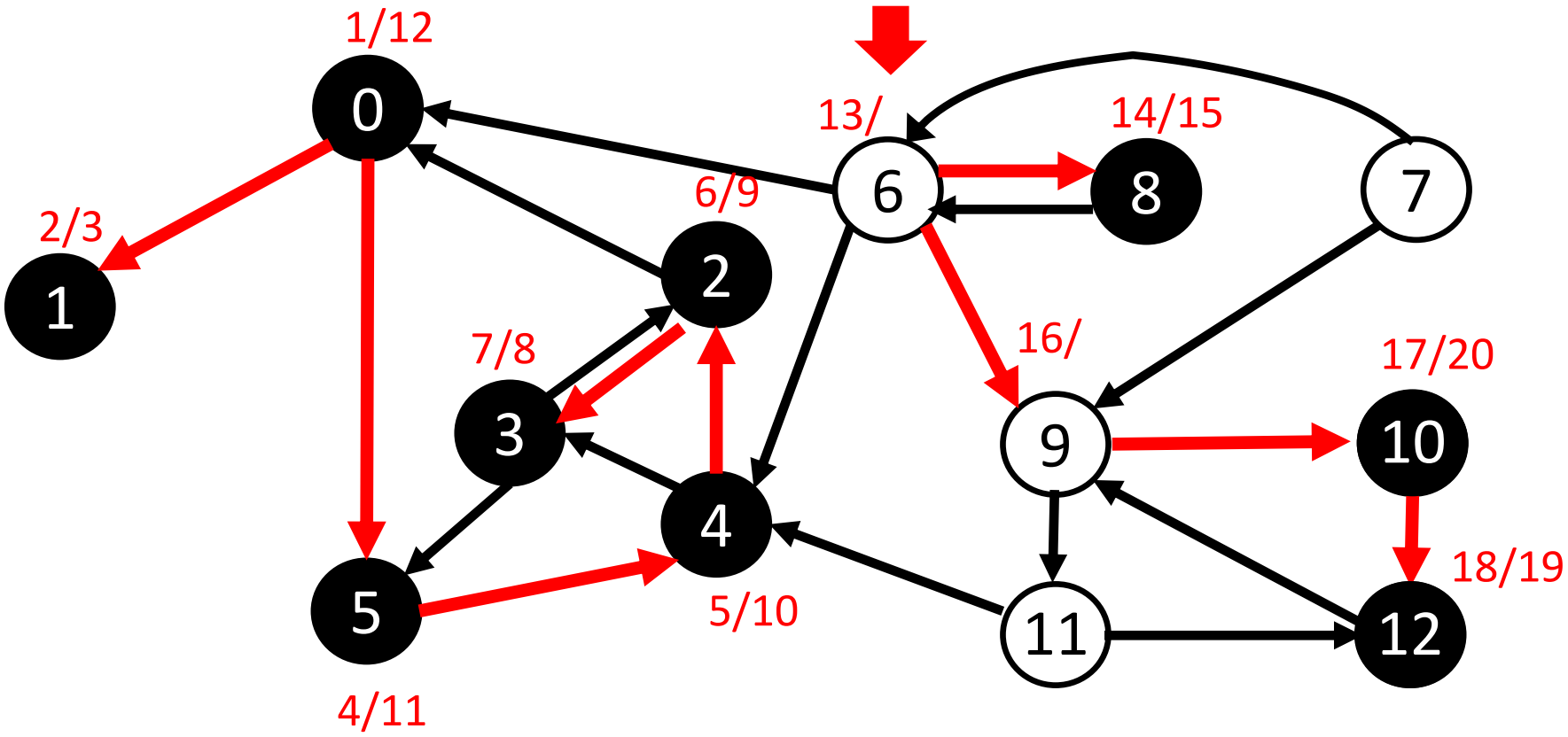

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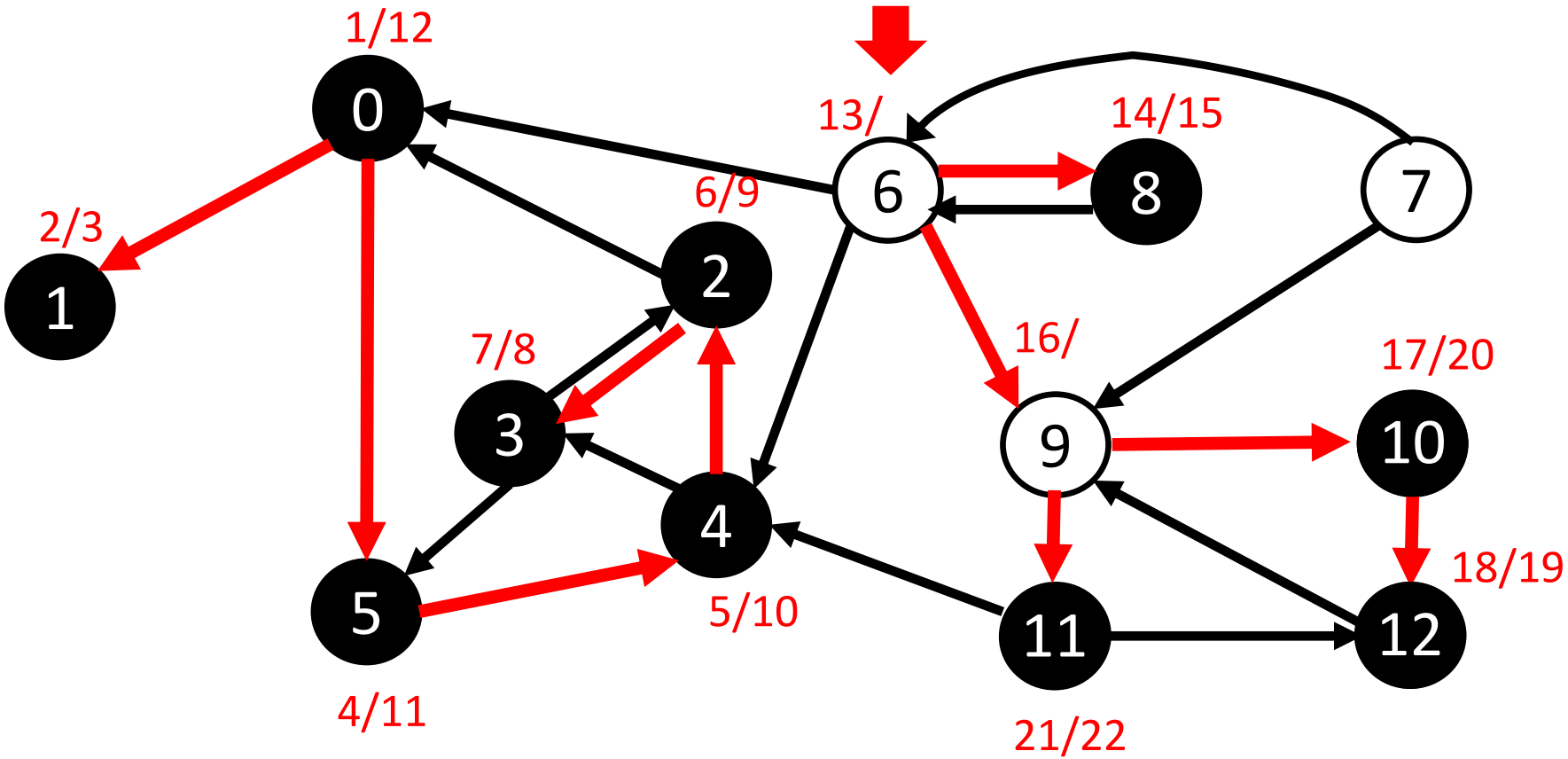

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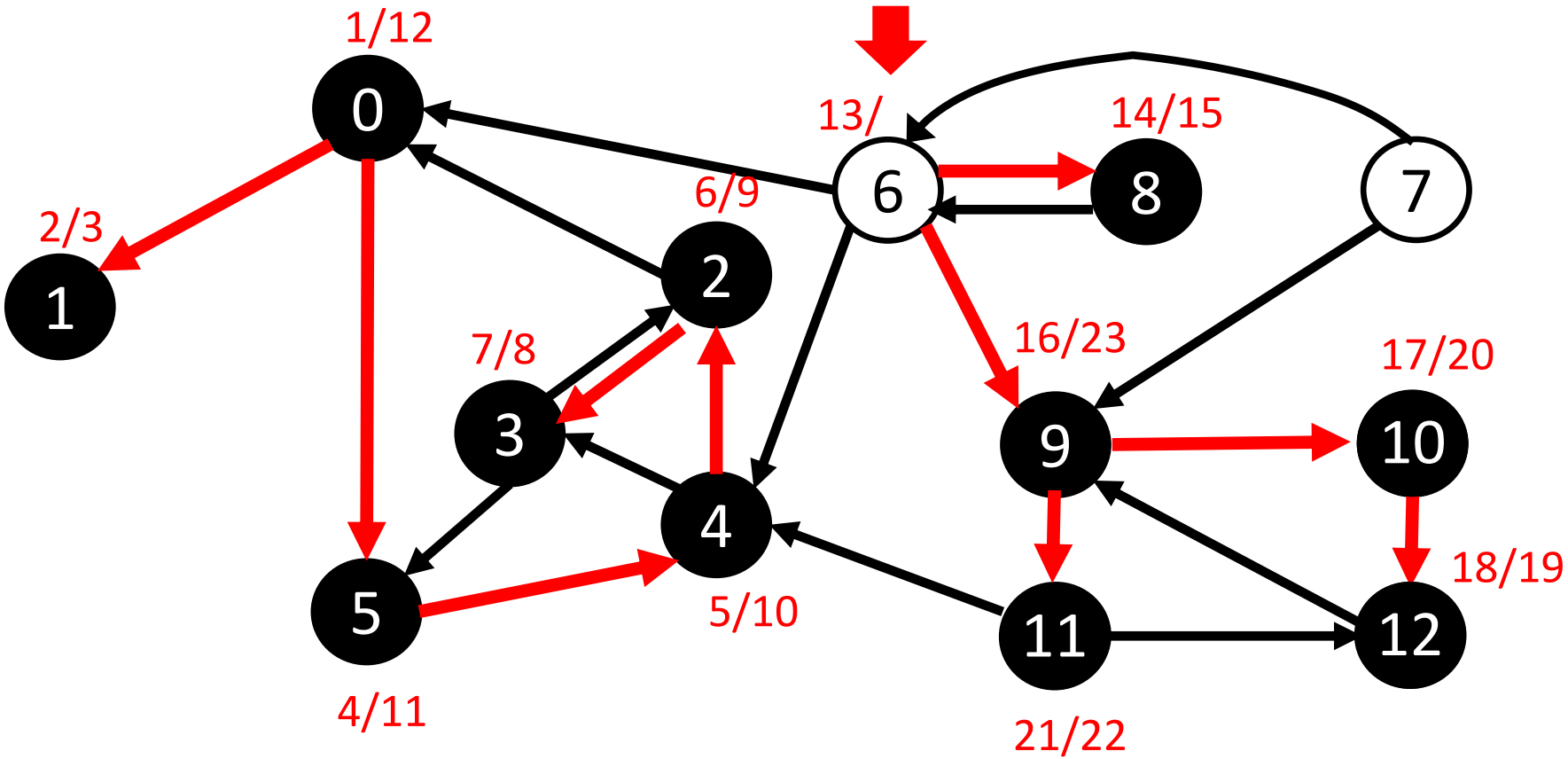

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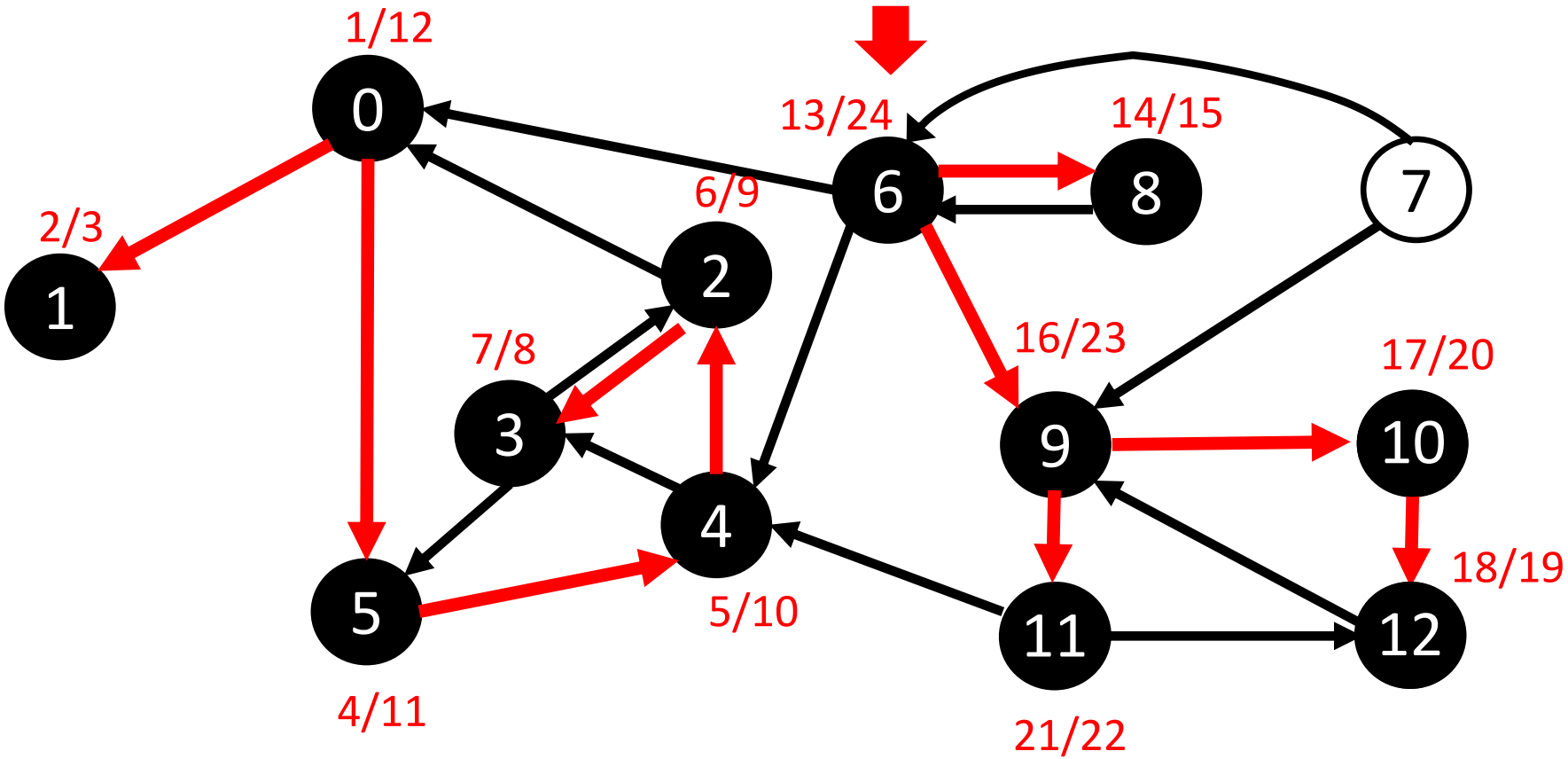

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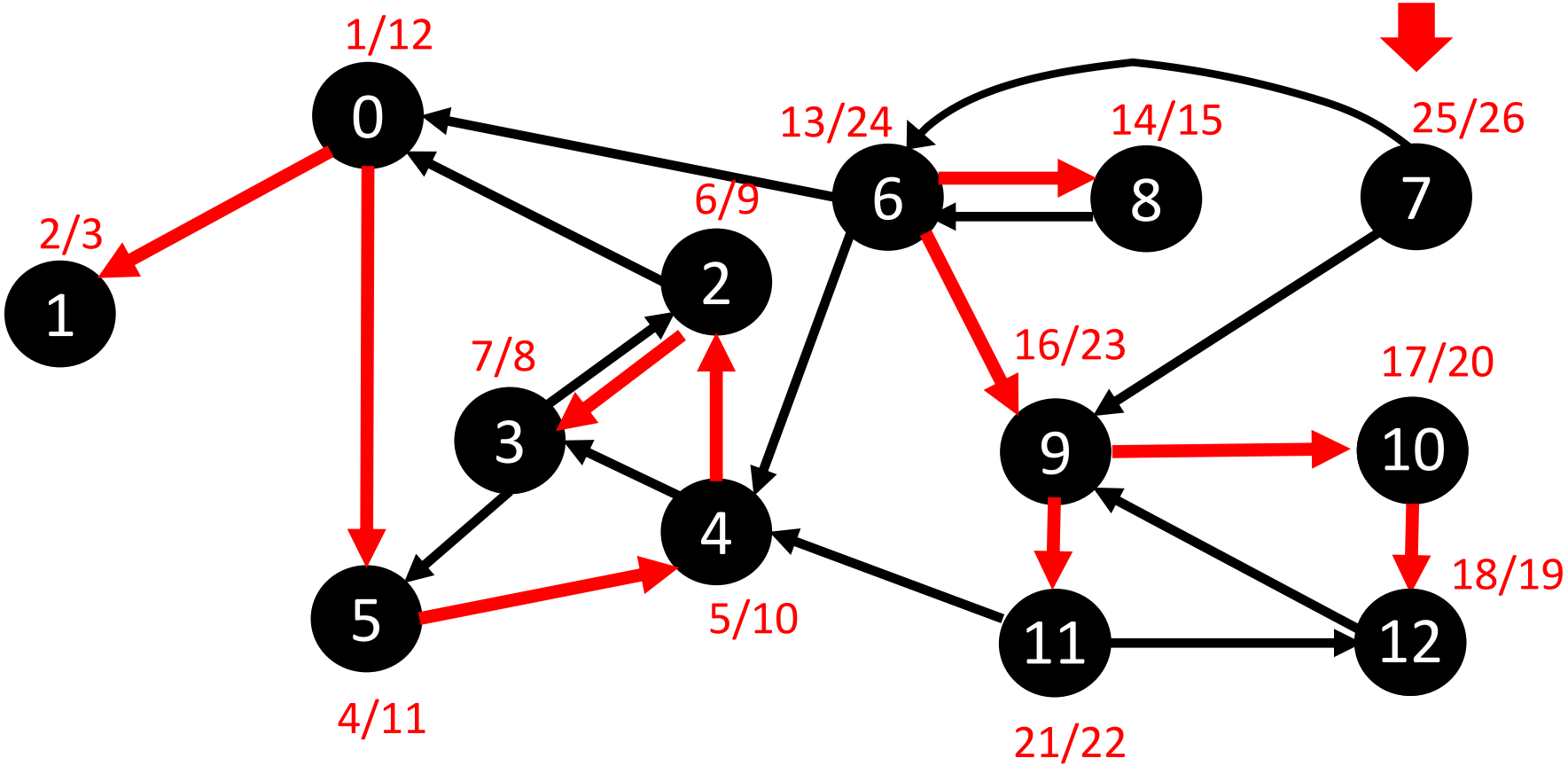

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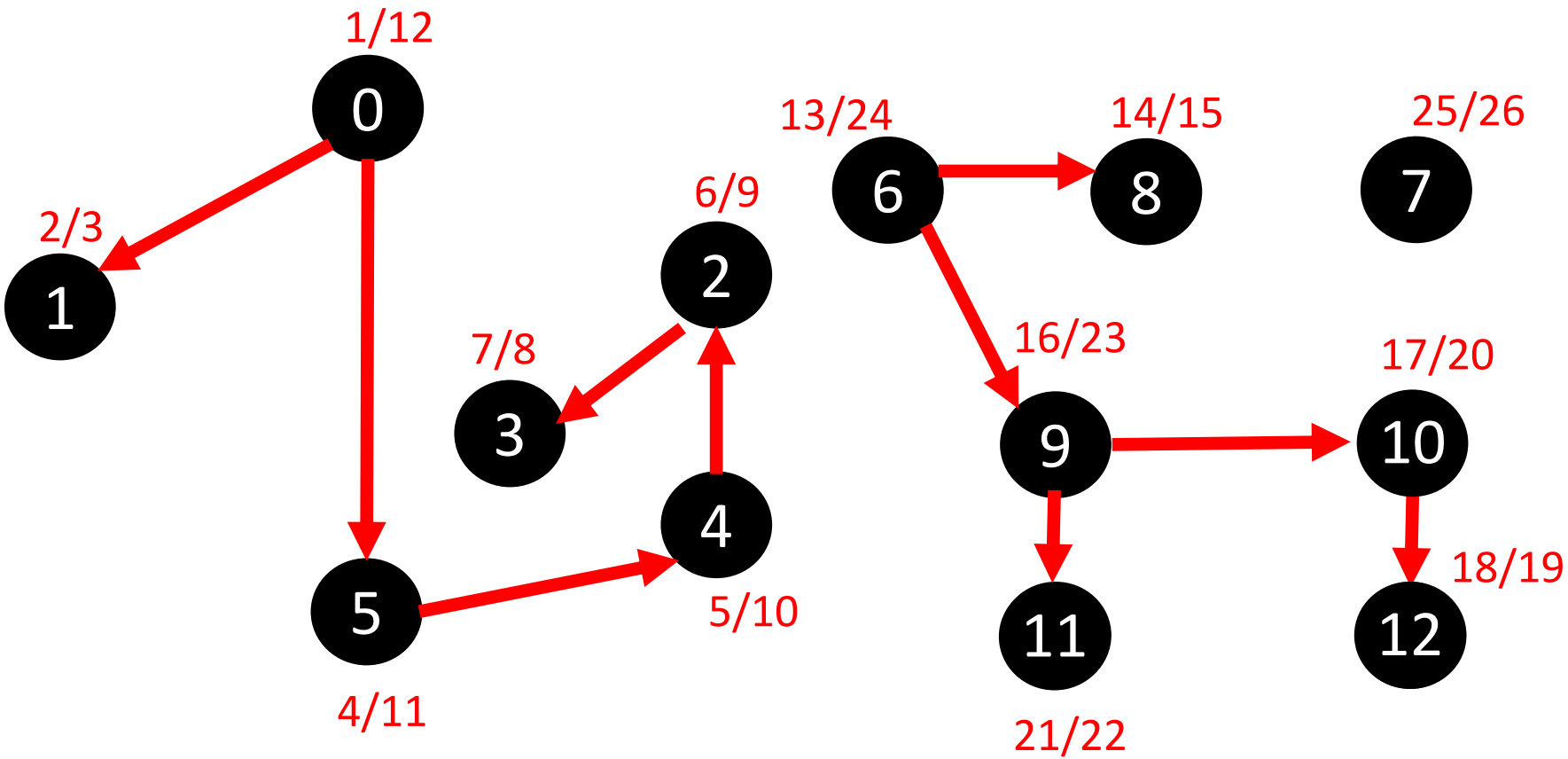

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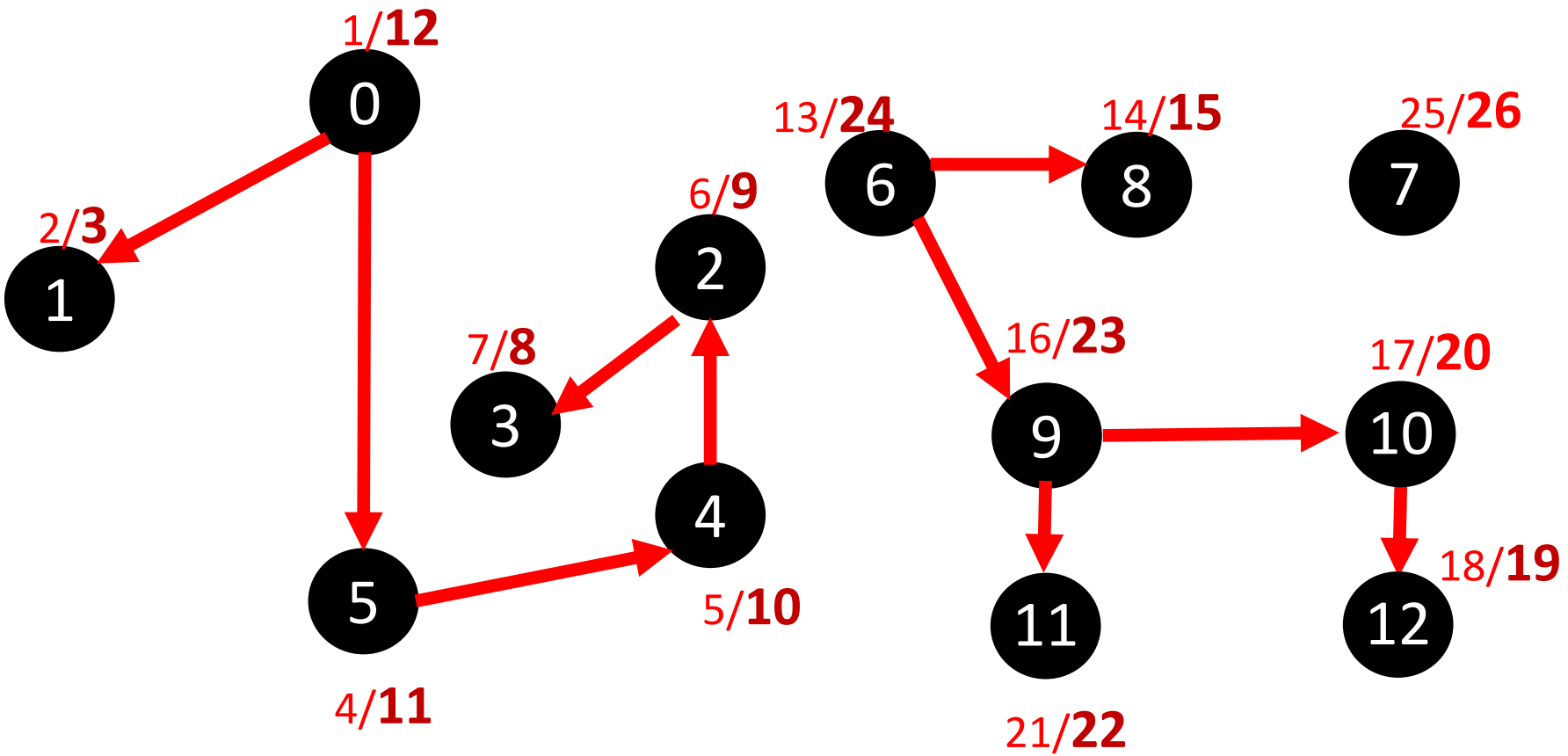
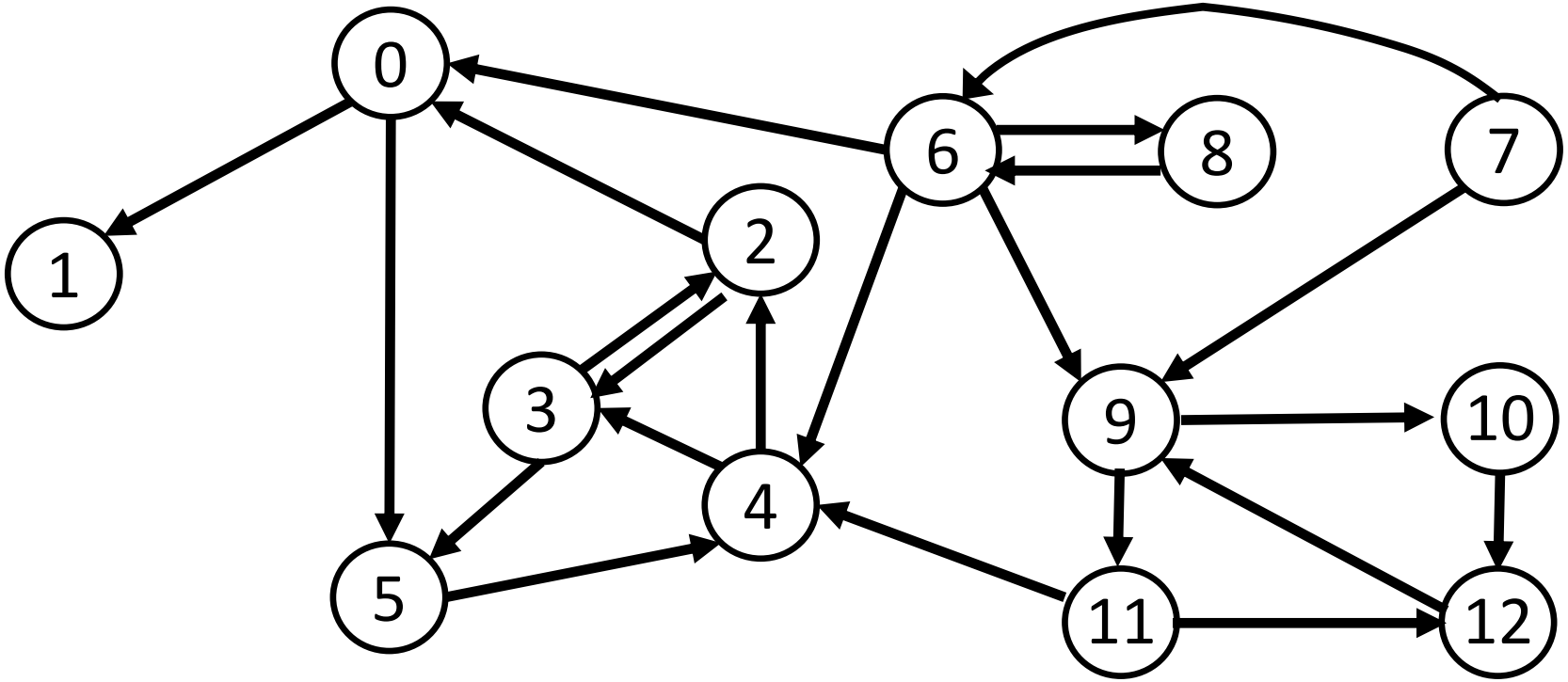


Strong components

26	24	23	22	20	19	15	12	11	10	9	8	3
7	6	9	11	10	12	8	0	5	4	2	3	1

Strong components

26	24	23	22	20	19	15	12	11	10	9	8	3
7	6	9	11	10	12	8	0	5	4	2	3	1



Strong components

26	24	23	22	20	19	15	12	11	10	9	8	3
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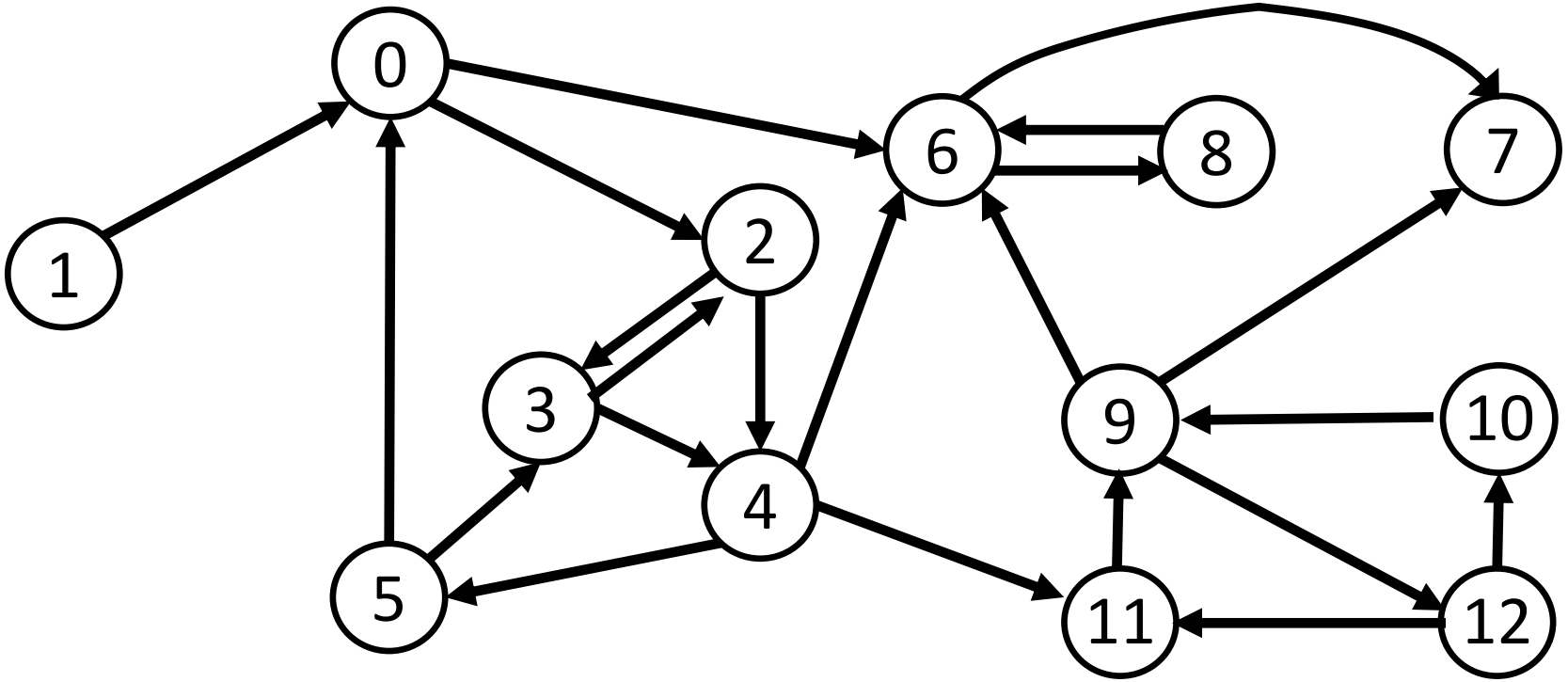

Strong components

26	24	23	22	20	19	15	12	11	10	9	8	3
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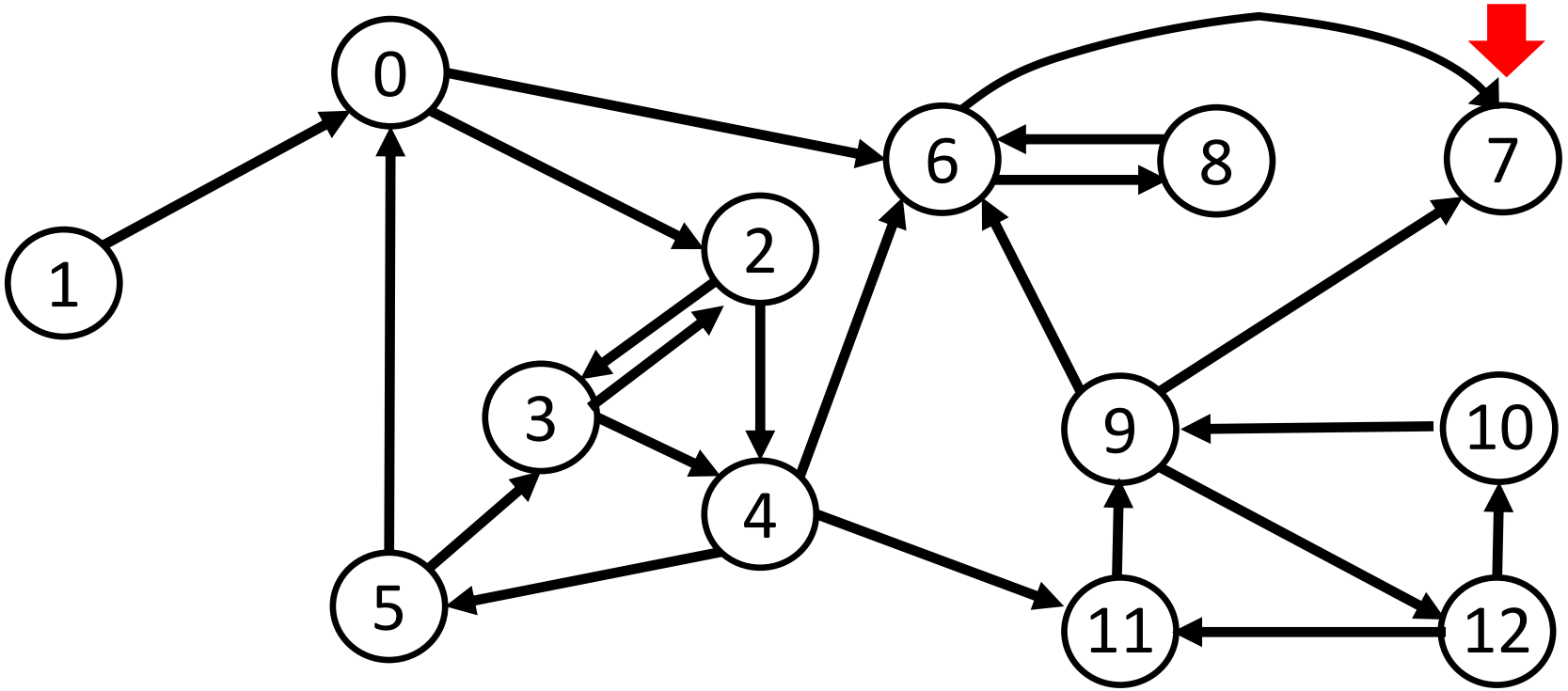

Strong components

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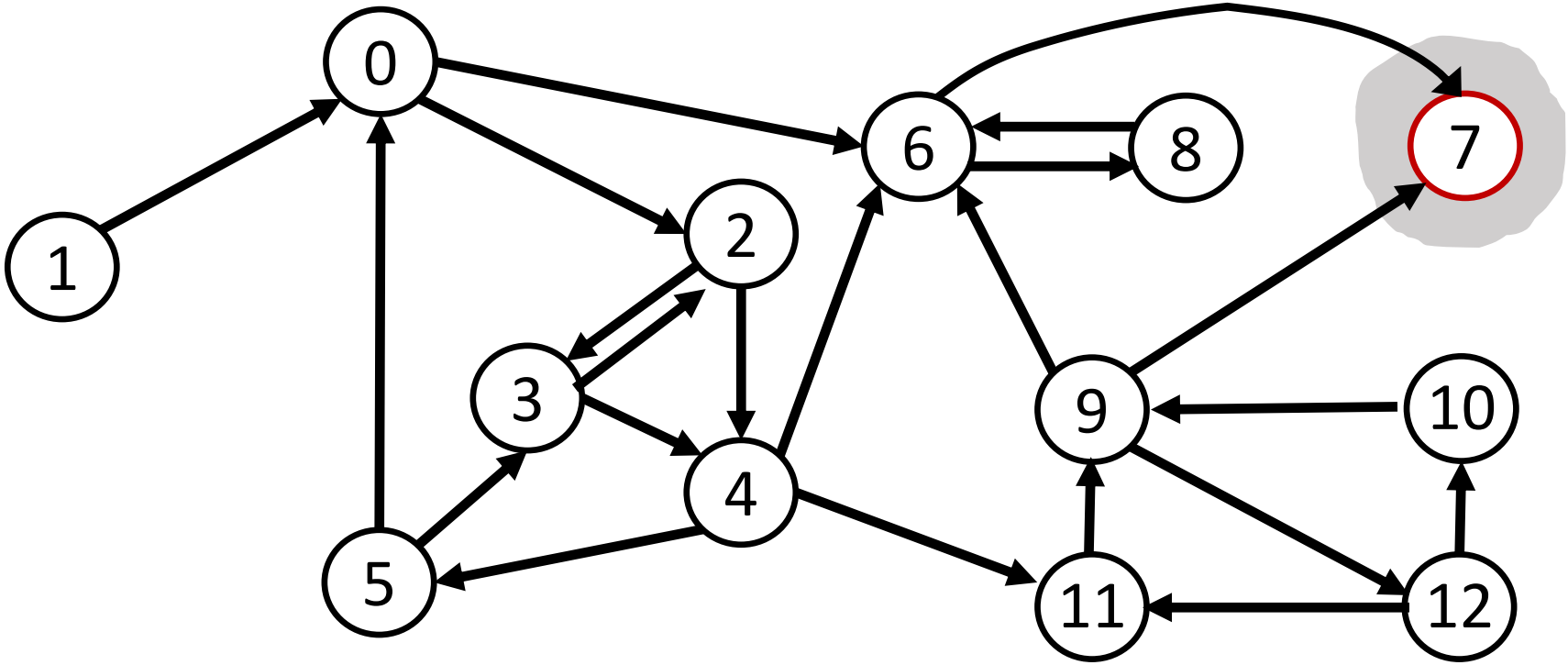

Strong components

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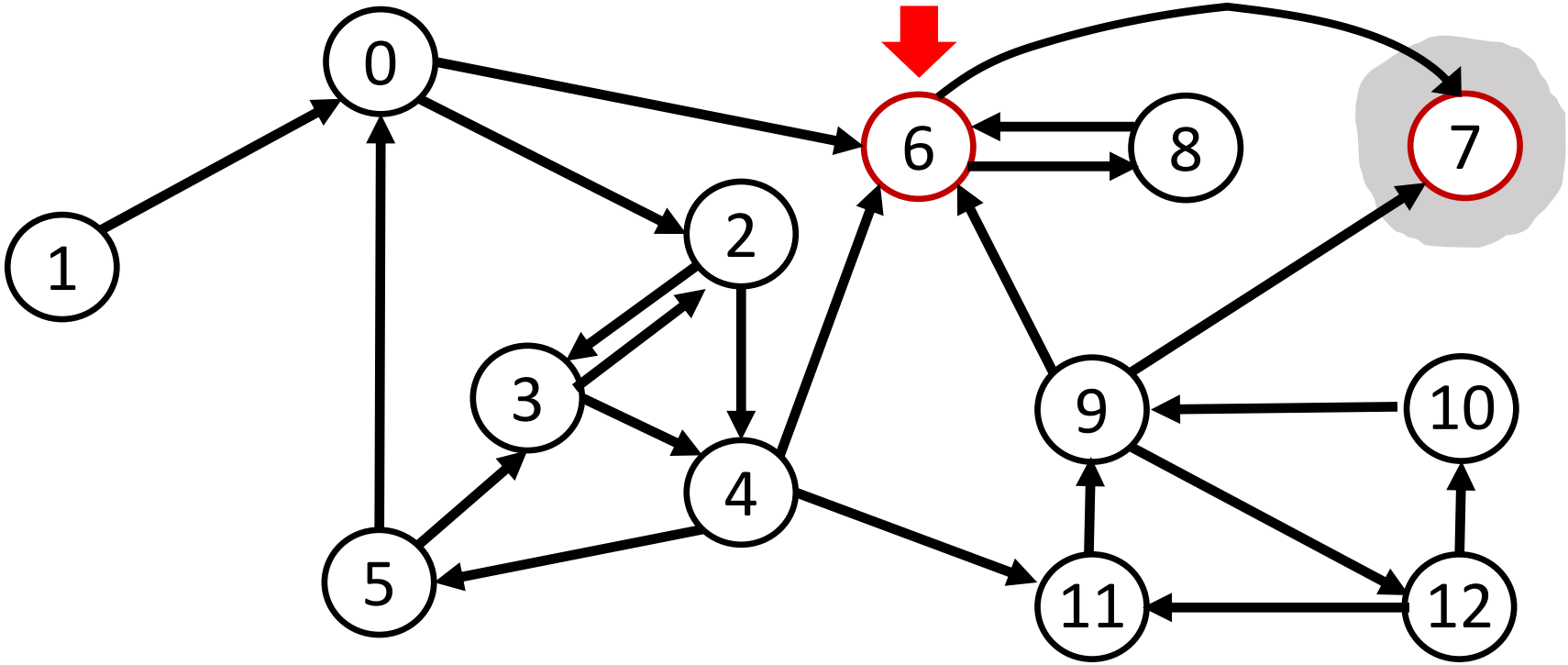

Strong components

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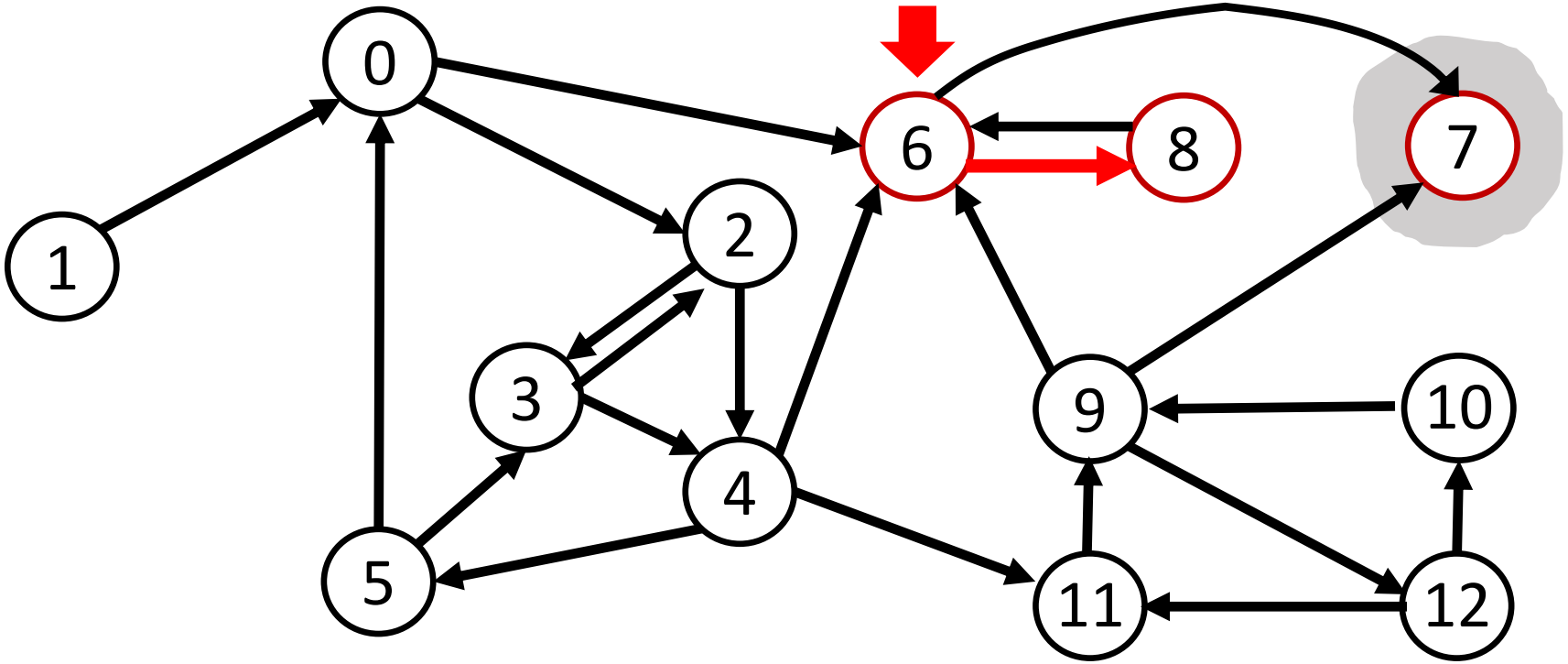

Strong components

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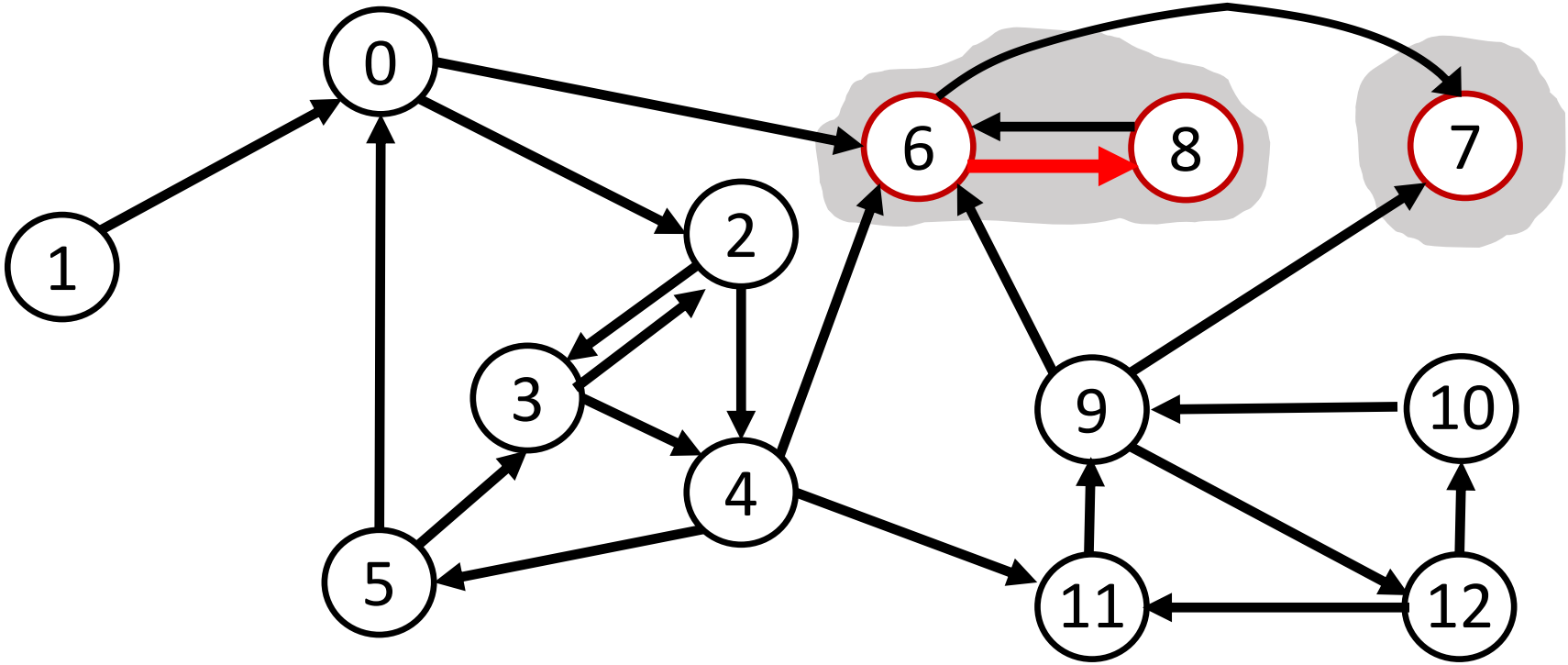
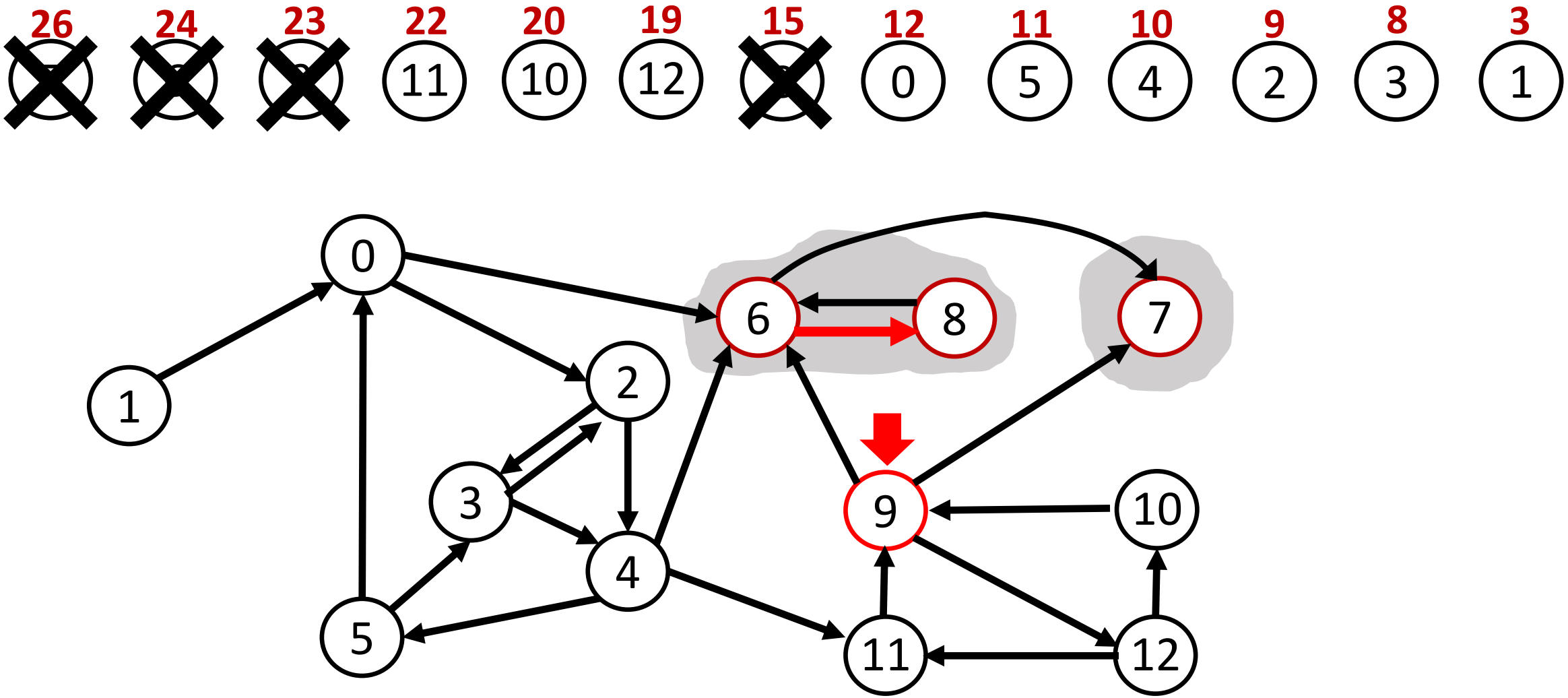
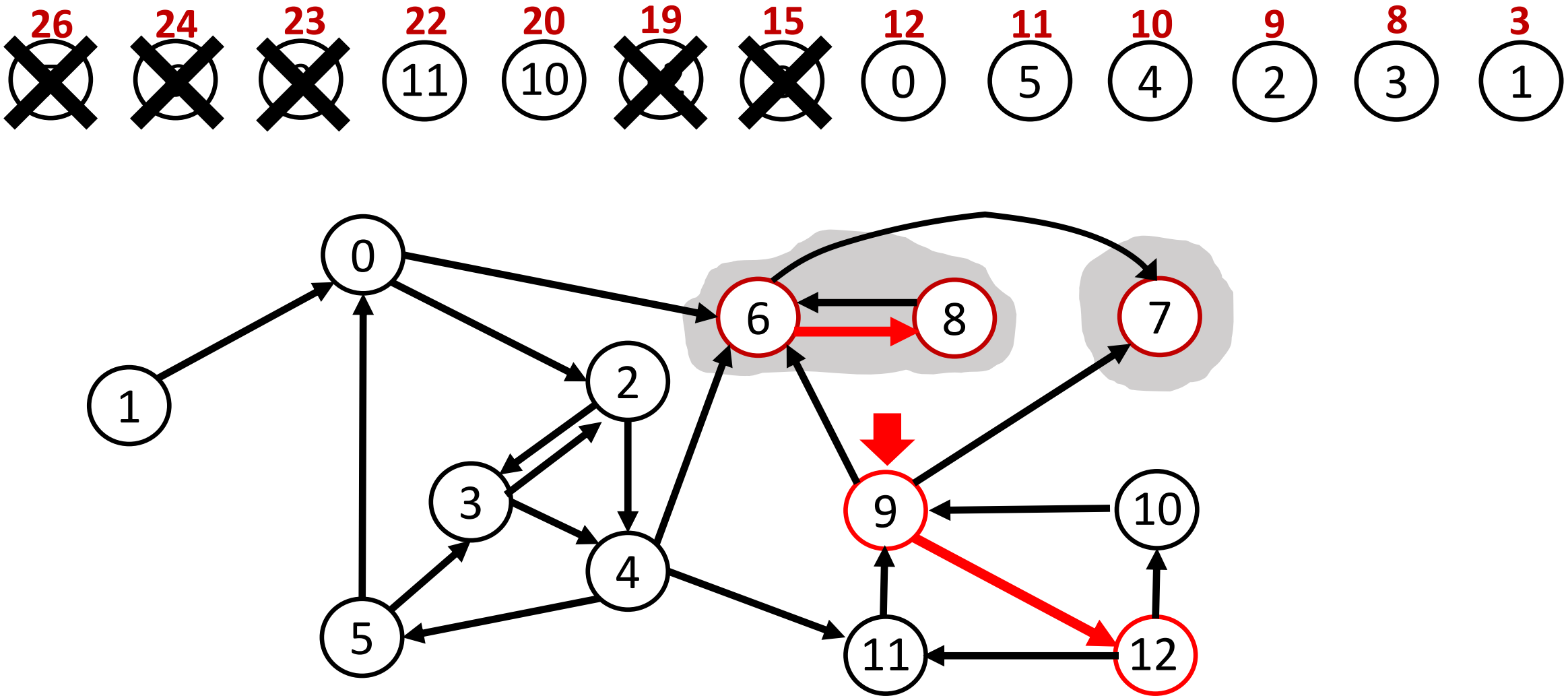
~~1~~	~~6~~	9	11	10	12	~~7~~	0	5	4	2	3	1



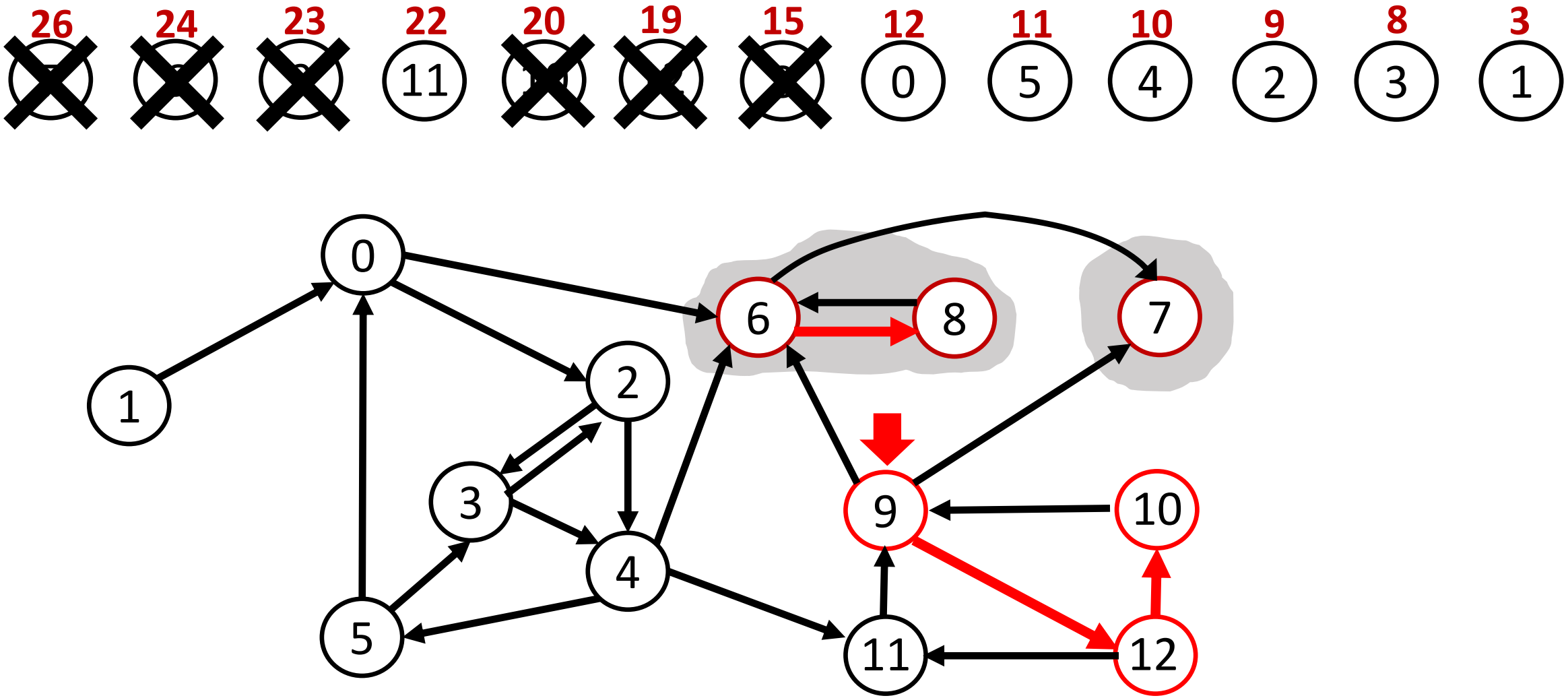
Strong components



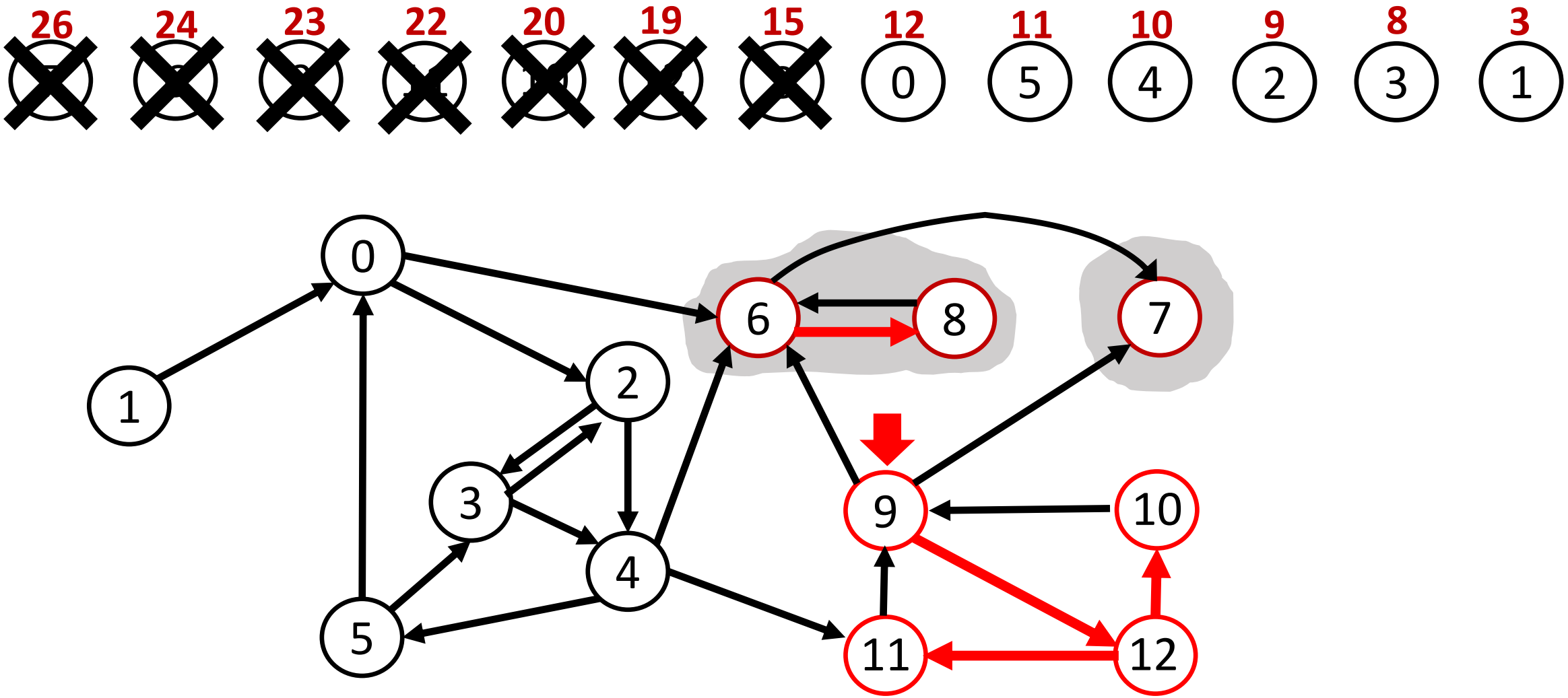
Strong components



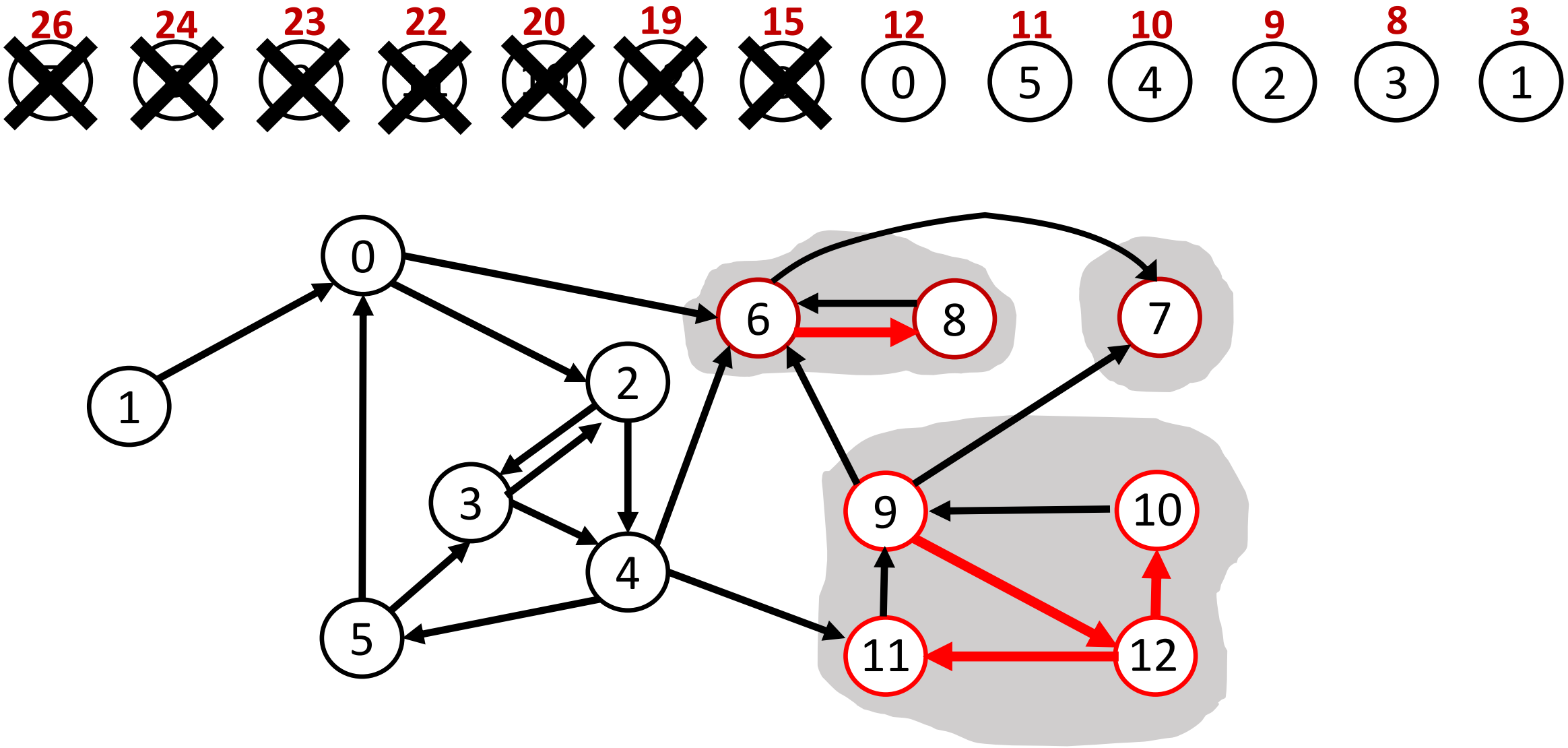
Strong components



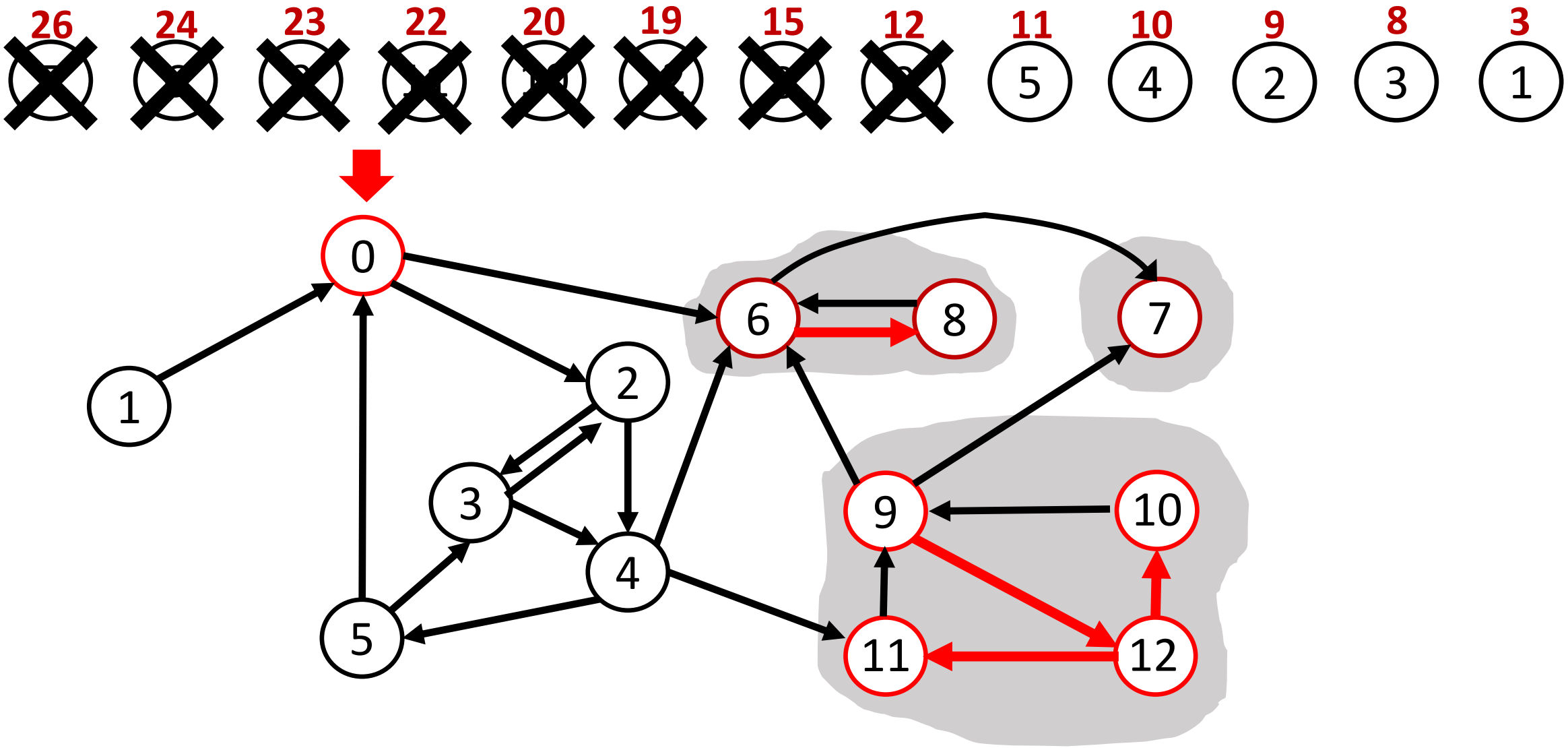
Strong components



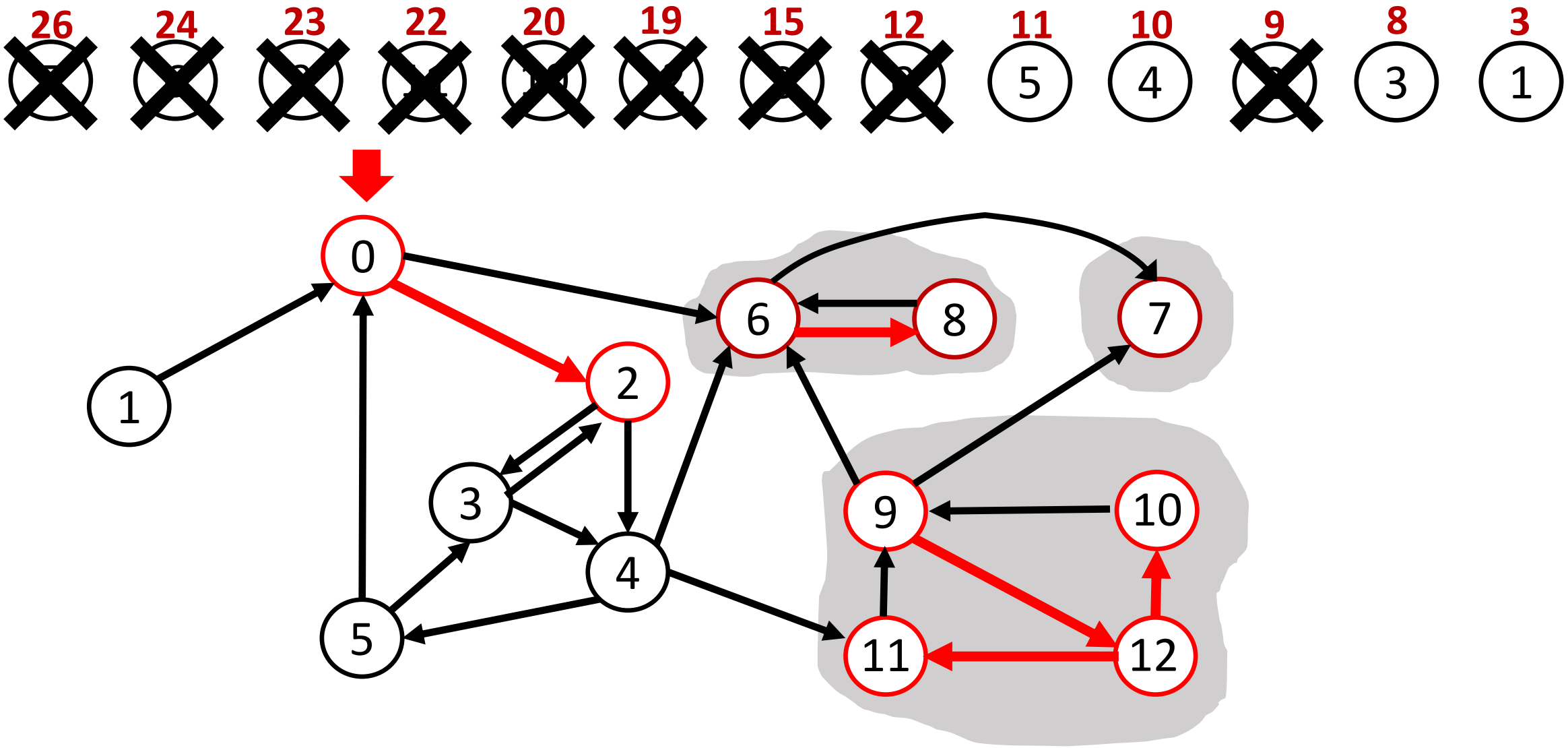
Strong components



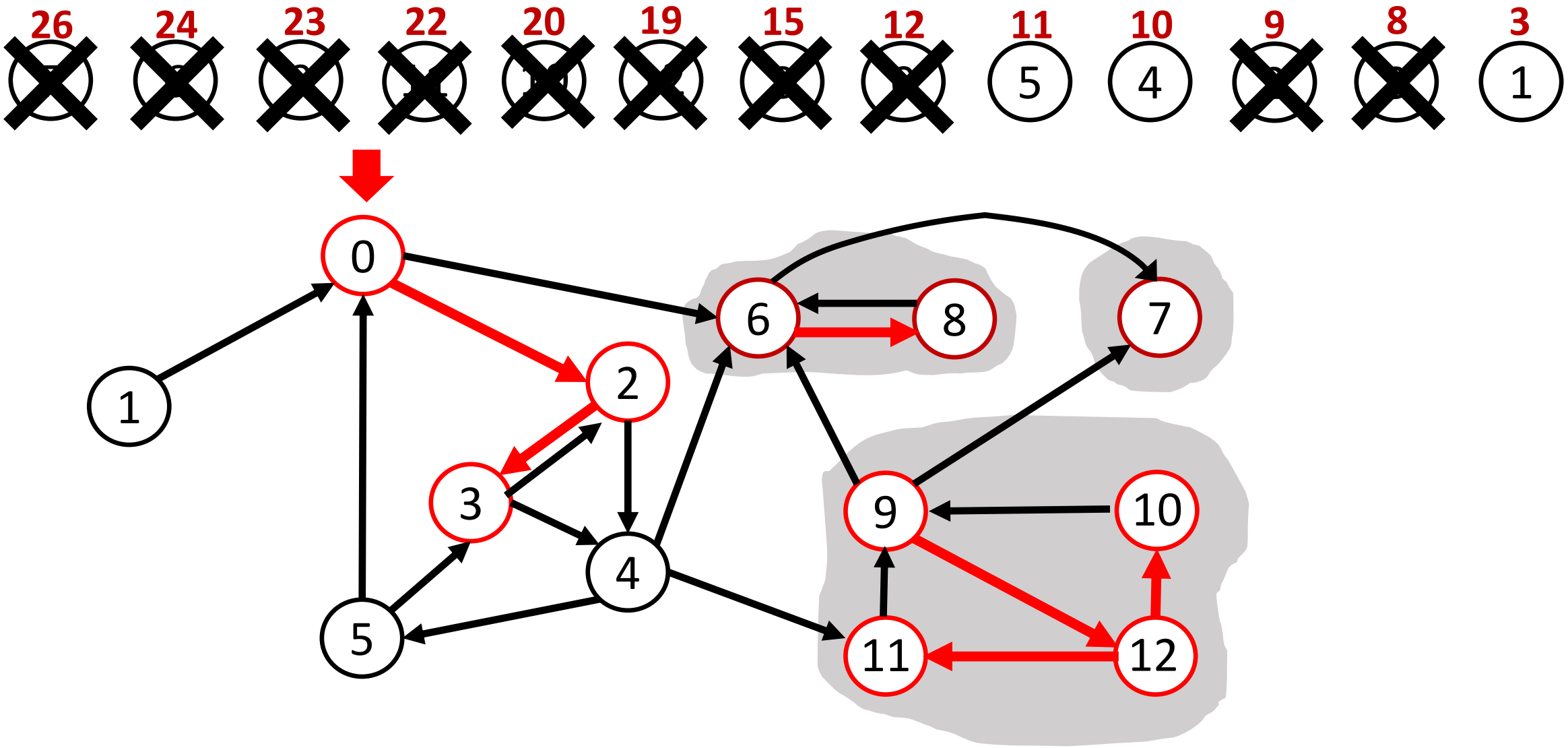
Strong components



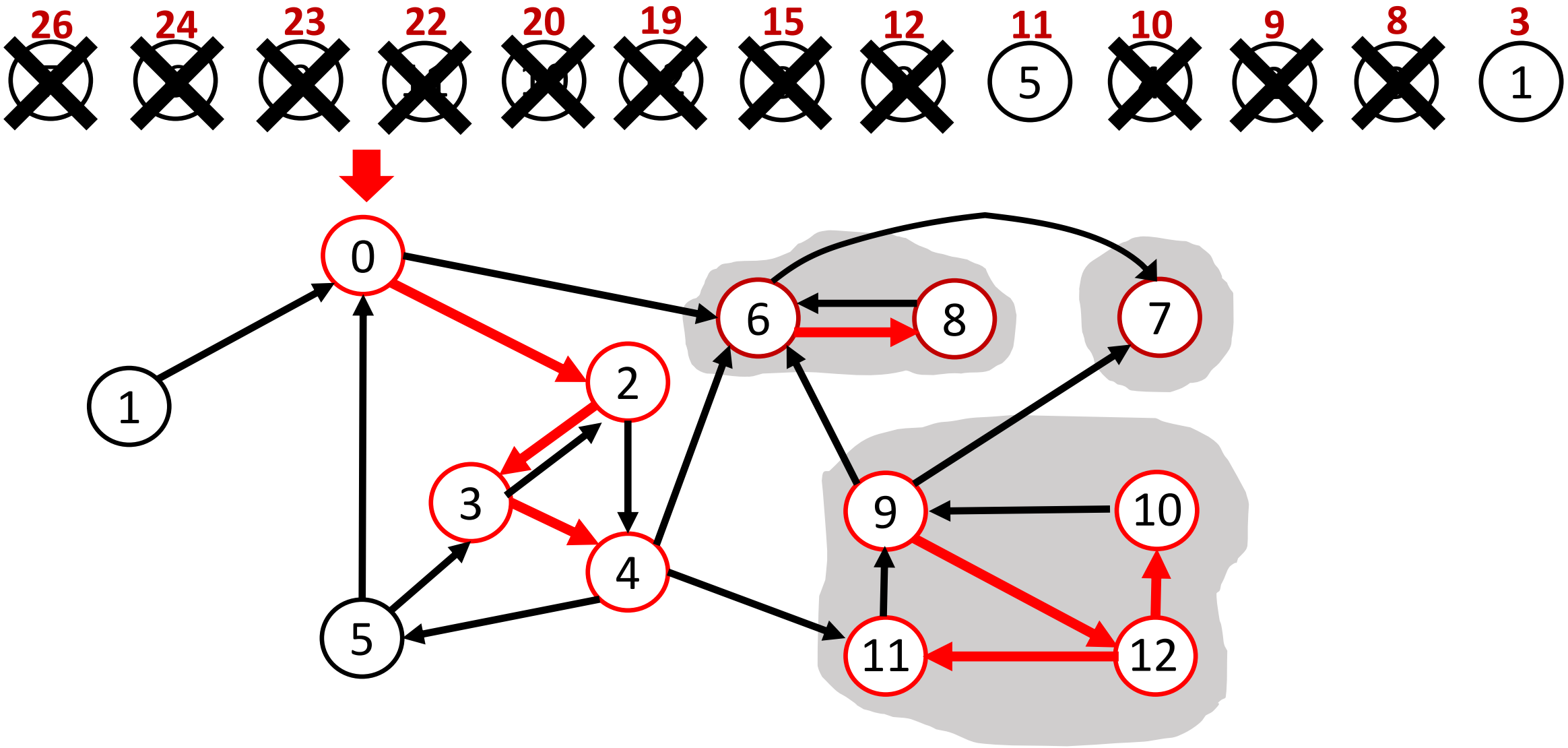
Strong components



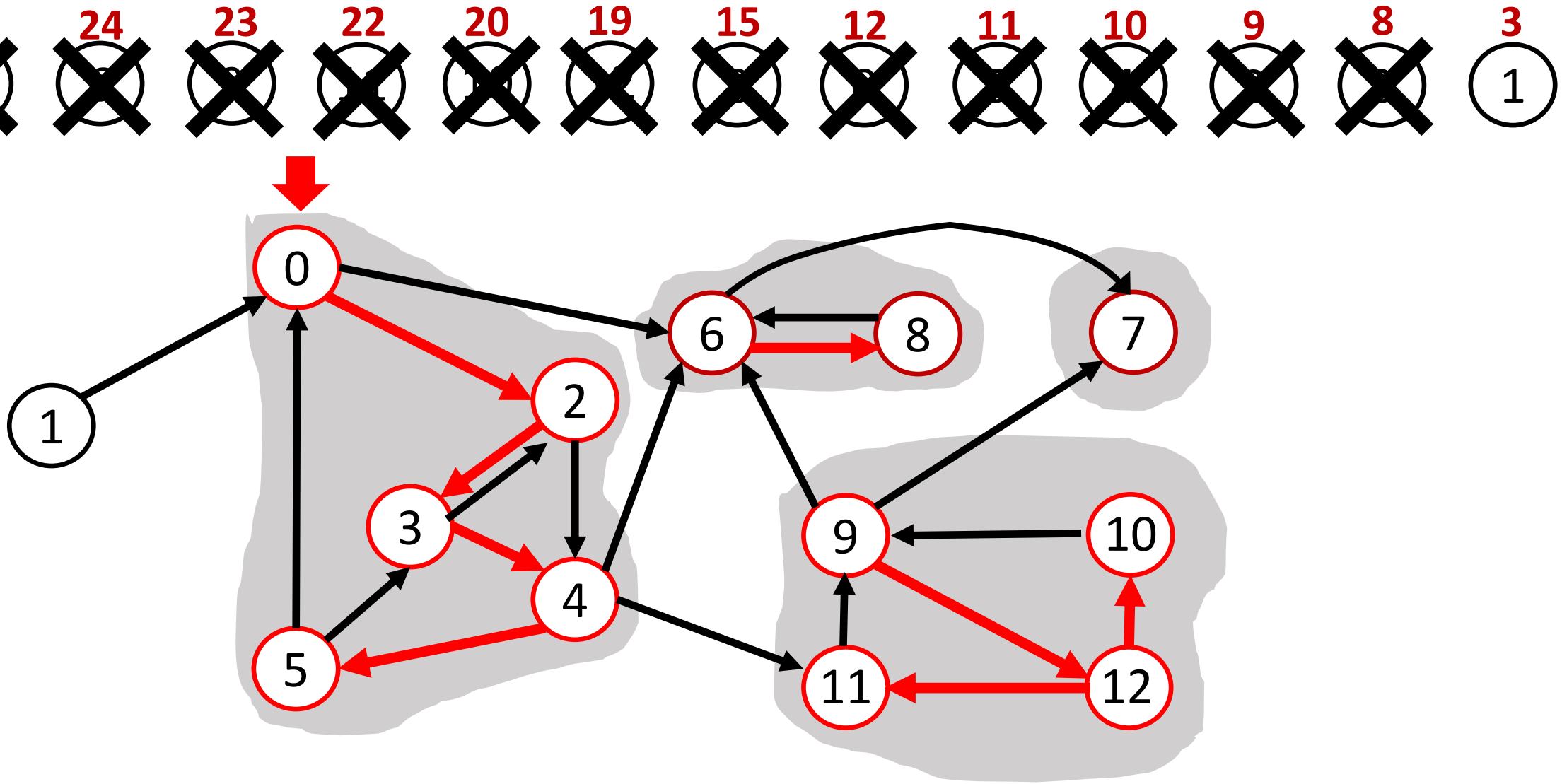
Strong components



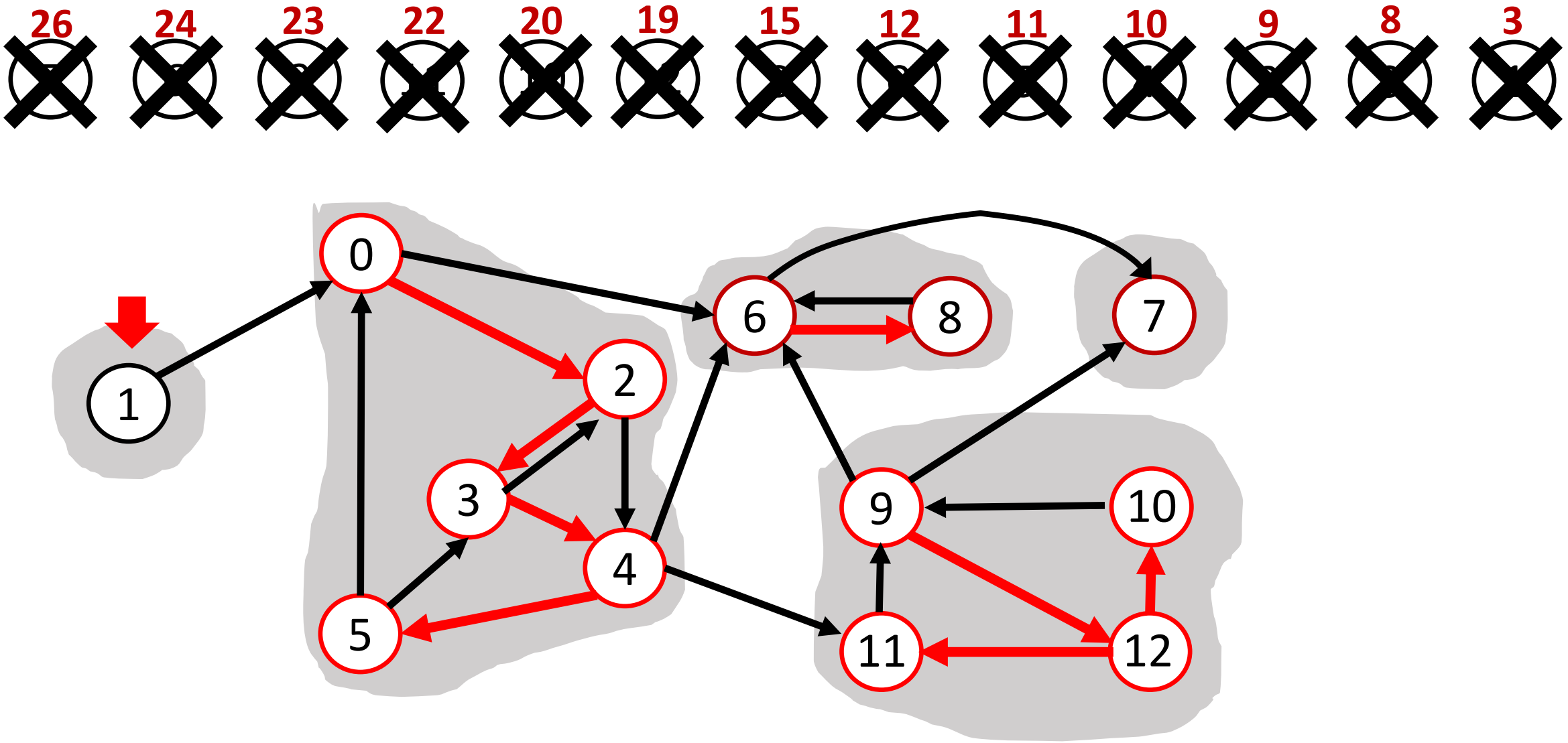
Strong components



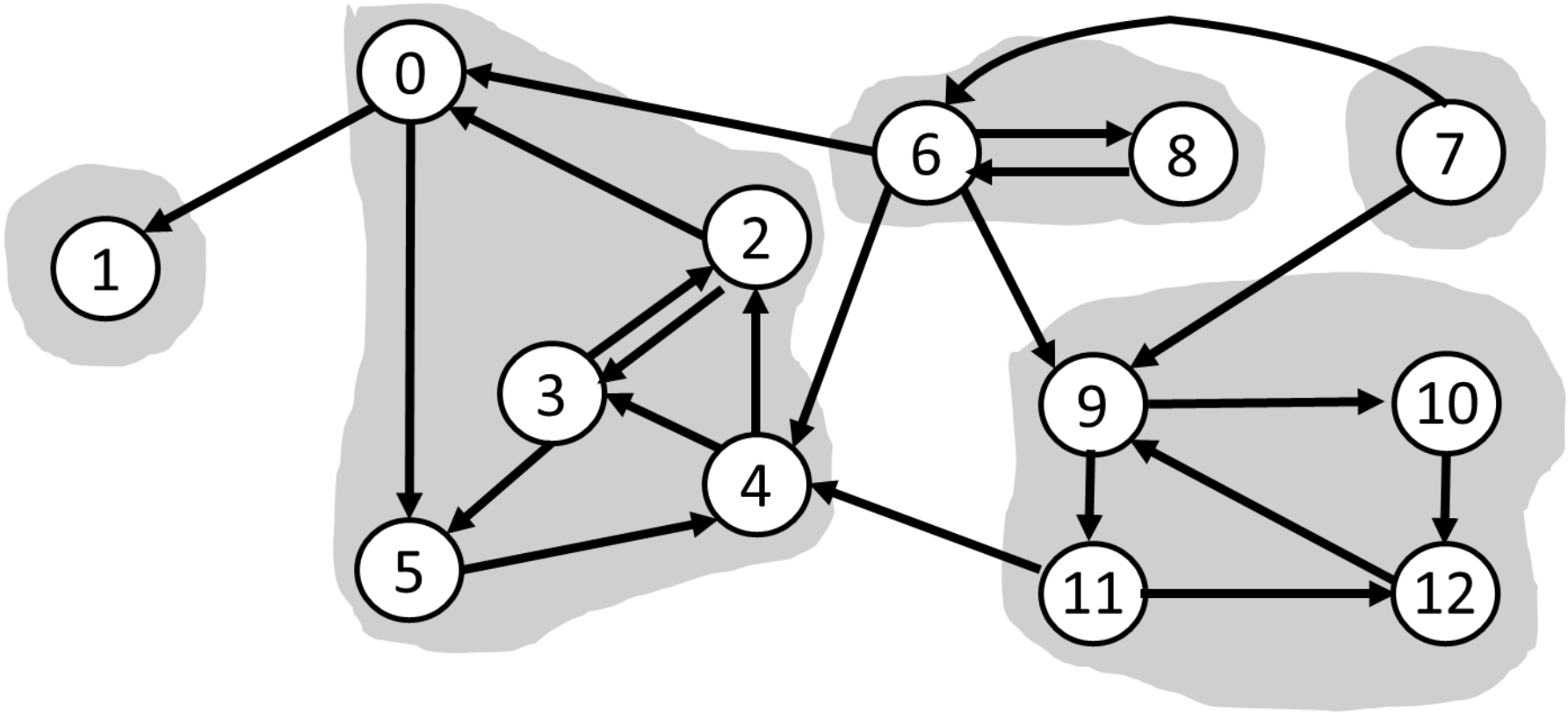
Strong components



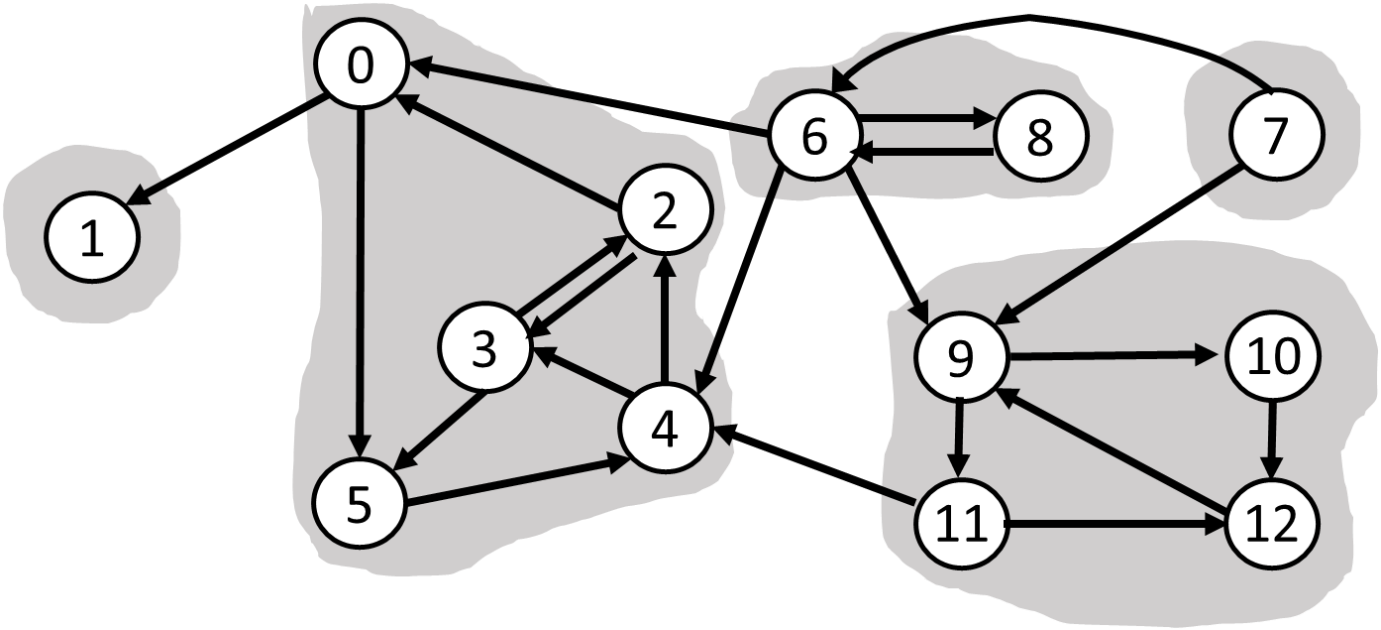
Strong components



Strong components



Strong components



Theorem: [Tarjan 1972] Can find all strong components in $O(m + n)$ time.

SIAM J. COMPUT.
Vol. 1, No. 2, June 1972

DEPTH-FIRST SEARCH AND LINEAR GRAPH ALGORITHMS*

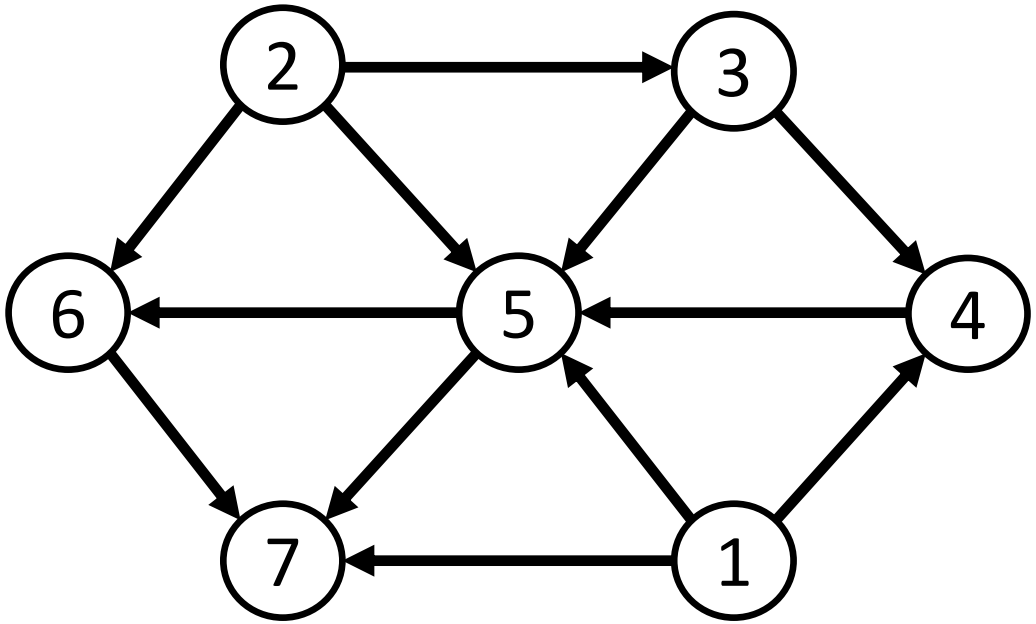
ROBERT TARJAN†

Abstract. The value of depth-first search or "backtracking" as a technique for solving problems is illustrated by two examples. An improved version of an algorithm for finding the strongly connected components of a directed graph and an algorithm for finding the biconnected components of an undirect graph are presented. The space and time requirements of both algorithms are bounded by $k_1V + k_2E + k_3$ for some constants k_1, k_2, and k_3, where V is the number of vertices and E is the number of edges of the graph being examined.

Section 3.6:
DAGs and Topological Ordering

Directed acyclic graphs

- **Def:** A DAG is a directed graph that contains no directed cycles.

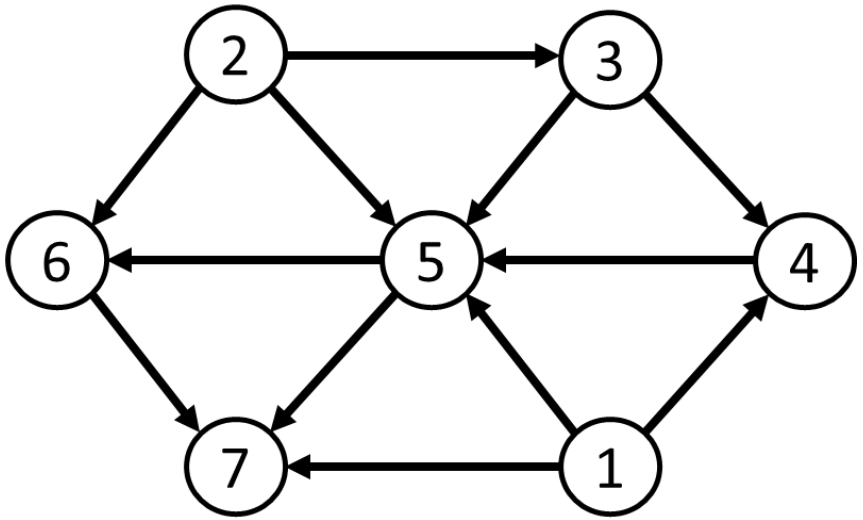


DAG

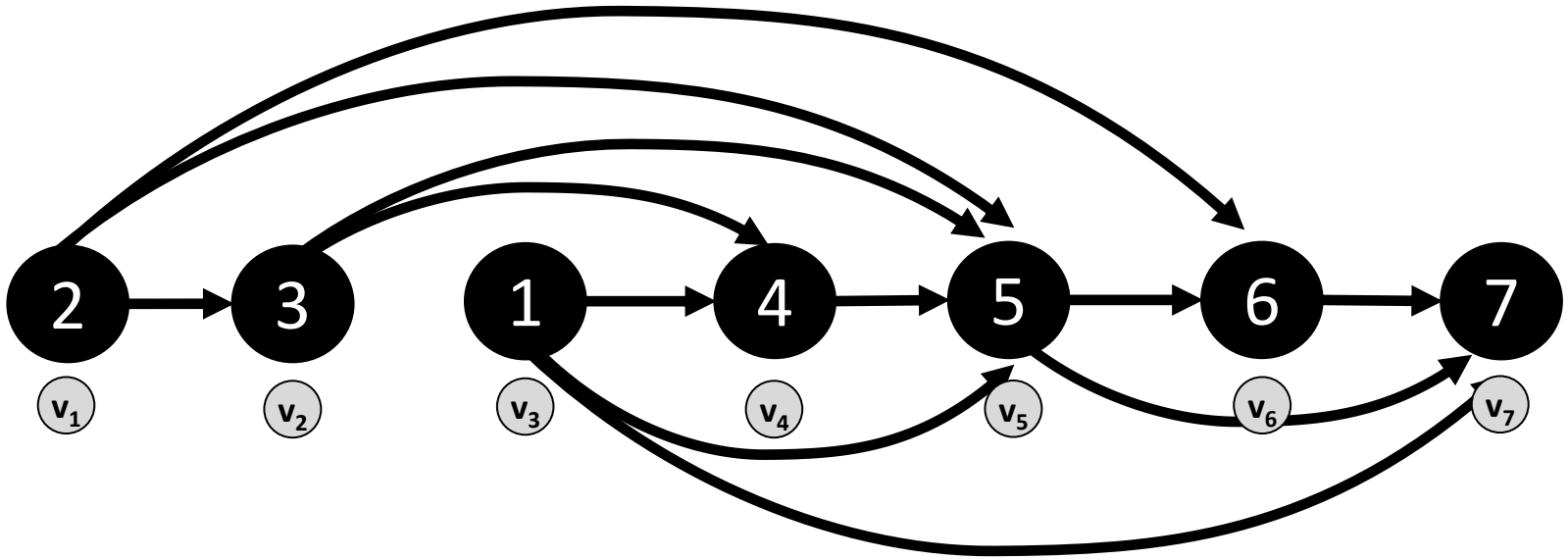
Directed acyclic graphs

- **Def:** A DAG is a directed graph that contains no directed cycles.

- **Def:** A topological order of a directed graph $G = (V, E)$ is an ordering of its nodes as $v_1, v_2, \ldots, v_n$ so that for every edge (v_i, v_j) we have $i < j$.



DAG



Topological Ordering

Precedence constraints

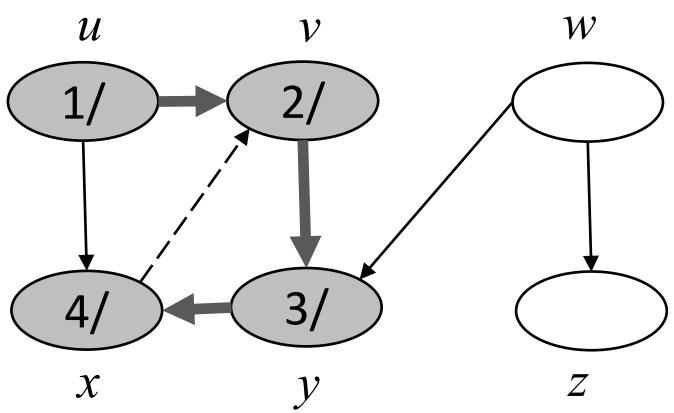
- **Precedence constraints:** Edge (v_i, v_j) means task v_i must occur before v_j.

- Applications.
  - **Course prerequisite graph**: course v_i must be taken before v_j.
  - **Pipeline of computing jobs**: output of job v_i needed to determine input of job v_j.

Topological sorting algorithm: DFS, *the boy savior*

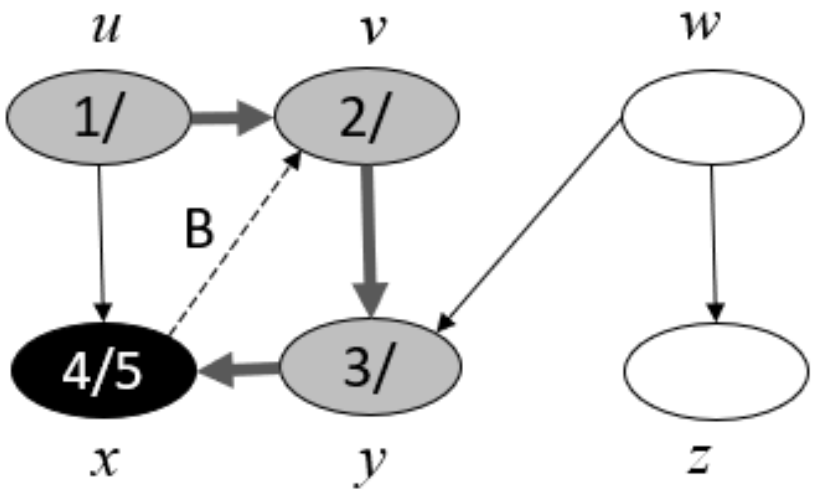
Remember, last time:

Back Edge



Back edges are those edges (u, v) connecting a vertex u to an ancestor vertex v in a depth-first tree. Self-loop (edge (u, u)), is also considered as Back edge

Three colors



Topological sorting algorithm: DFS, *the boy savior*

DFS(G)

```
for each vertex u ∈ V
    u.color = WHITE
    ~~u.π = NIL~~
~~time = 0~~
for each vertex u ∈ V
    if u.color == WHITE
        DFS-VISIT(G, u)
```

DFS-VISIT(G, u)

```
~~time = time + 1~~
~~u.d = time~~
u.color = GRAY
for each v ∈ Adj[u]
    if v.color == WHITE
        ~~v.π = u~~
        DFS-VISIT(G, v)
~~u.color = BLACK~~
~~time = time + 1~~
~~u.f = time~~
```

New ***if*** block to check if it a **BACK Edge**

Topological sorting algorithm: DFS, *the boy savior*

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for each vertex u ∈ V
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L=[ ]
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DFS-VISIT(G, u)

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u.color = GRAY
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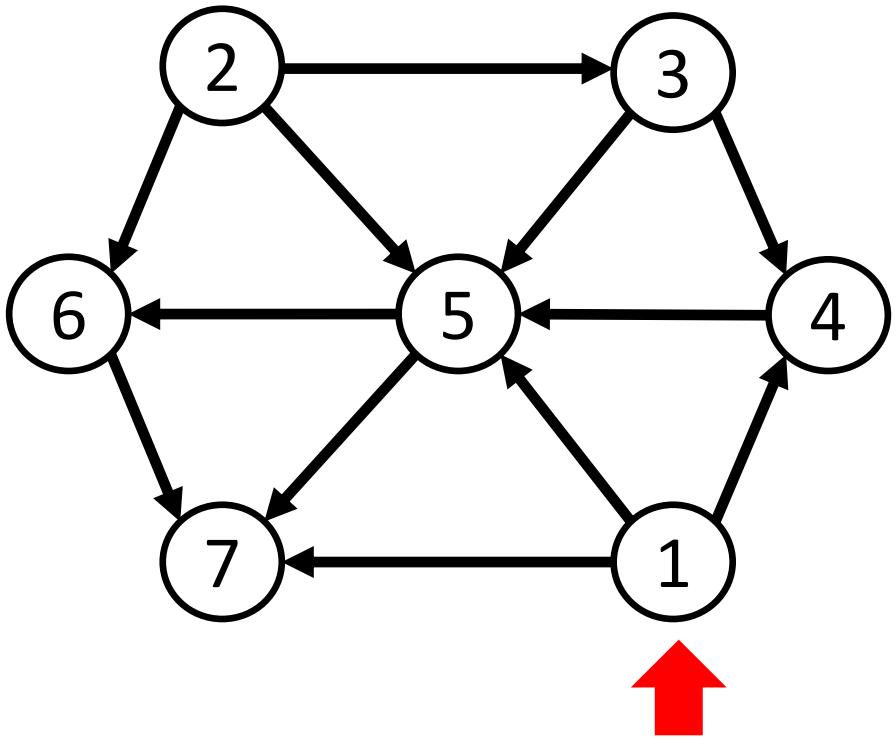
                DFS-VISIT(G, v)
        if v.color == GRAY
                Break
L-> add it to the front (u)
```



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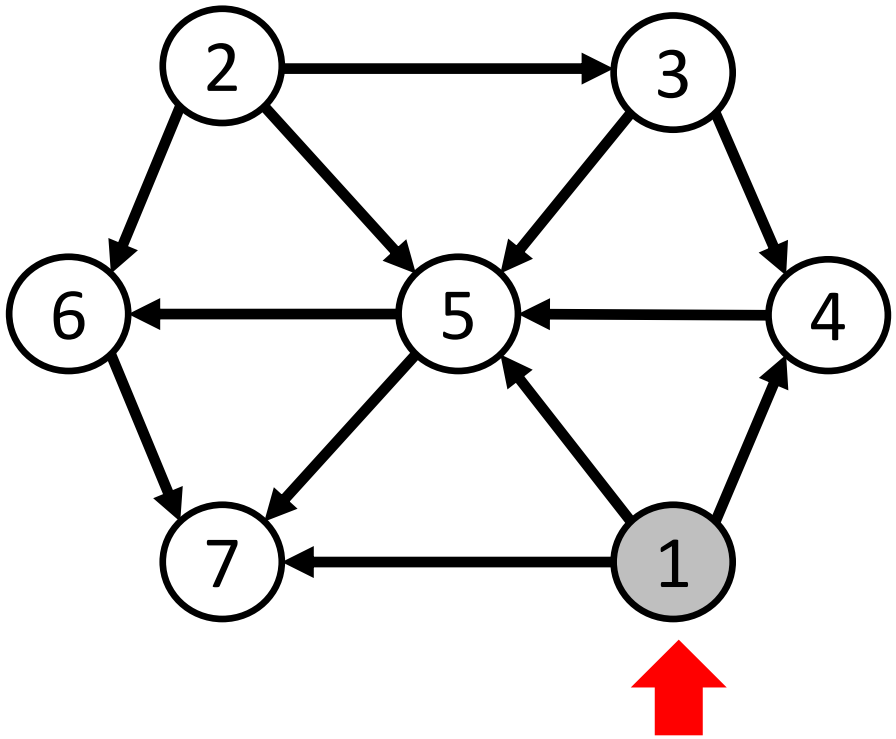

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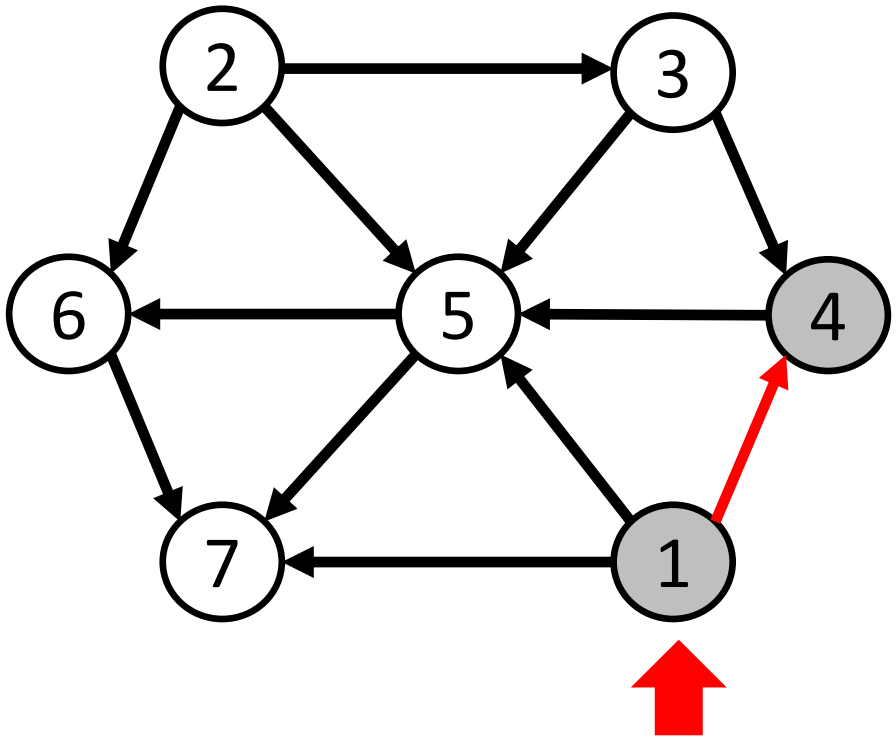

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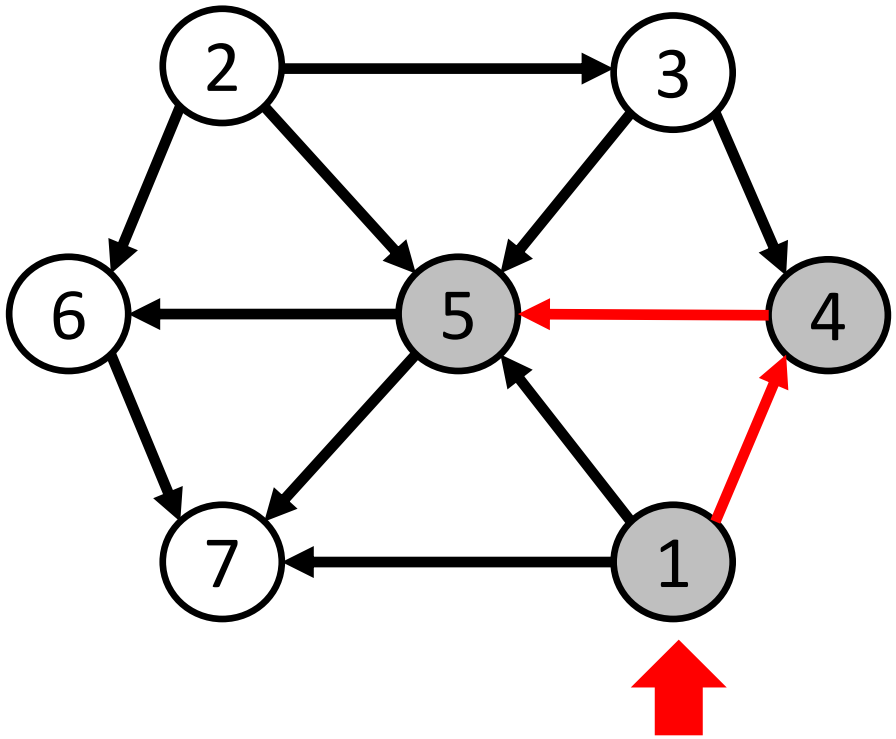

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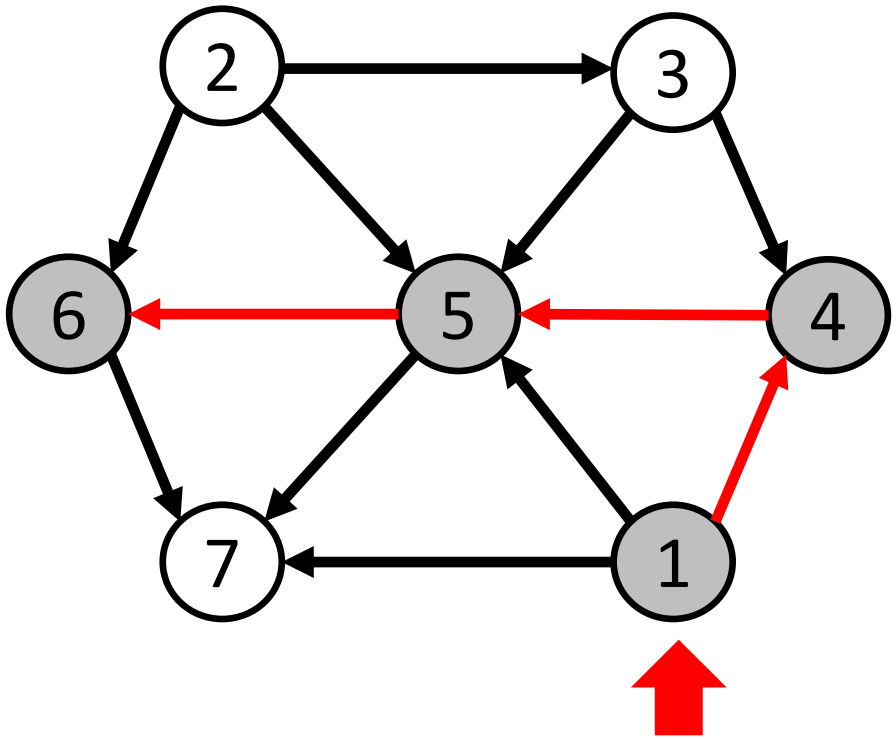

Topological sorting algorithm: DFS, *the boy savior*

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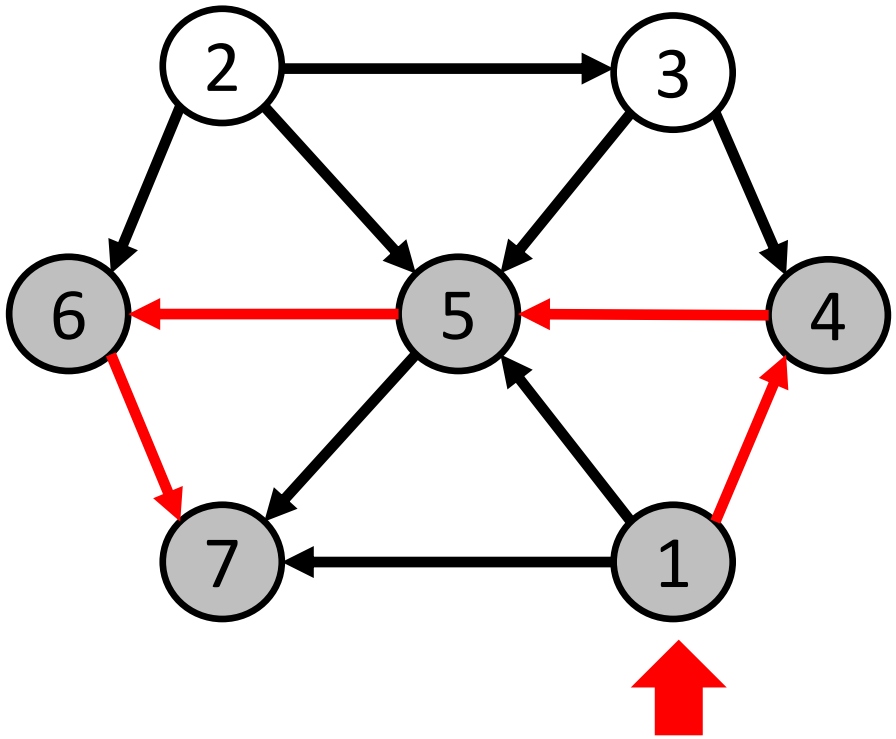


Topological sorting algorithm: DFS, *the boy savior*

L 7

DFS(G)

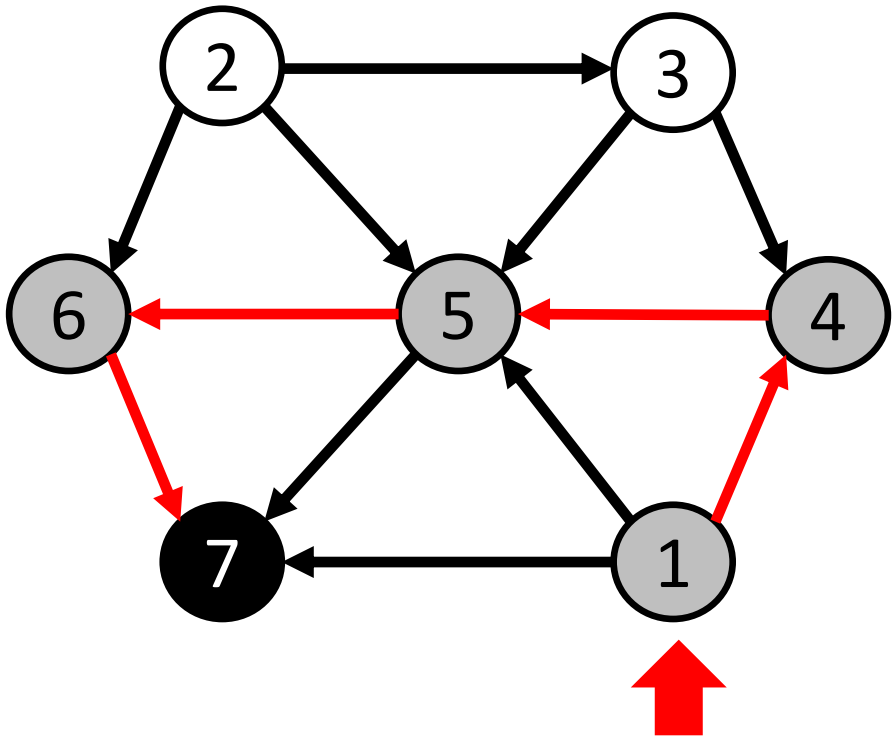
```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
for each vertex u ∈ V
        if u.color == WHITE
                DFS-VISIT(G, u)
```

DFS-VISIT(G, u)

```

u.color = GRAY
for each v ∈ Adj[u]
        if v.color == WHITE
                DFS-VISIT(G, v)
        if v.color == GRAY
                Break
L-> add it to the front (u)
```



Topological sorting algorithm: DFS, *the boy savior*

L 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
for each vertex u ∈ V
        if u.color == WHITE
                DFS-VISIT(G, u)
```

DFS-VISIT(G, u)

```

u.color = GRAY
for each v ∈ Adj[u]
        if v.color == WHITE
                DFS-VISIT(G, v)
        if v.color == GRAY
                Break
L-> add it to the front (u)
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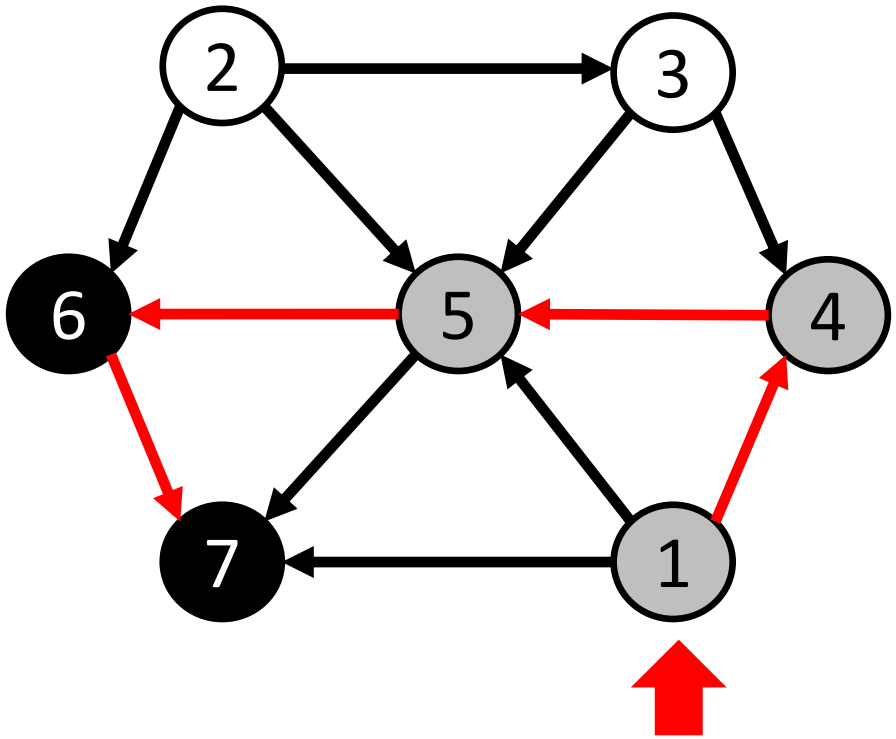

Topological sorting algorithm: DFS, *the boy savior*

L 5 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
for each vertex u ∈ V
        if u.color == WHITE
                DFS-VISIT(G, u)
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DFS-VISIT(G, u)

```

u.color = GRAY
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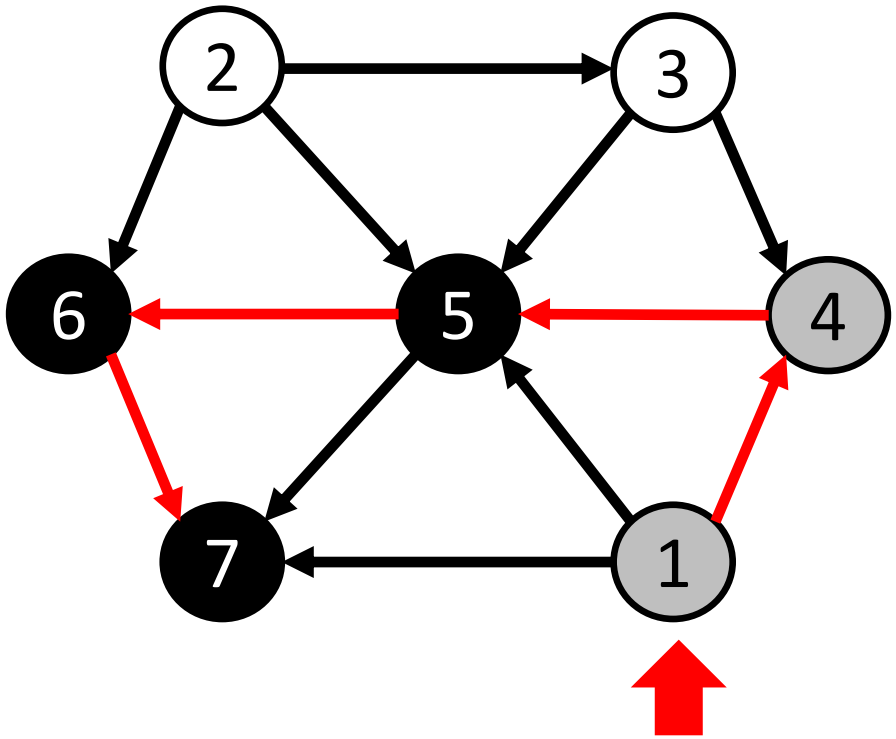

Topological sorting algorithm: DFS, *the boy savior*

L 4 5 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
for each vertex u ∈ V
        if u.color == WHITE
                DFS-VISIT(G, u)
```

DFS-VISIT(G, u)

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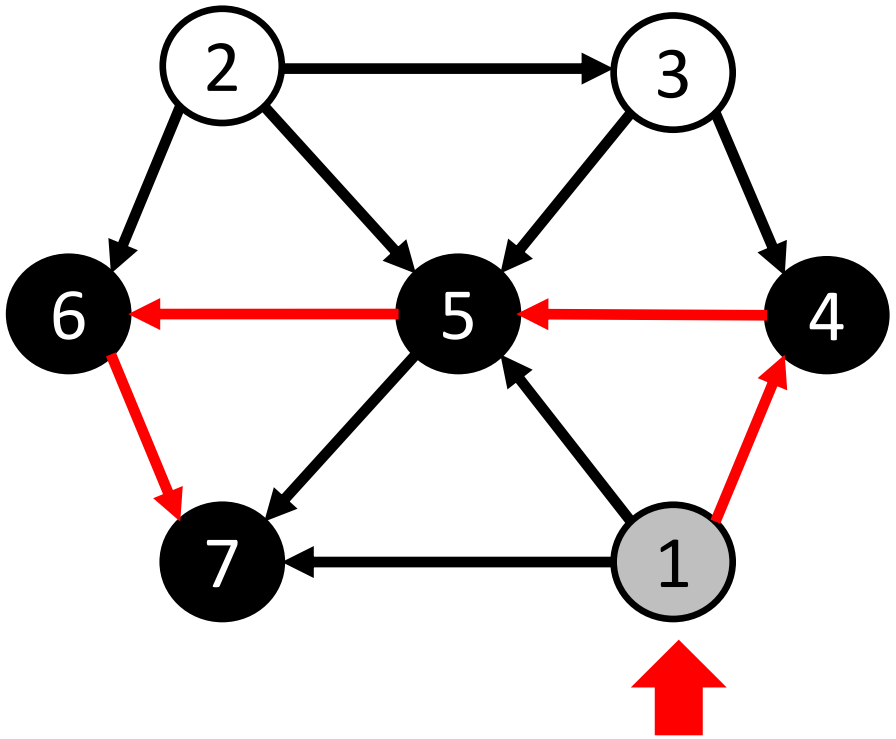

Topological sorting algorithm: DFS, *the boy savior*

L 1 4 5 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
for each vertex u ∈ V
        if u.color == WHITE
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```

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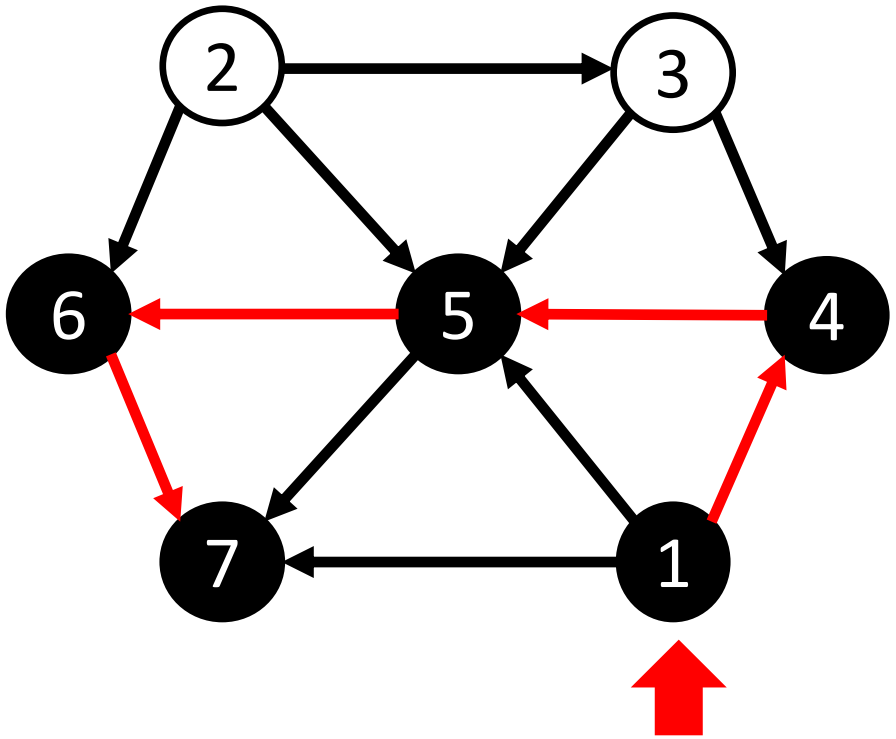

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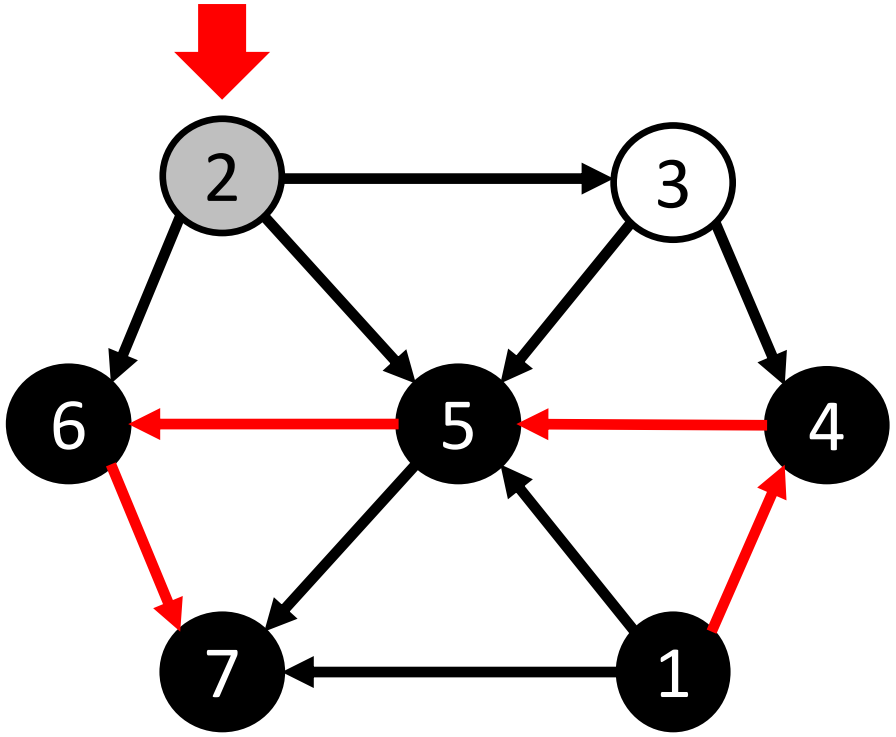

Topological sorting algorithm: DFS, *the boy savior*

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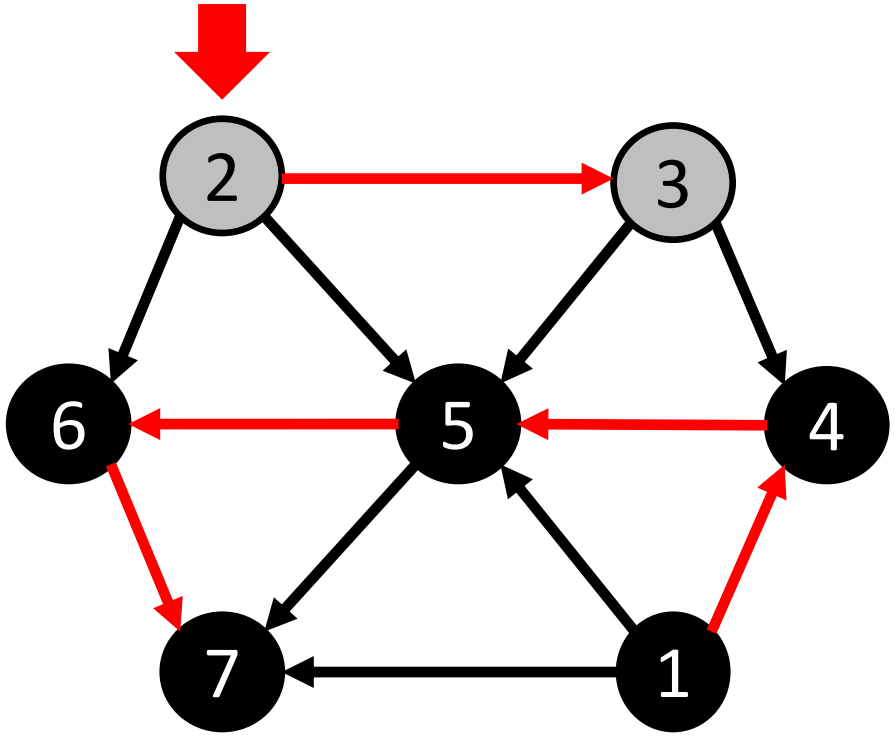

Topological sorting algorithm: DFS, *the boy savior*

L 3 1 4 5 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
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DFS-VISIT(G, u)

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                Break
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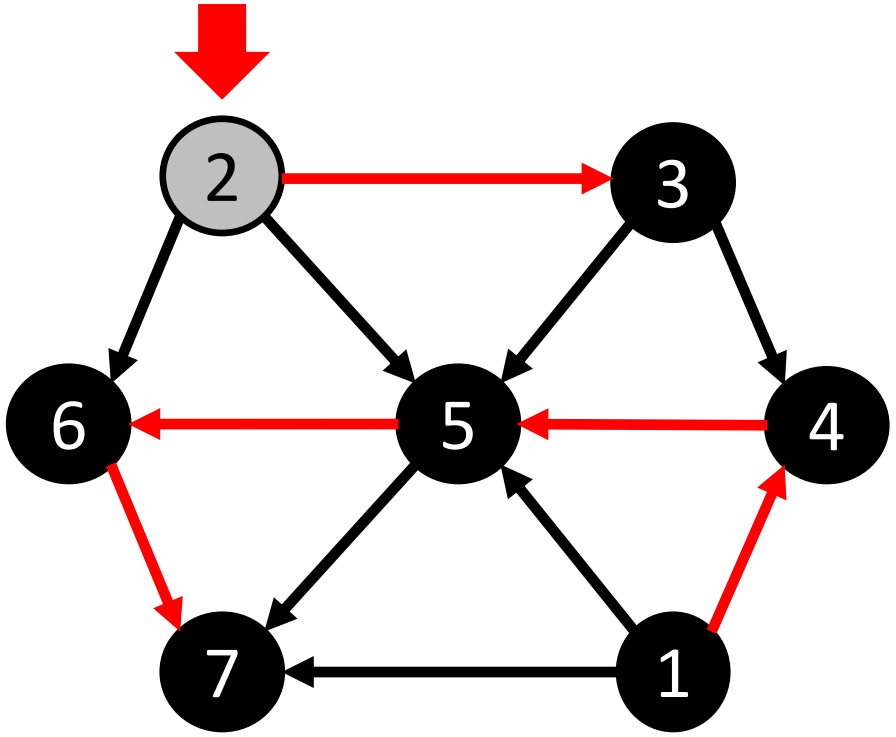
```



Topological sorting algorithm: DFS, *the boy savior*

L 2 3 1 4 5 6 7

DFS(G)

```
for each vertex u ∈ V
        u.color = WHITE

L=[ ]
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                DFS-VISIT(G, u)
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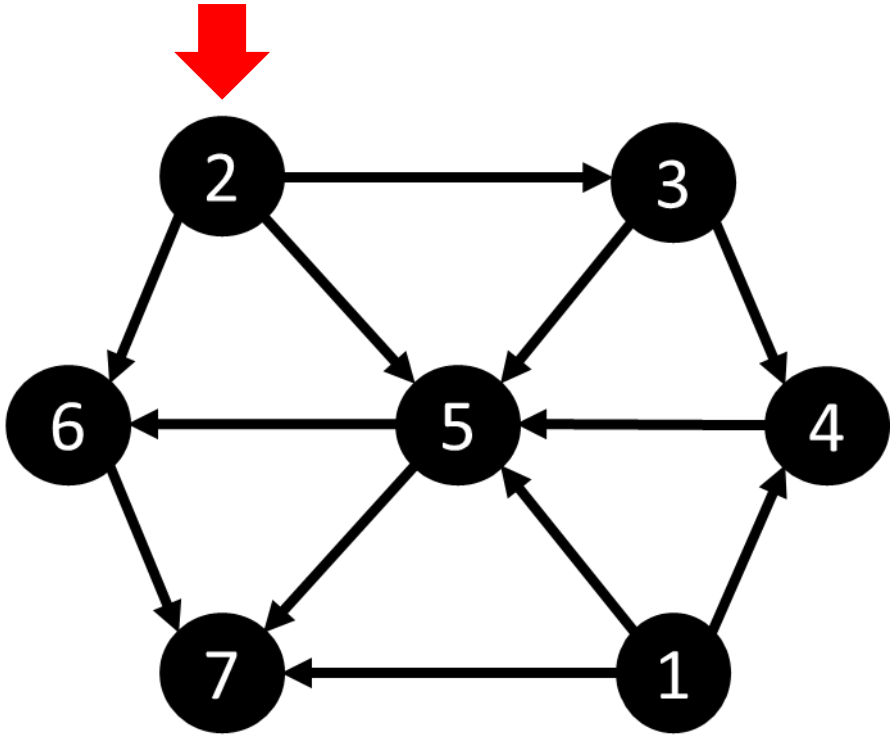
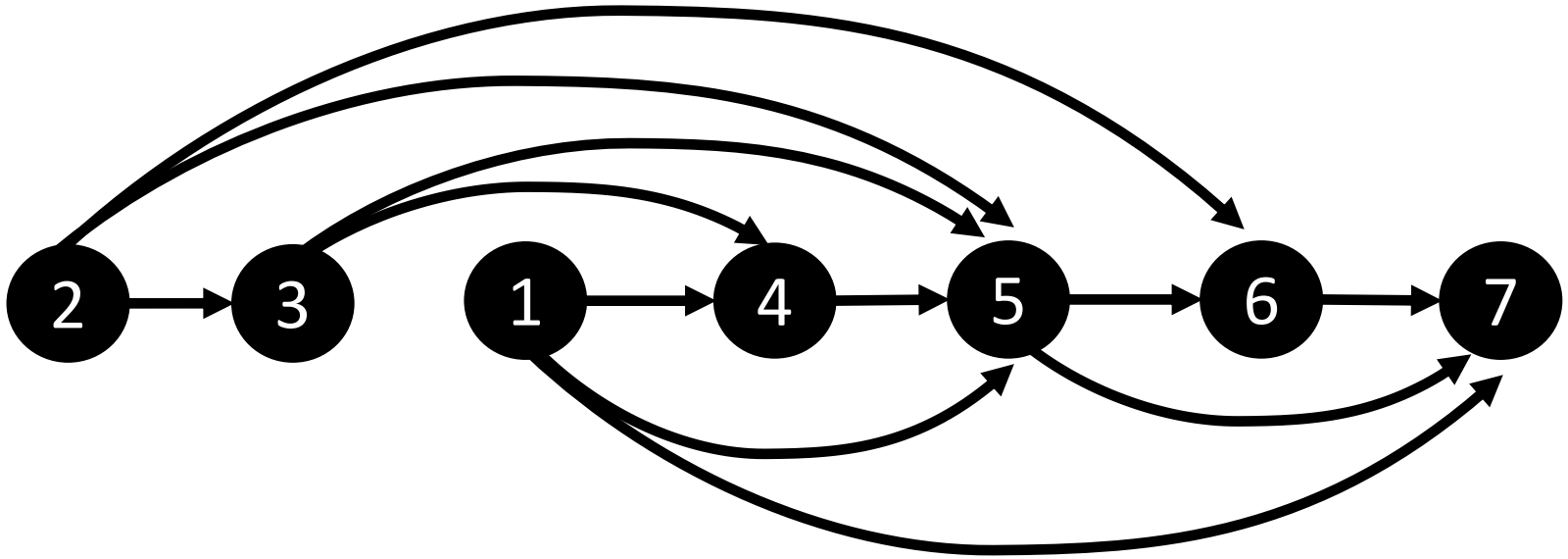
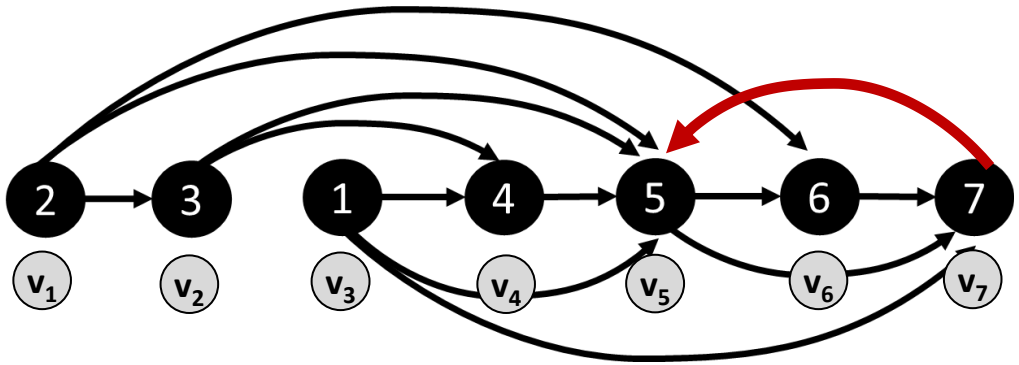


Topological sorting algorithm: DFS, *the boy savior*



Directed acyclic graphs

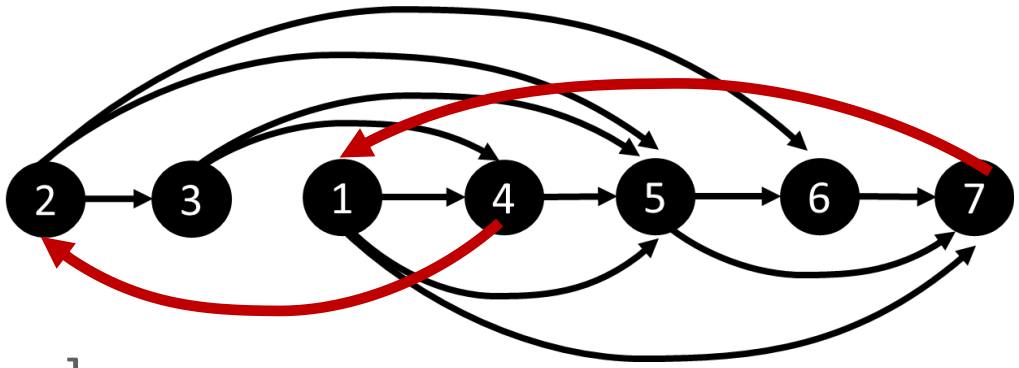
- **Lemma.** If G has a topological order, then G is a DAG.



- **Pf.** [by contradiction]
  - Suppose that G has a topological order, and that G also has a directed cycle C. Let's see what happens.
  - By definition, every edge (v_i, v_j) in topological order, i < j.
  - On the other hand, since (v_7, v_5) is an edge, we must have $j < i$, **a contradiction**.

Directed acyclic graphs

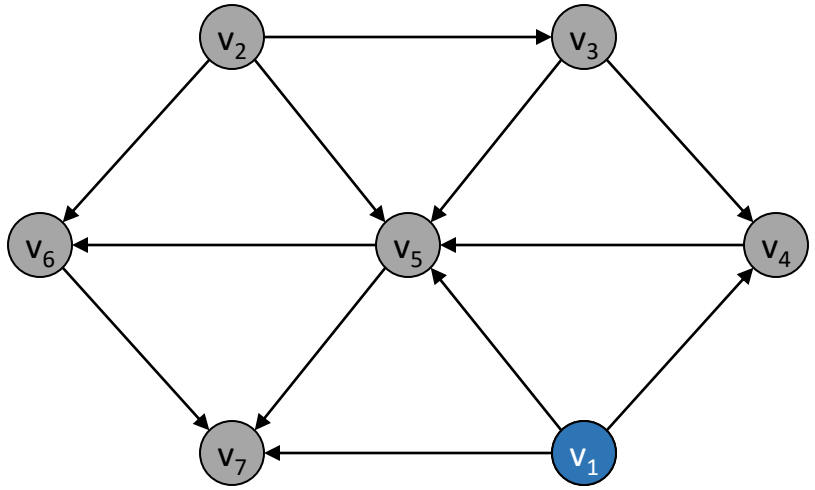
- Lemma. If G is a DAG, then G has a node with no entering edges.



- Pf. [by contradiction]
  - Suppose that G is a DAG and every node has at least one entering edge. Let's see what happens.
  - Graph **G** will have a **cycle**

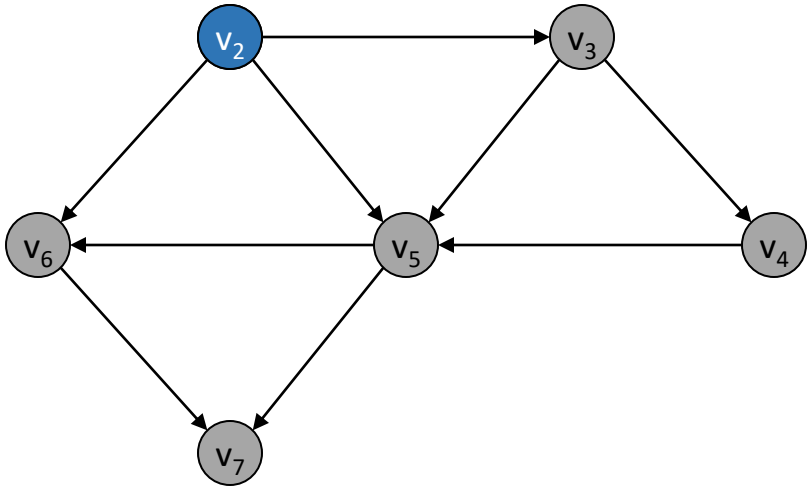
Another Topological Ordering Algorithm: Example

```
To compute a topological ordering of G:
Find a node v with no incoming edges and order it first
Delete v from G
Recursively compute a topological ordering of G−{v}
  and append this order after v
```



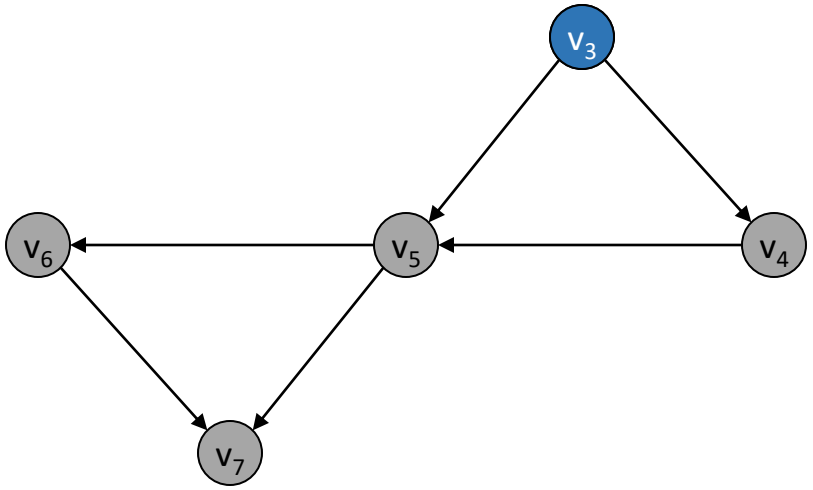
Topological order:

Topological Ordering Algorithm: Example



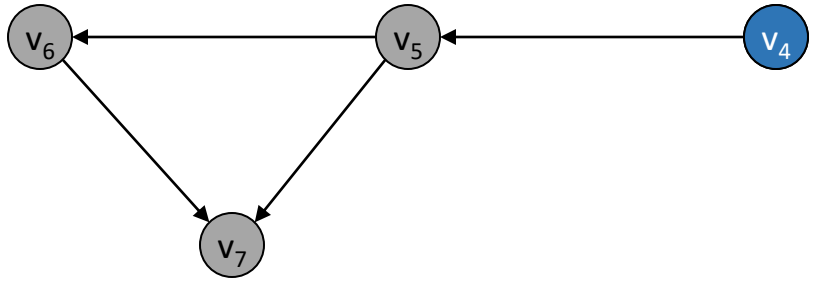
Topological order: v_1

Topological Ordering Algorithm: Example



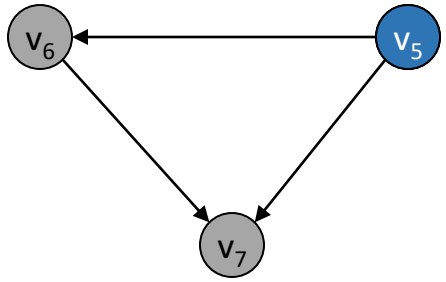
Topological order: v_1, v_2

Topological Ordering Algorithm: Example



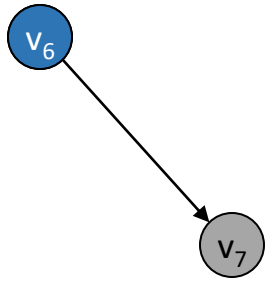
Topological order: v_1, v_2, v_3

Topological Ordering Algorithm: Example



Topological order: v_1, v_2, v_3, v_4

Topological Ordering Algorithm: Example

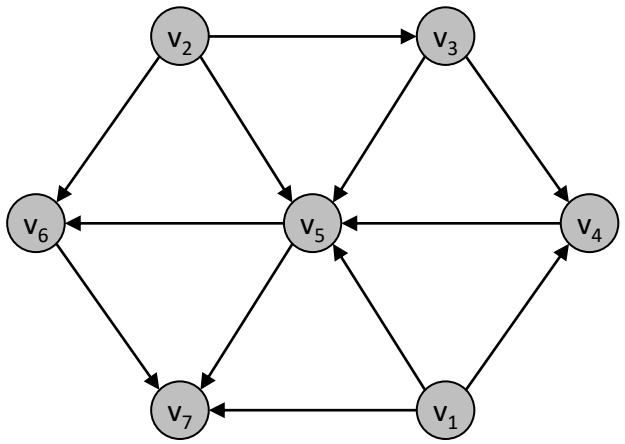


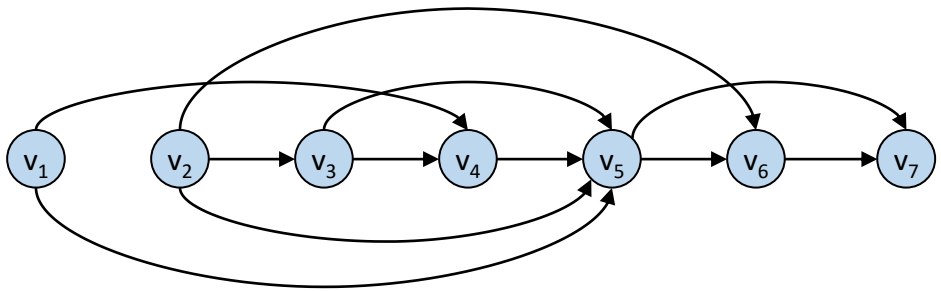
Topological order: v_1, v_2, v_3, v_4, v_5

Topological Ordering Algorithm: Example

Topological order: $v_1, v_2, v_3, v_4, v_5, v_6$

Topological Ordering Algorithm: Example





Topological order: $v_1, v_2, v_3, v_4, v_5, v_6, v_7$.

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over